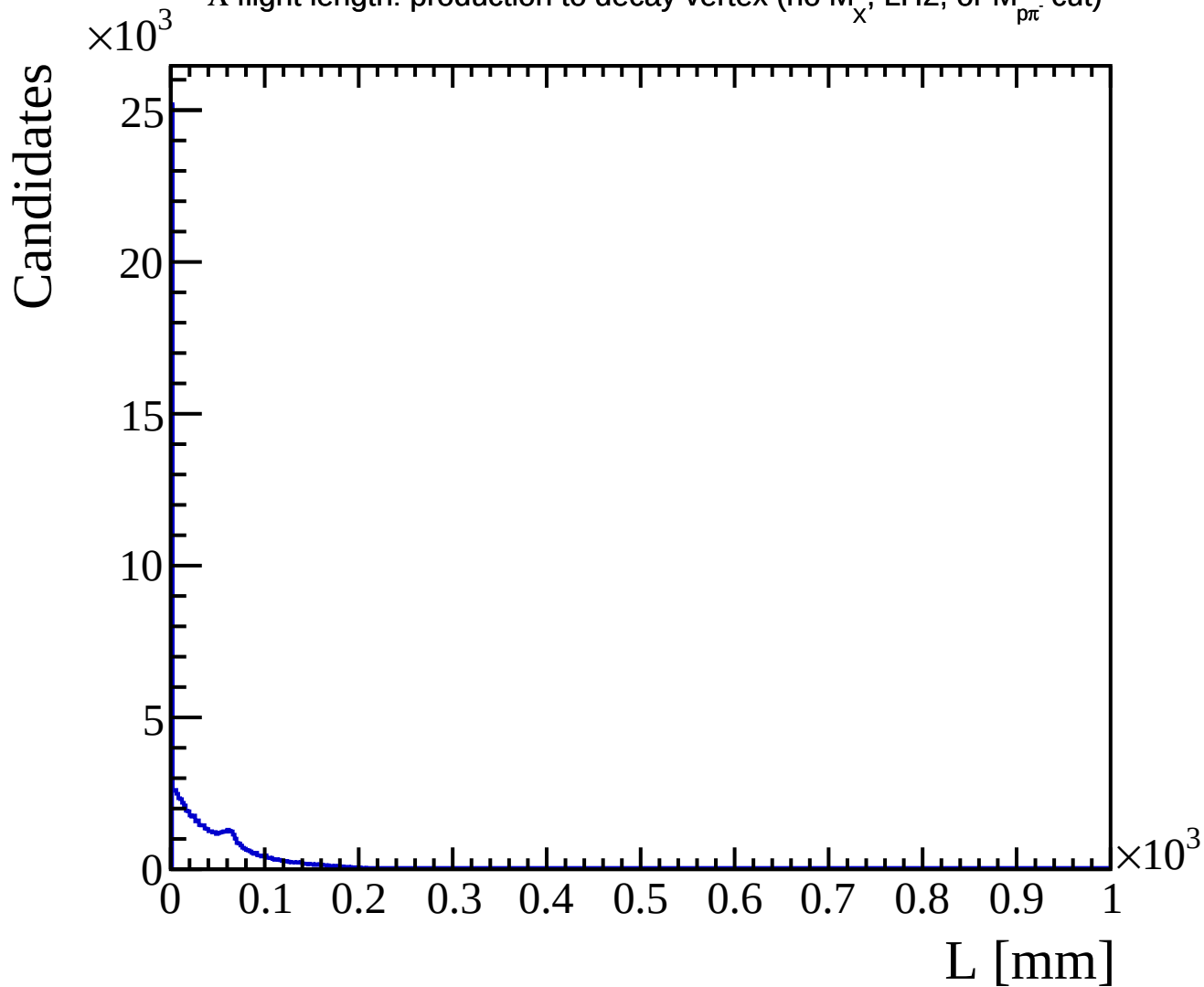
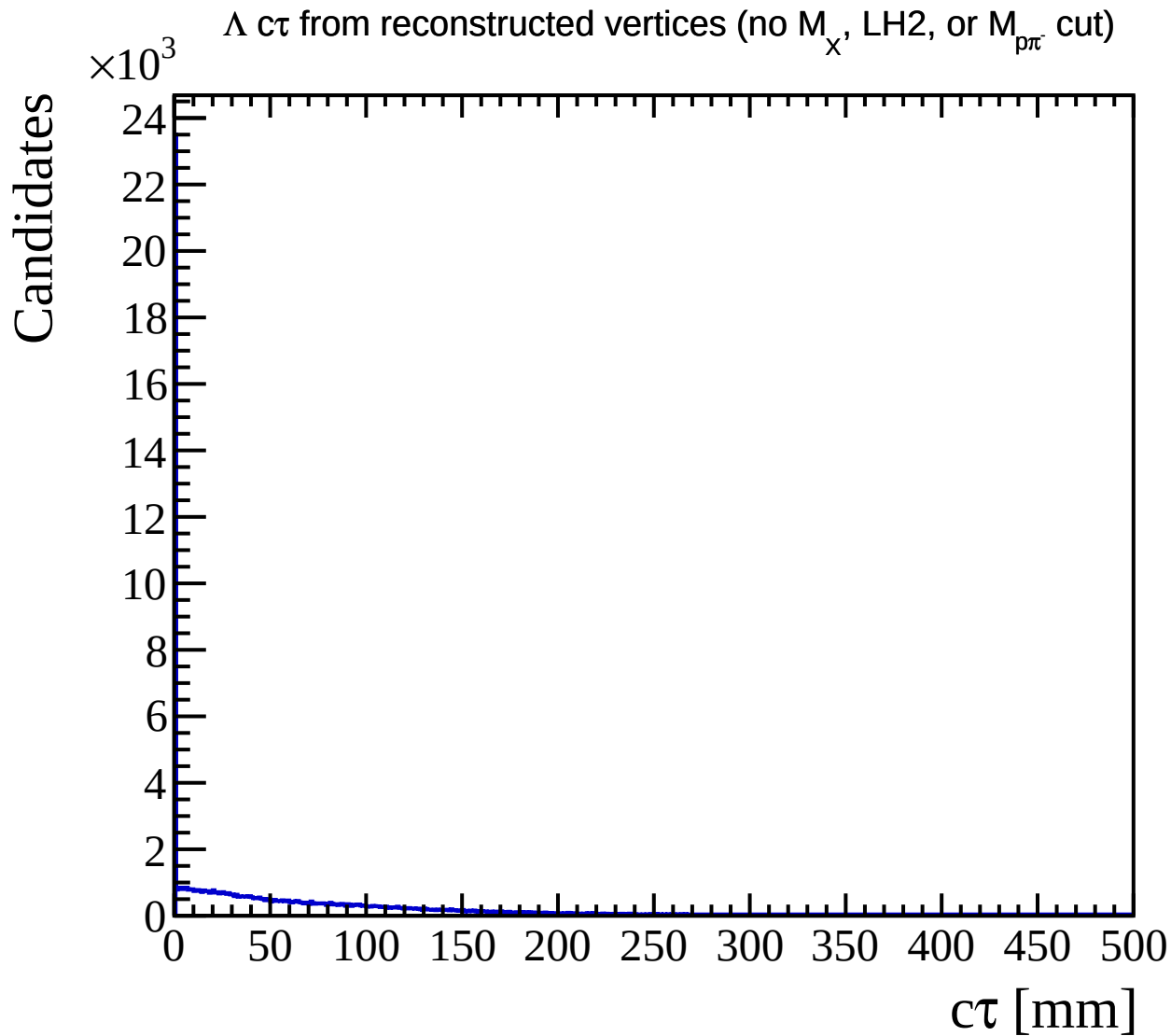
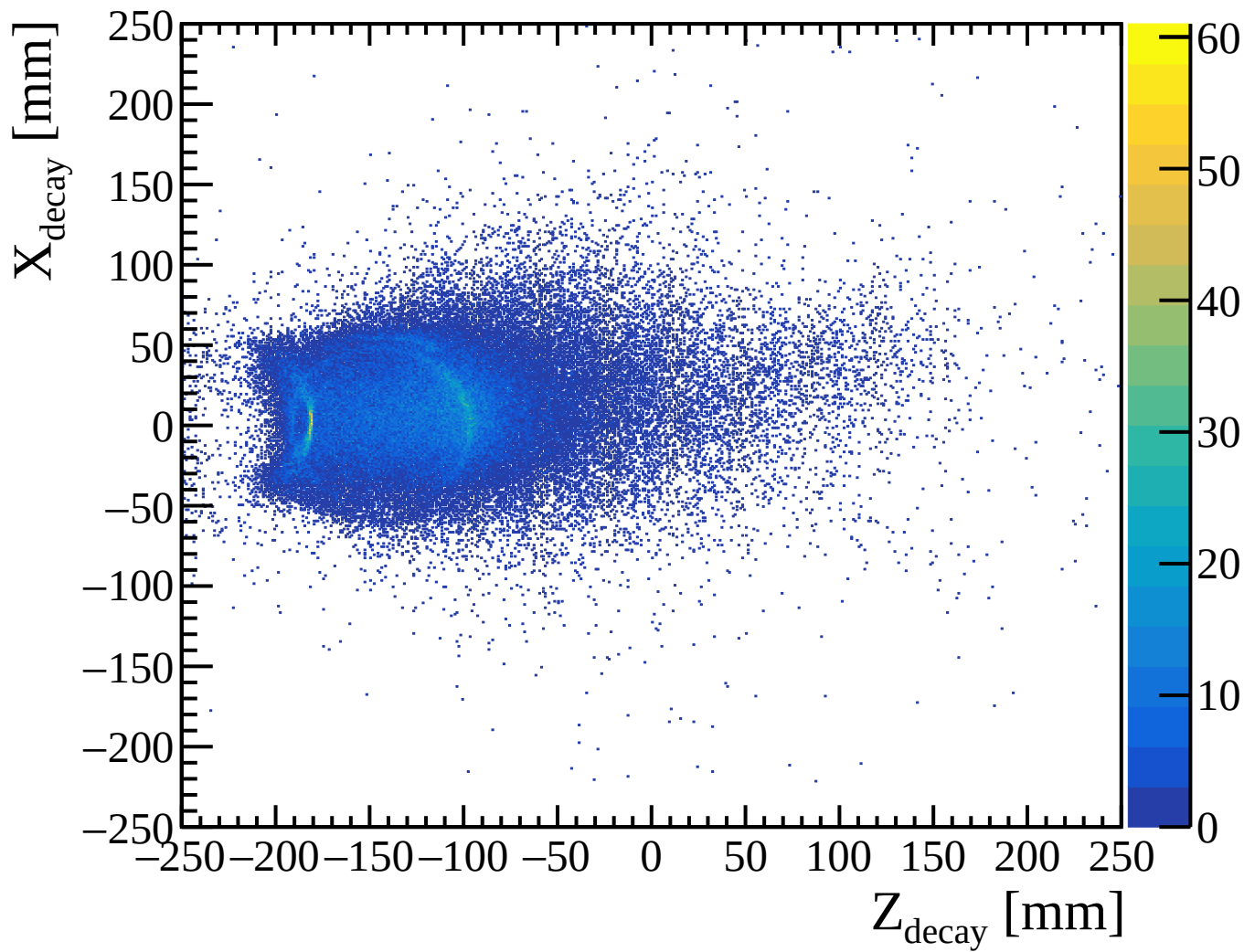


Λ flight length: production to decay vertex (no M_{χ} , LH2, or $M_{\rho\pi^-}$ cut)

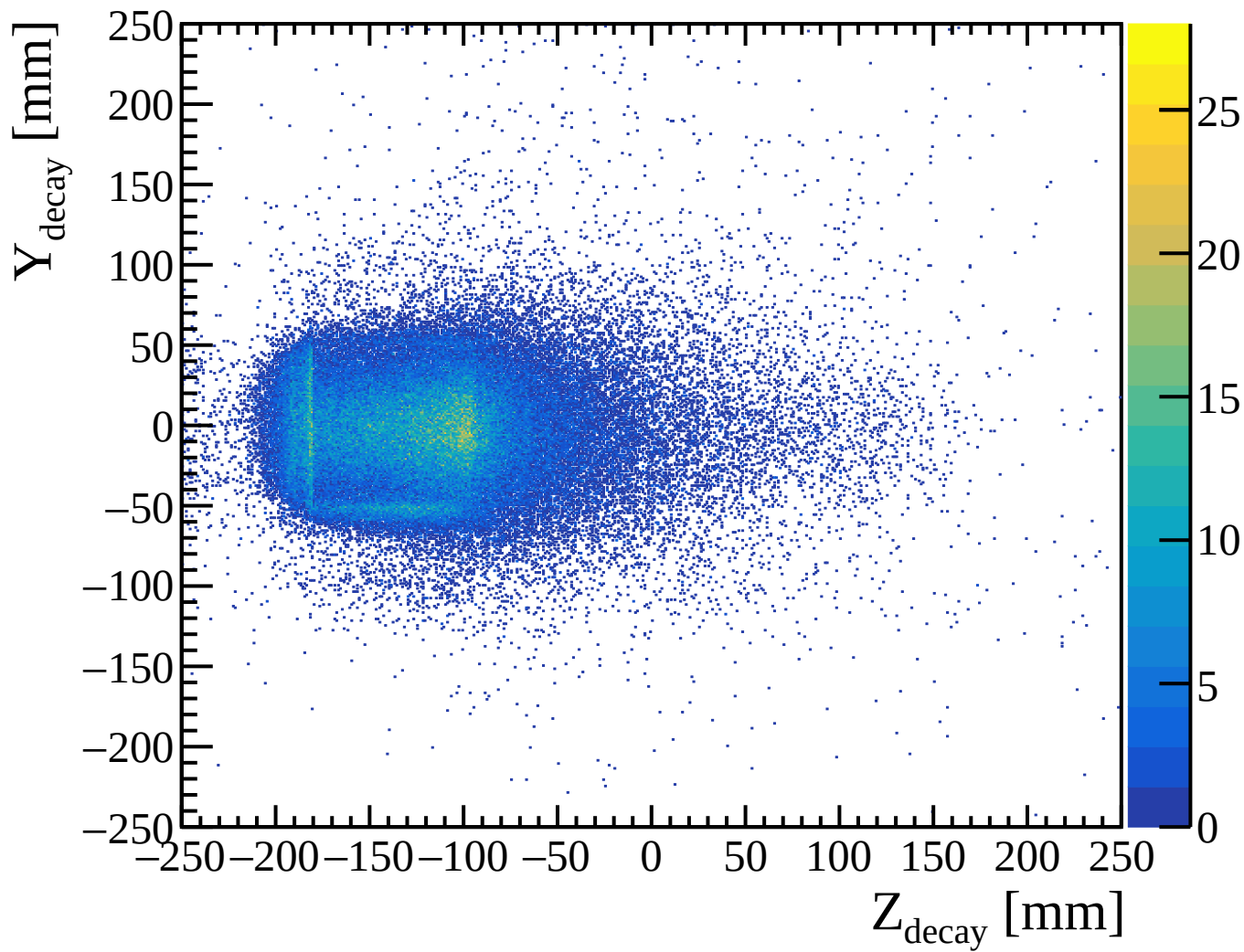




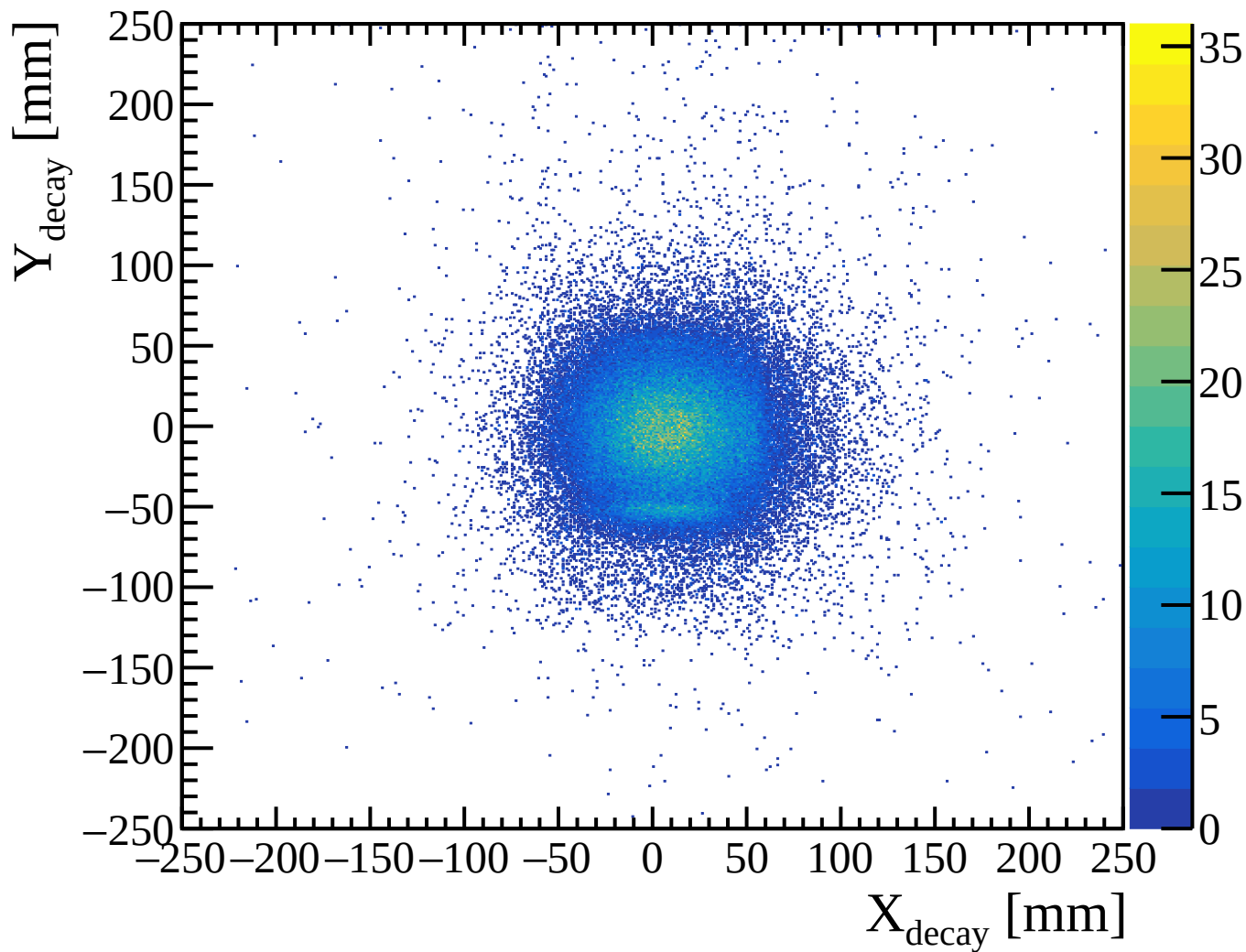
Λ decay vertex distribution



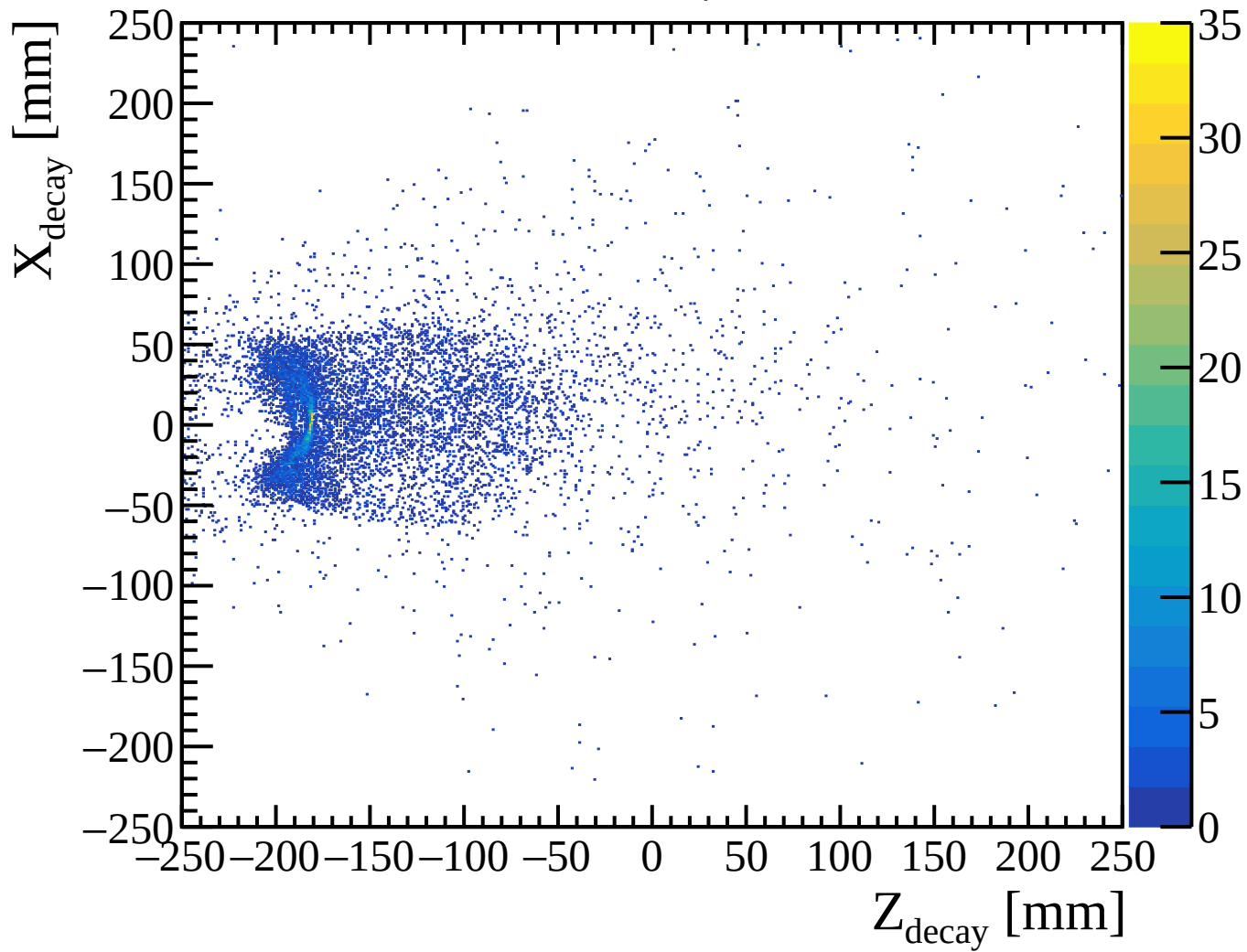
Λ decay vertex distribution



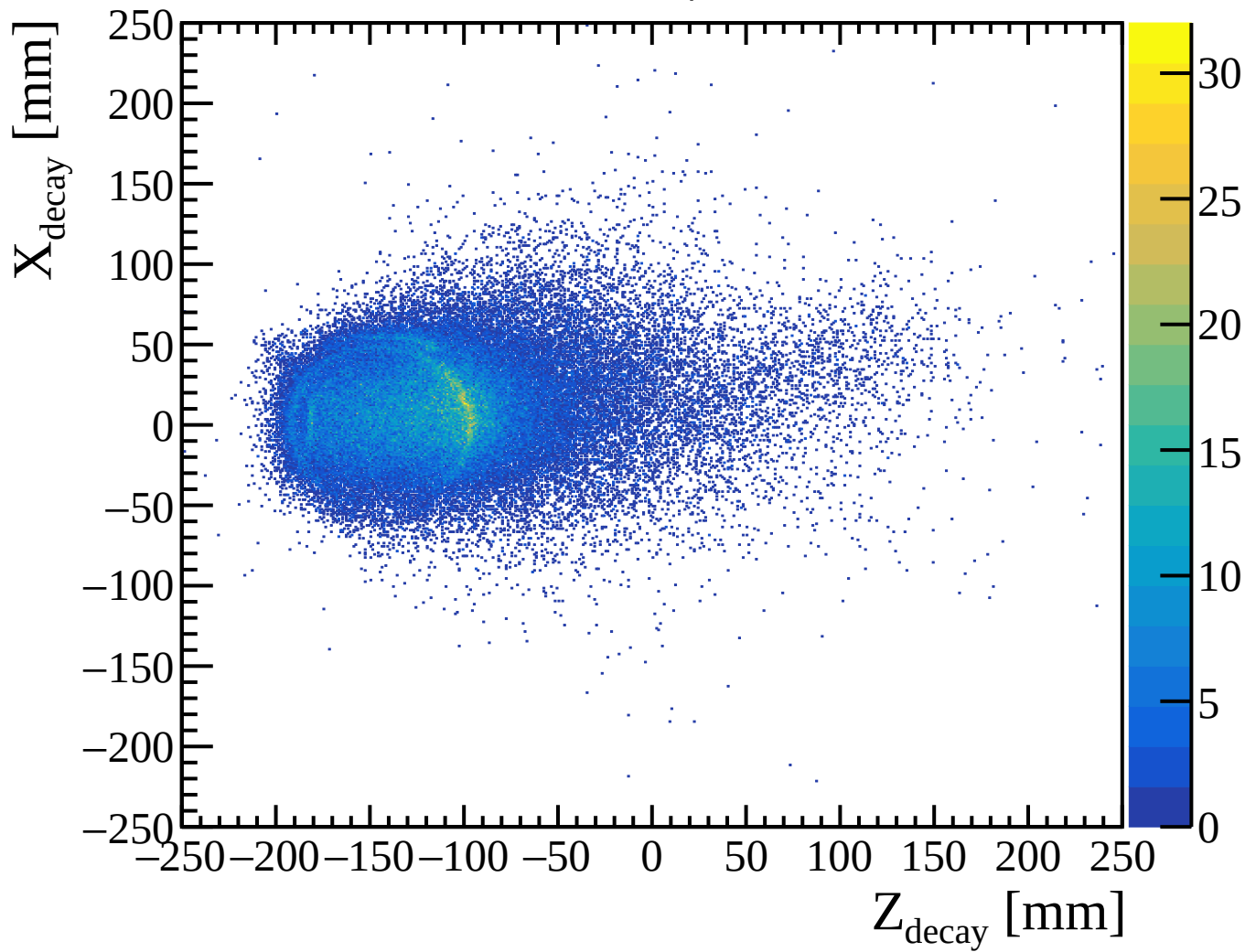
Λ decay vertex distribution



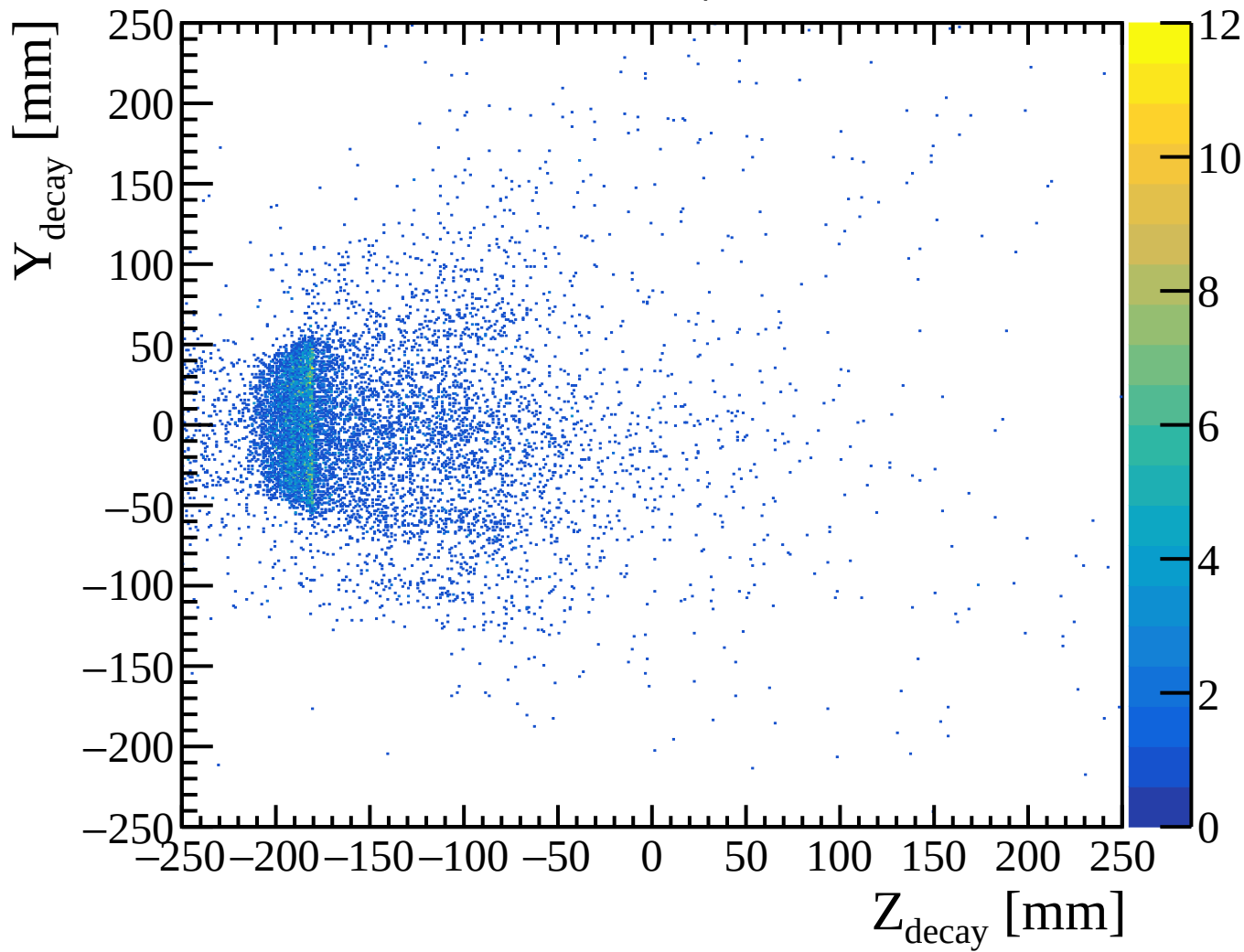
Λ decay vertex, $Z_{\text{prod}} < -220$ mm



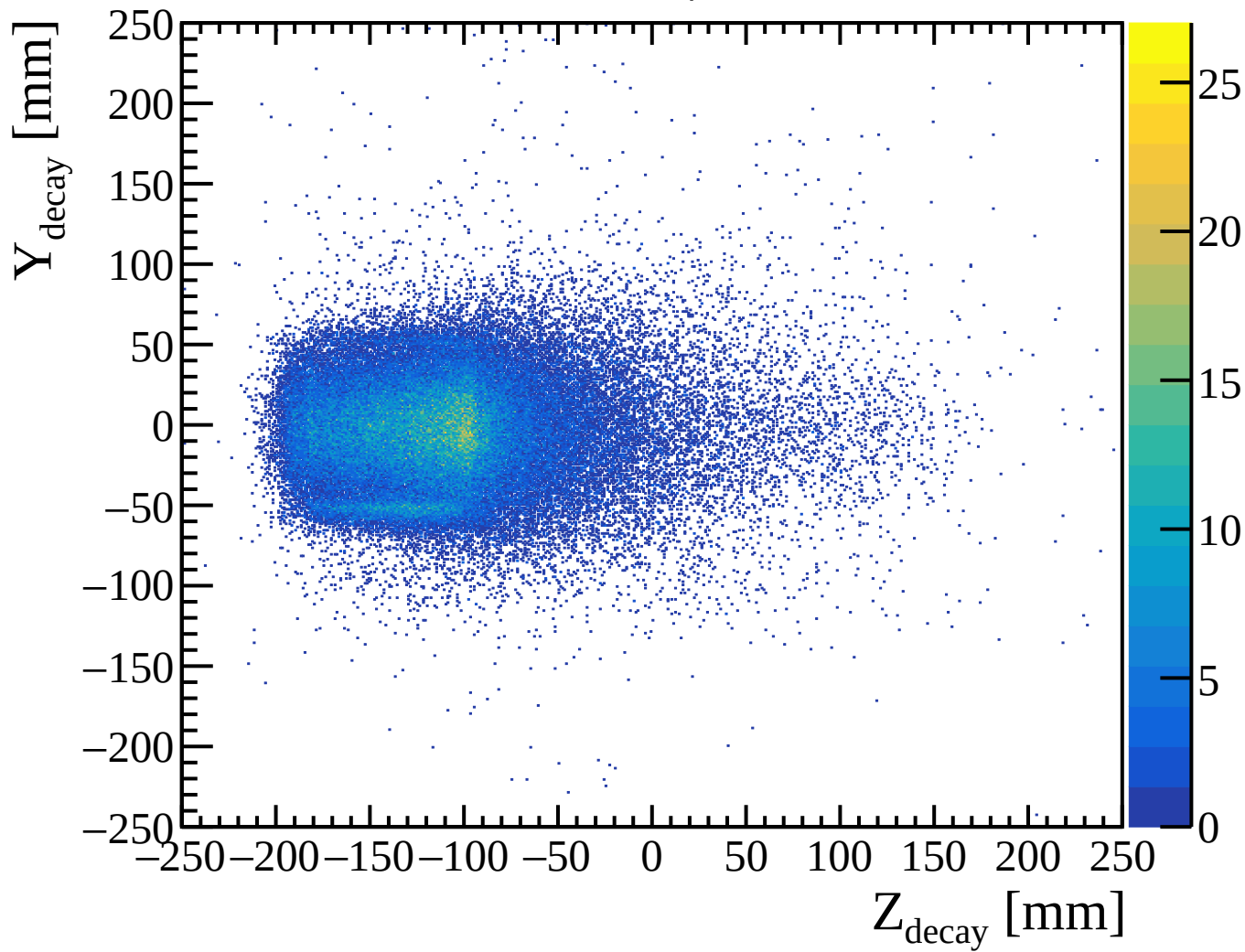
Λ decay vertex, $Z_{\text{prod}} \geq -220$ mm



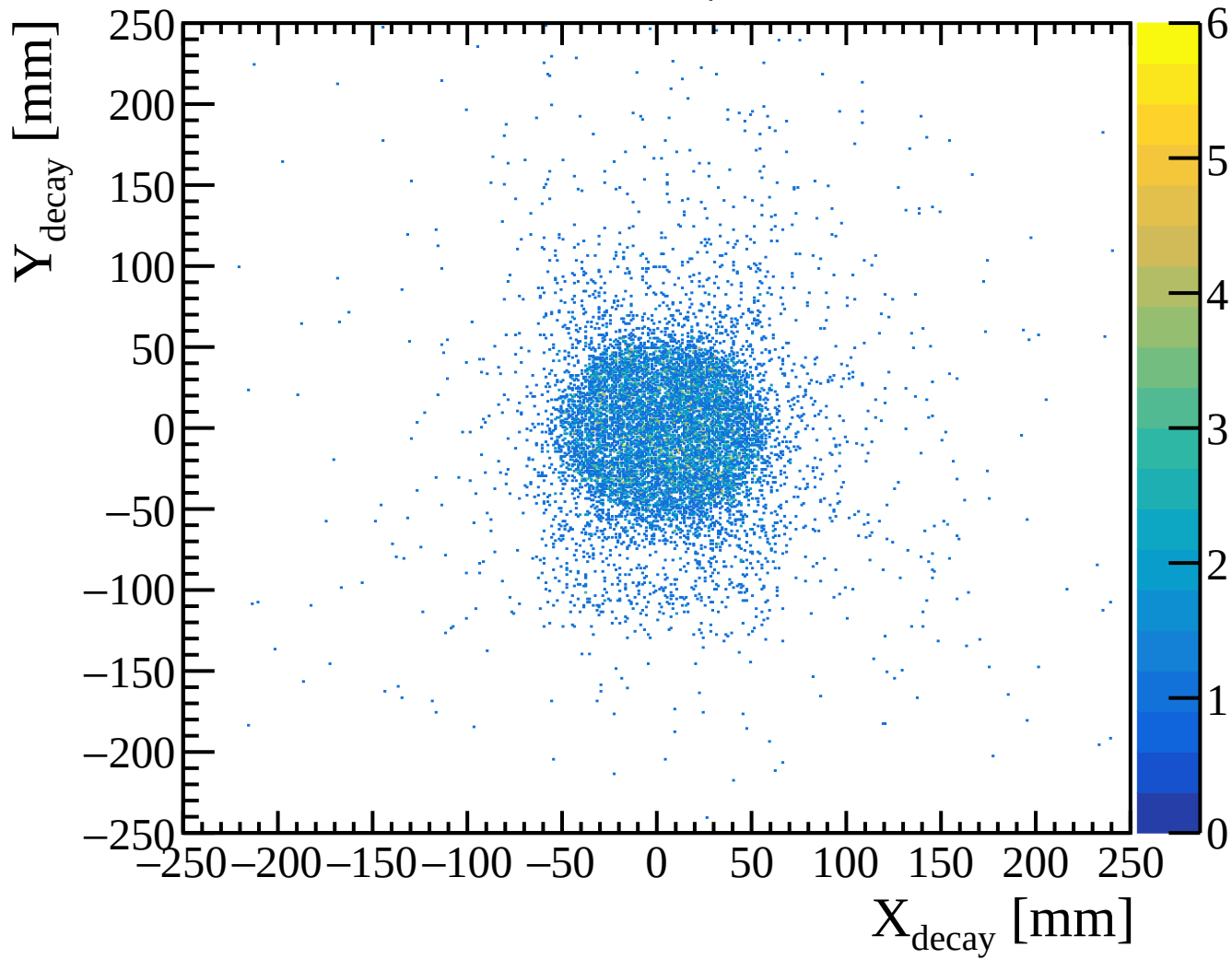
Λ decay vertex, $Z_{\text{prod}} < -220$ mm



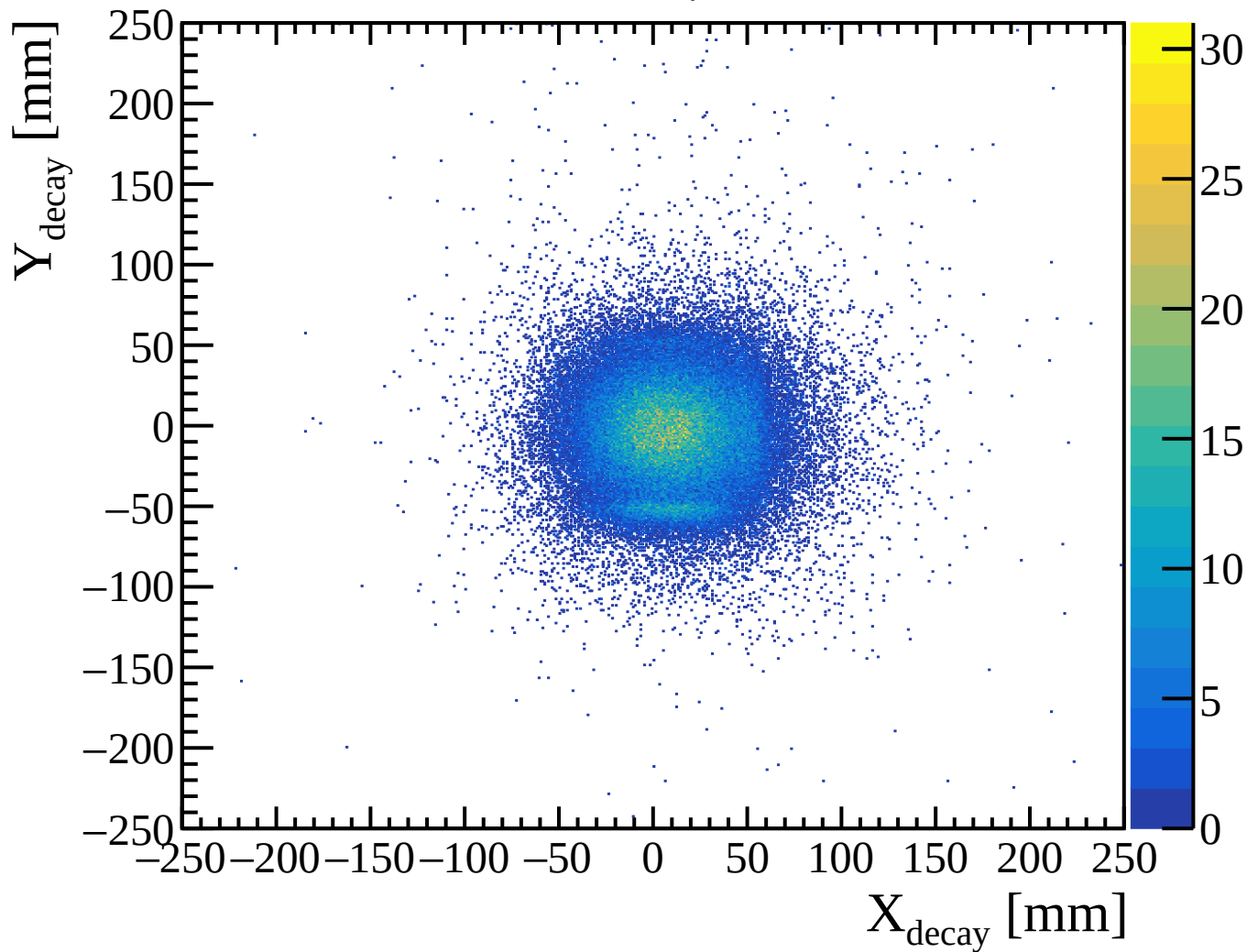
Λ decay vertex, $Z_{\text{prod}} \geq -220$ mm



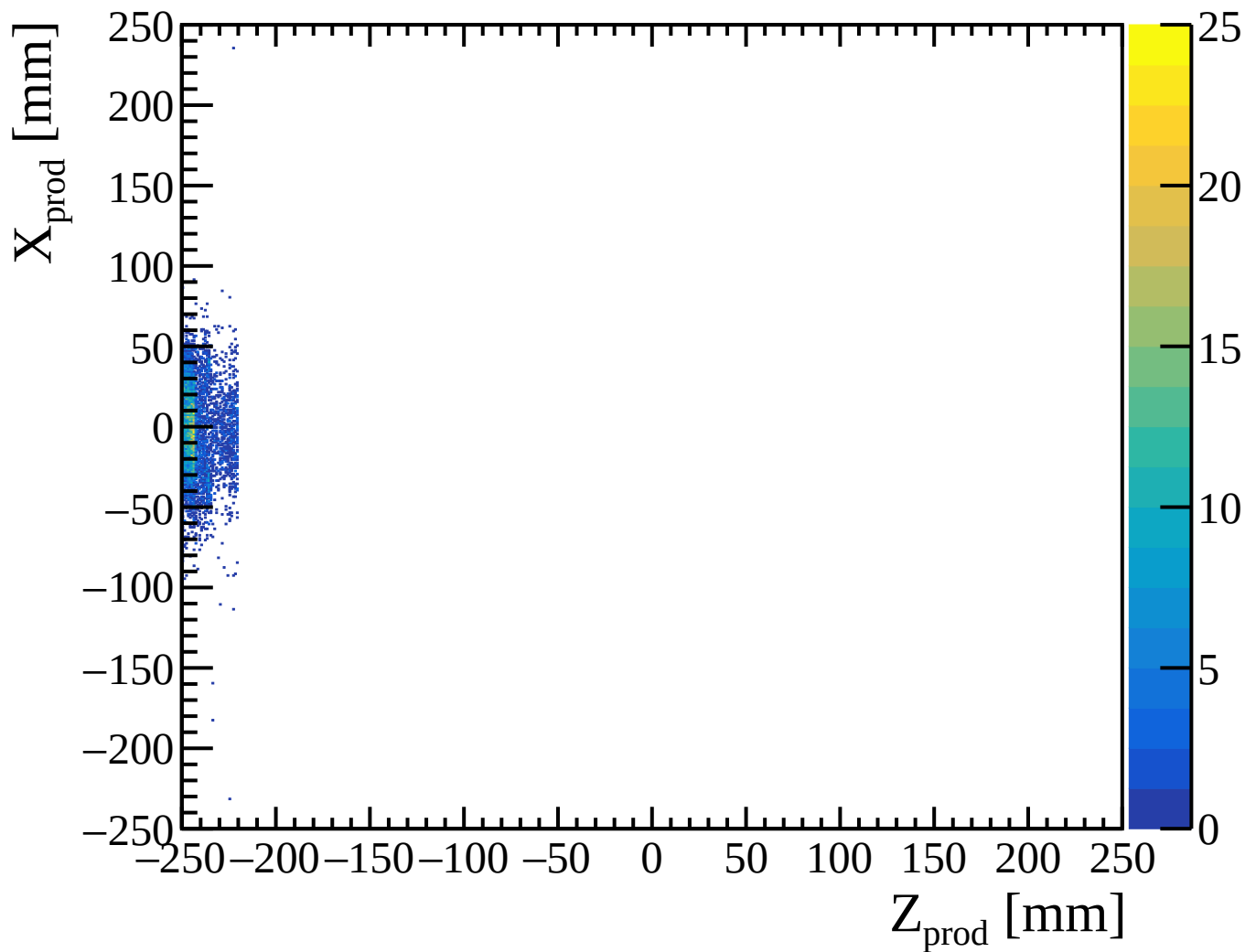
Λ decay vertex, $Z_{\text{prod}} < -220$ mm



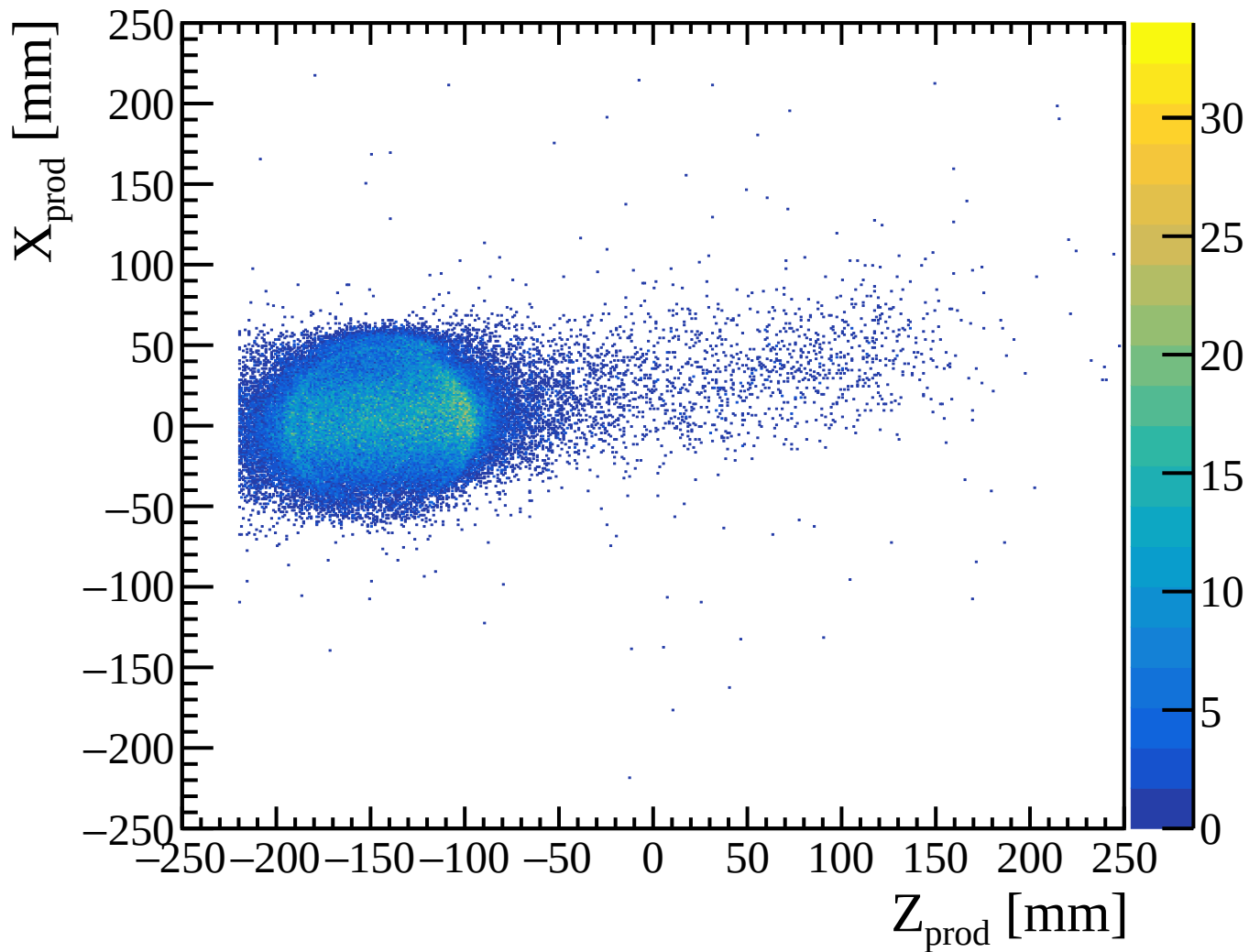
Λ decay vertex, $Z_{\text{prod}} \geq -220$ mm



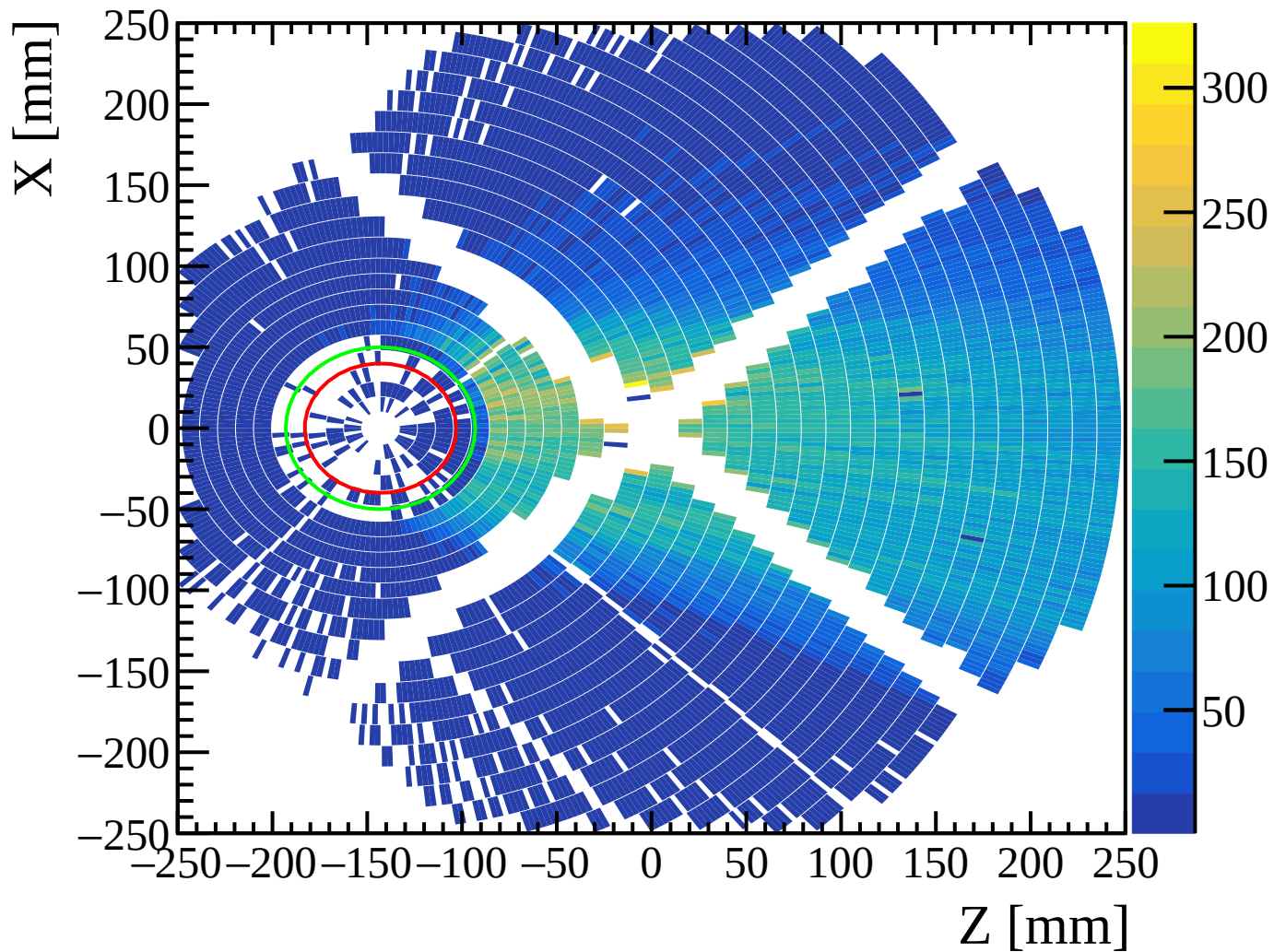
Production vertex, $Z_{\text{prod}} < -220$ mm



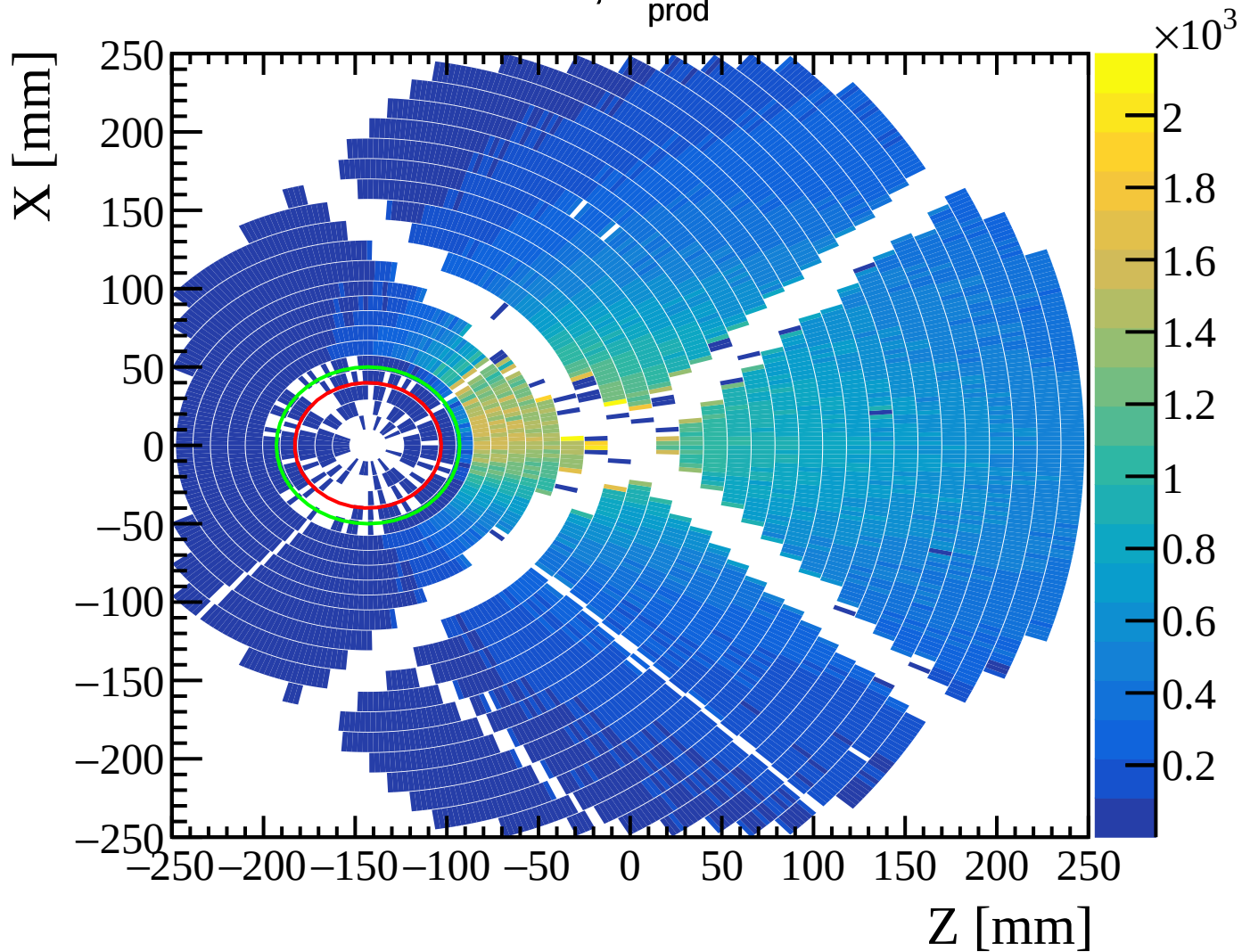
Production vertex, $Z_{\text{prod}} \geq -220$ mm



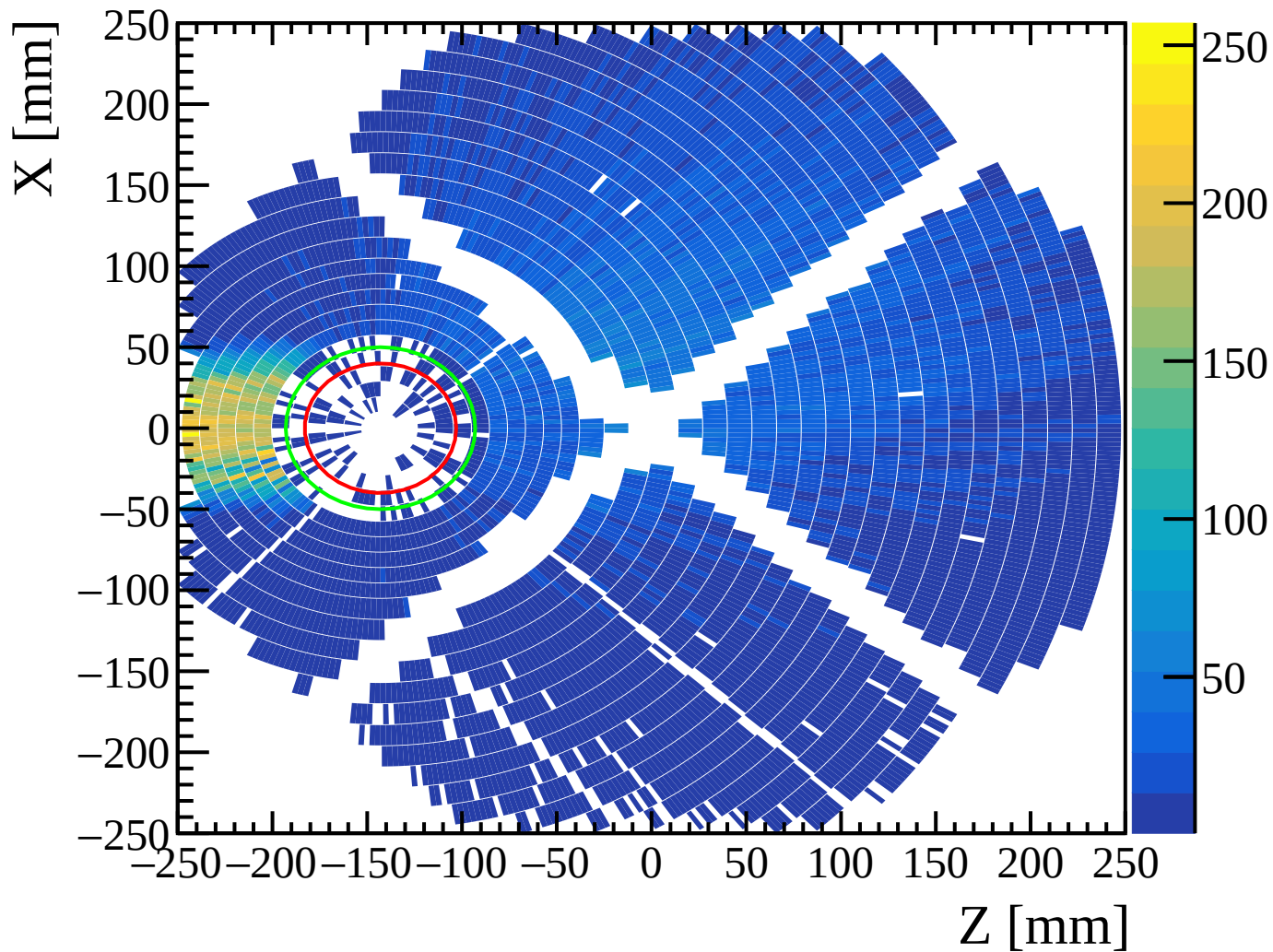
Proton tracks, $Z_{\text{prod}} < -220$ mm



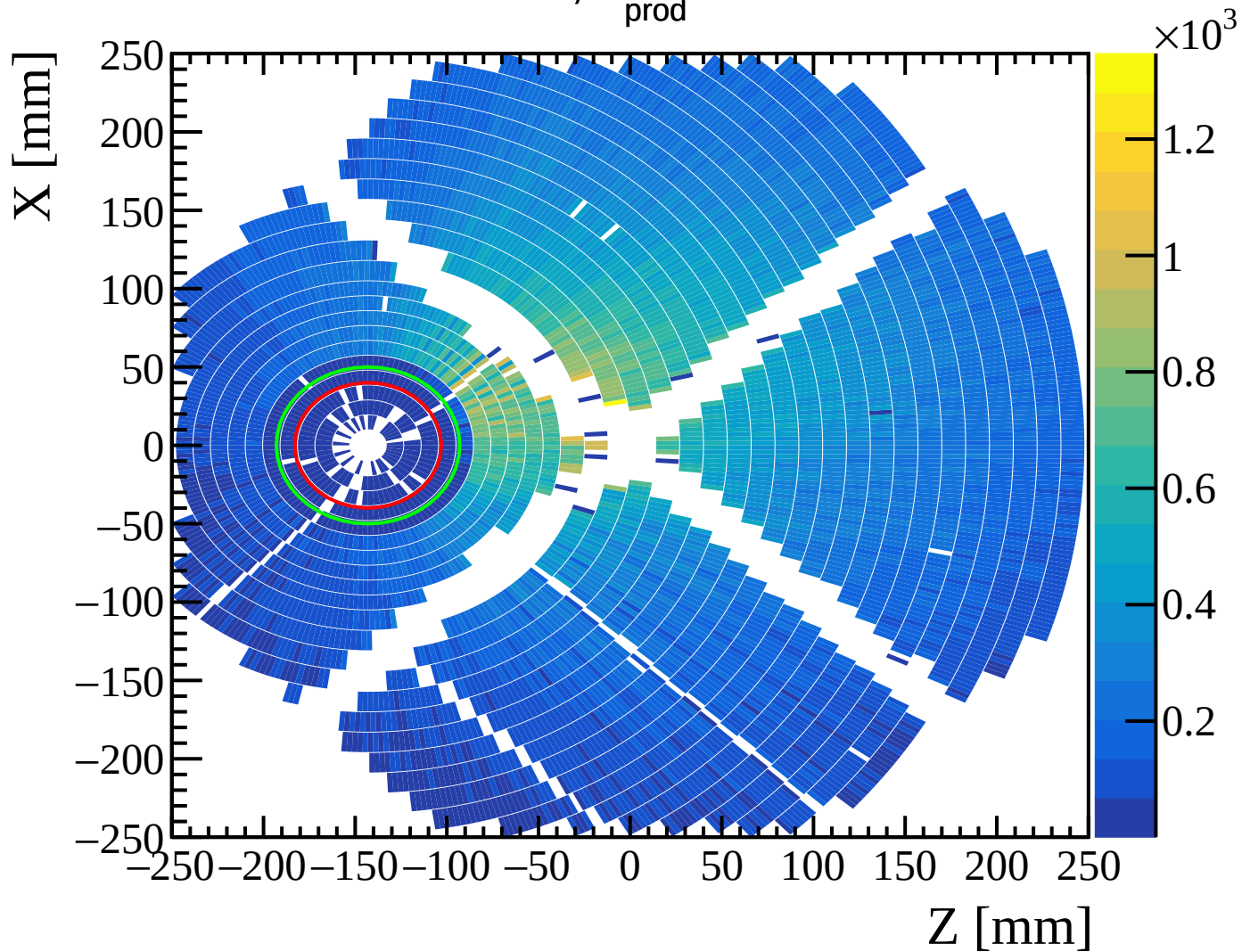
Proton tracks, $Z_{\text{prod}} \geq -220$ mm



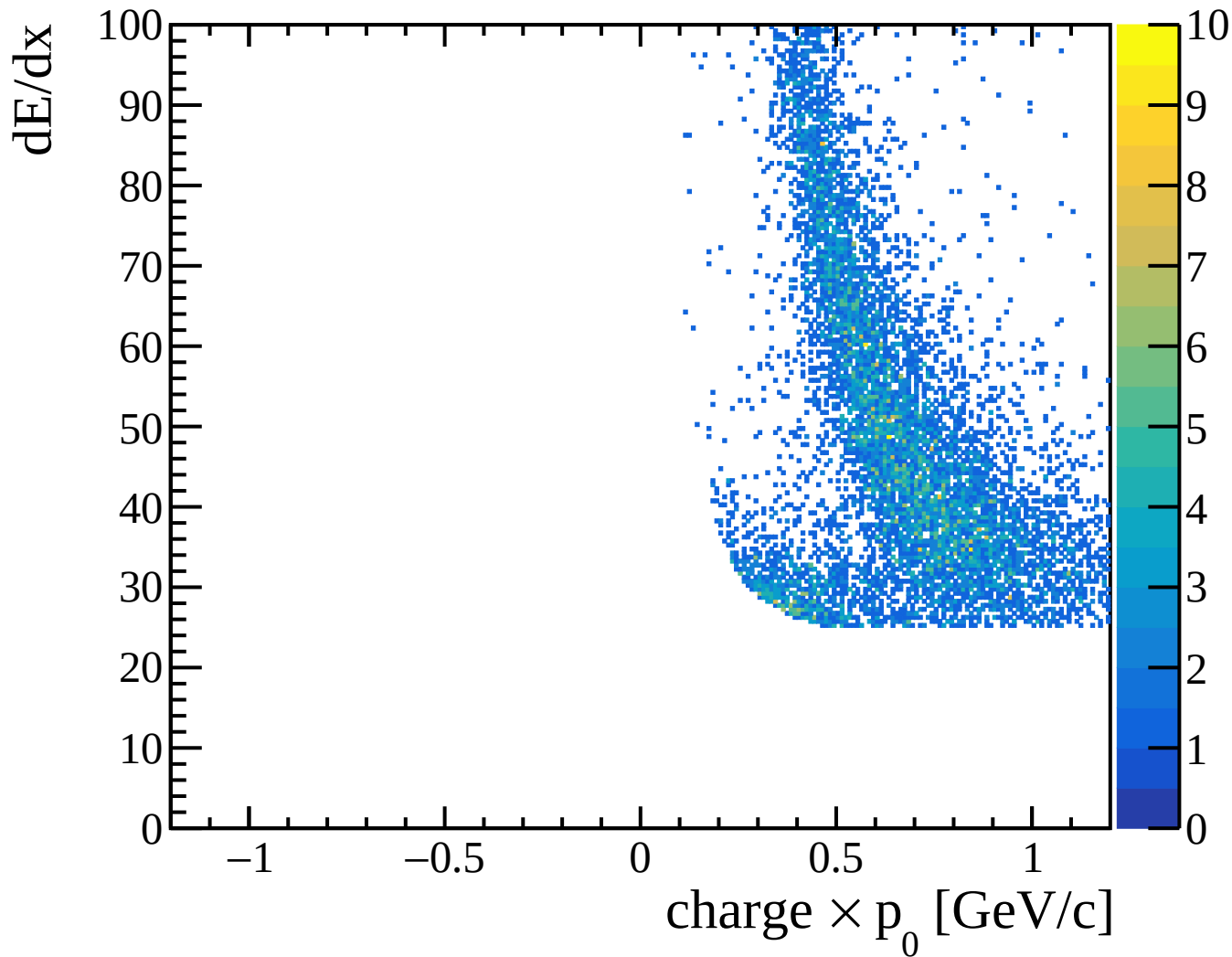
Pion tracks, $Z_{\text{prod}} < -220$ mm



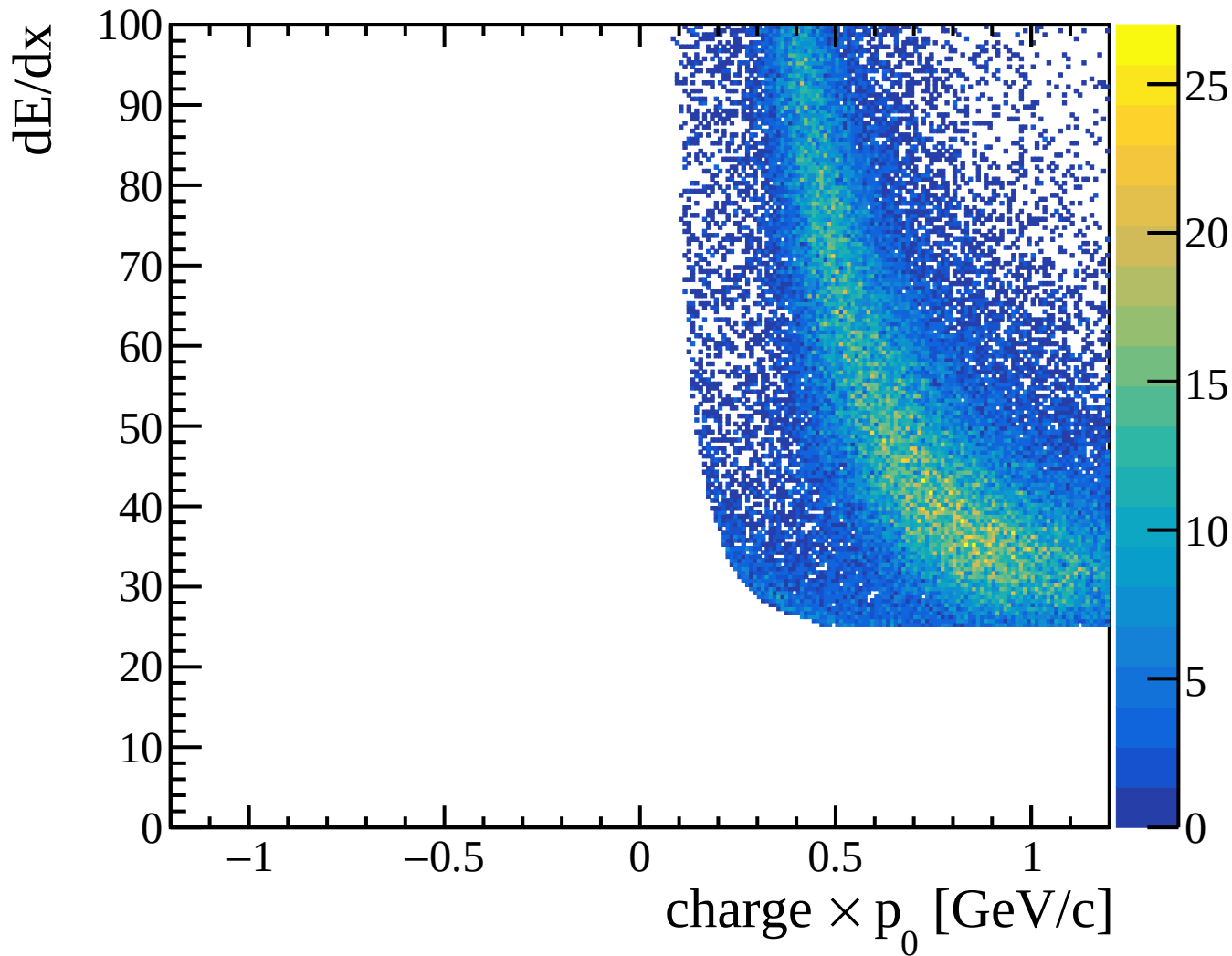
Pion tracks, $Z_{\text{prod}} \geq -220$ mm



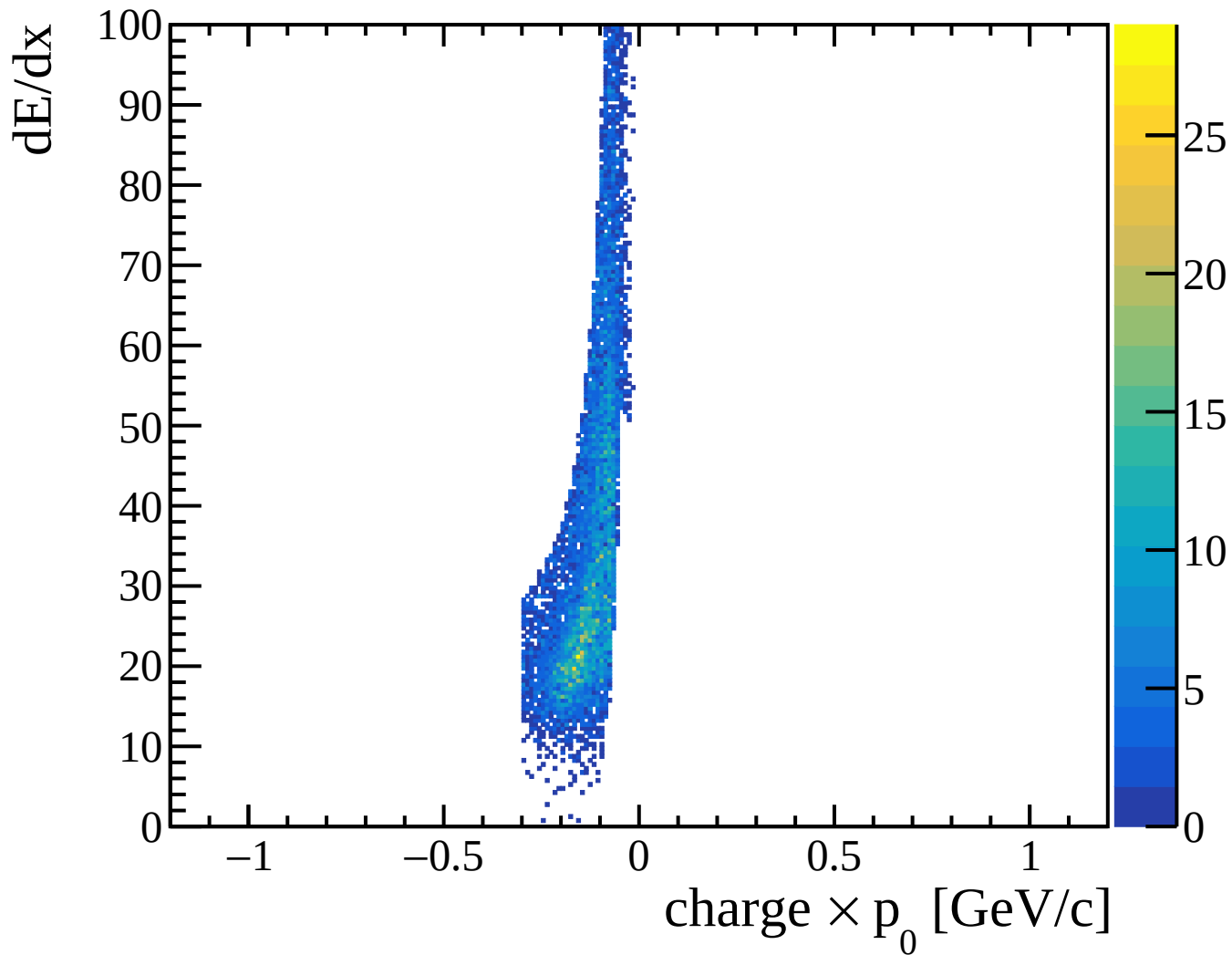
Proton PID, $Z_{\text{prod}} < -220$ mm



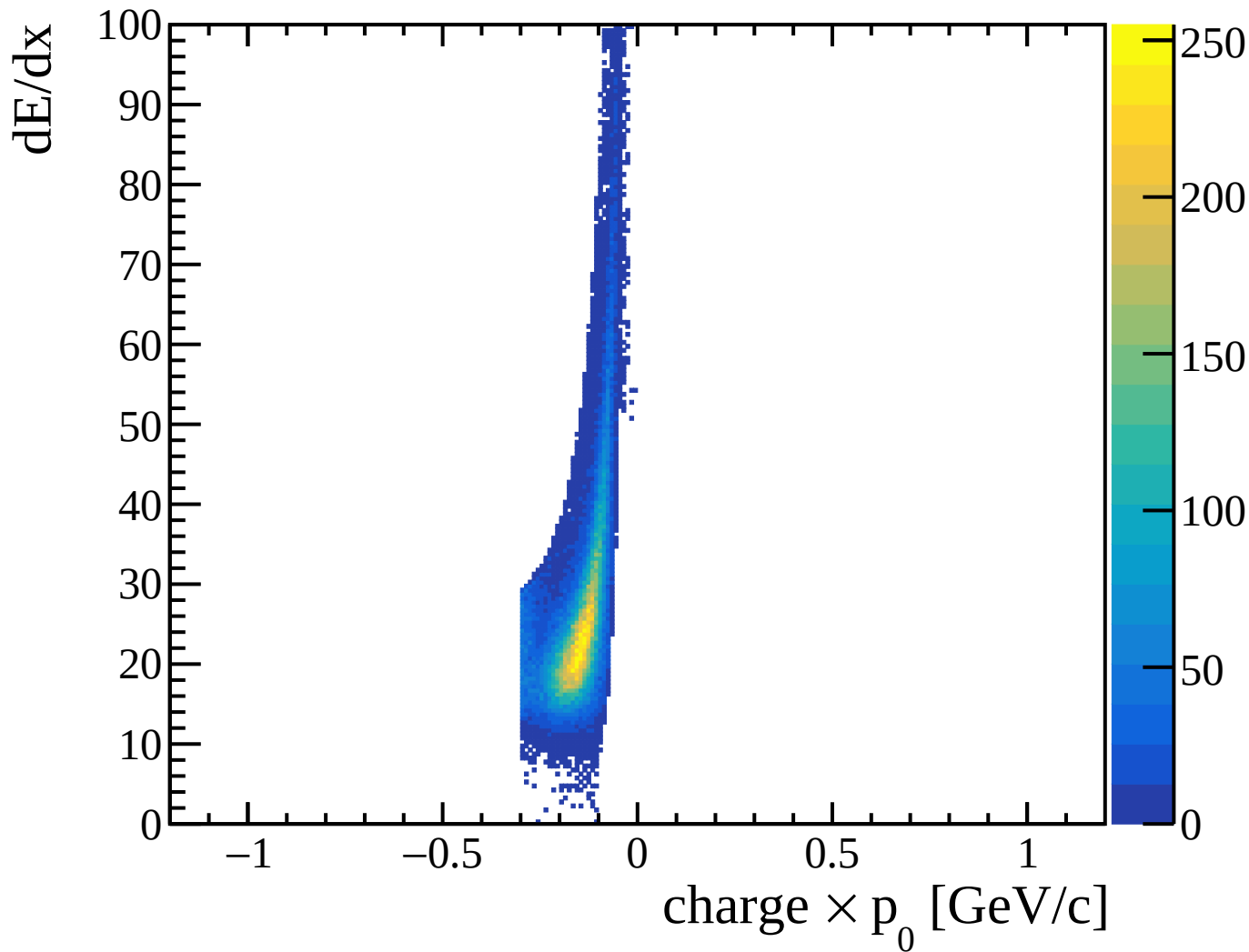
Proton PID, $Z_{\text{prod}} \geq -220$ mm



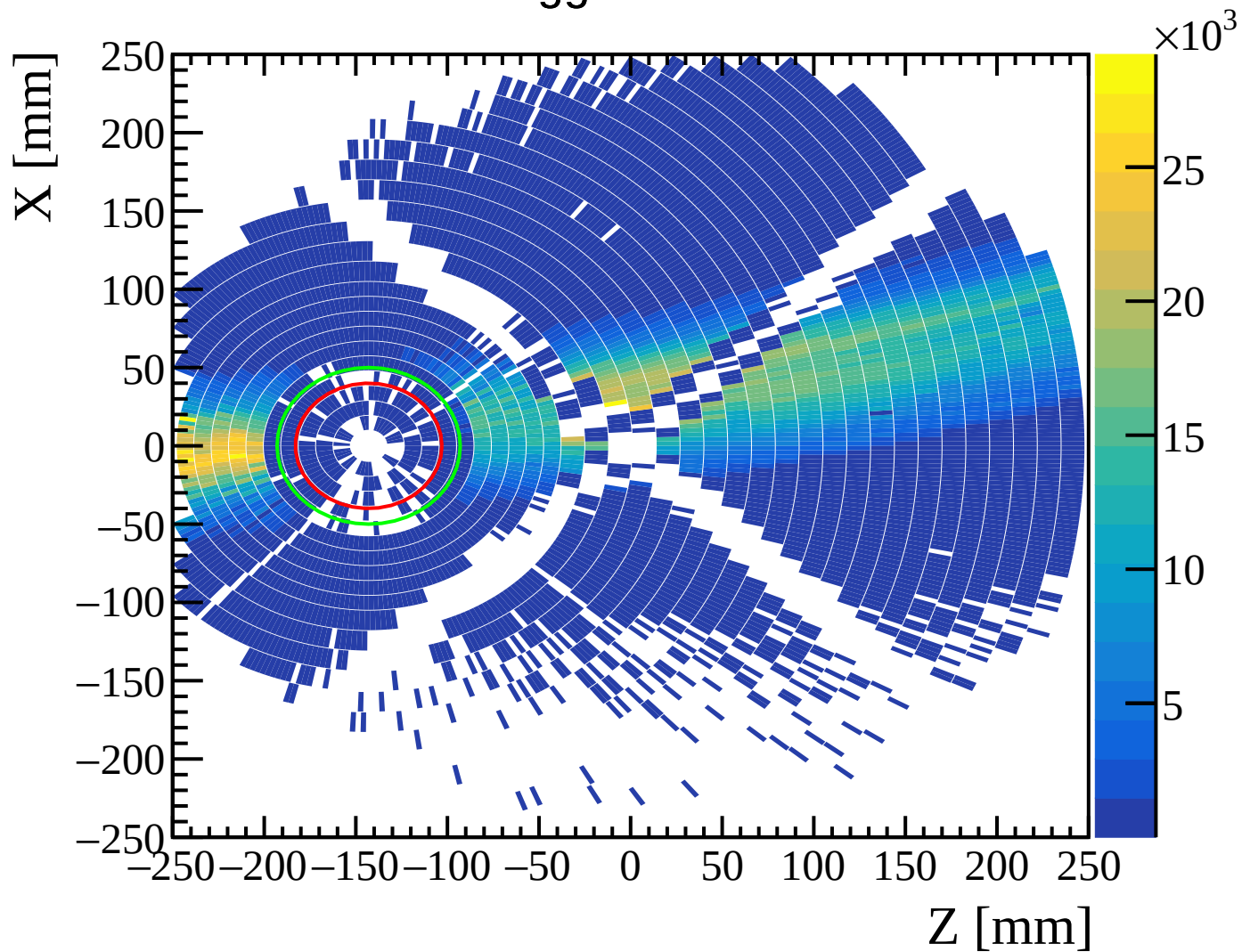
Pion PID, $Z_{\text{prod}} < -220$ mm



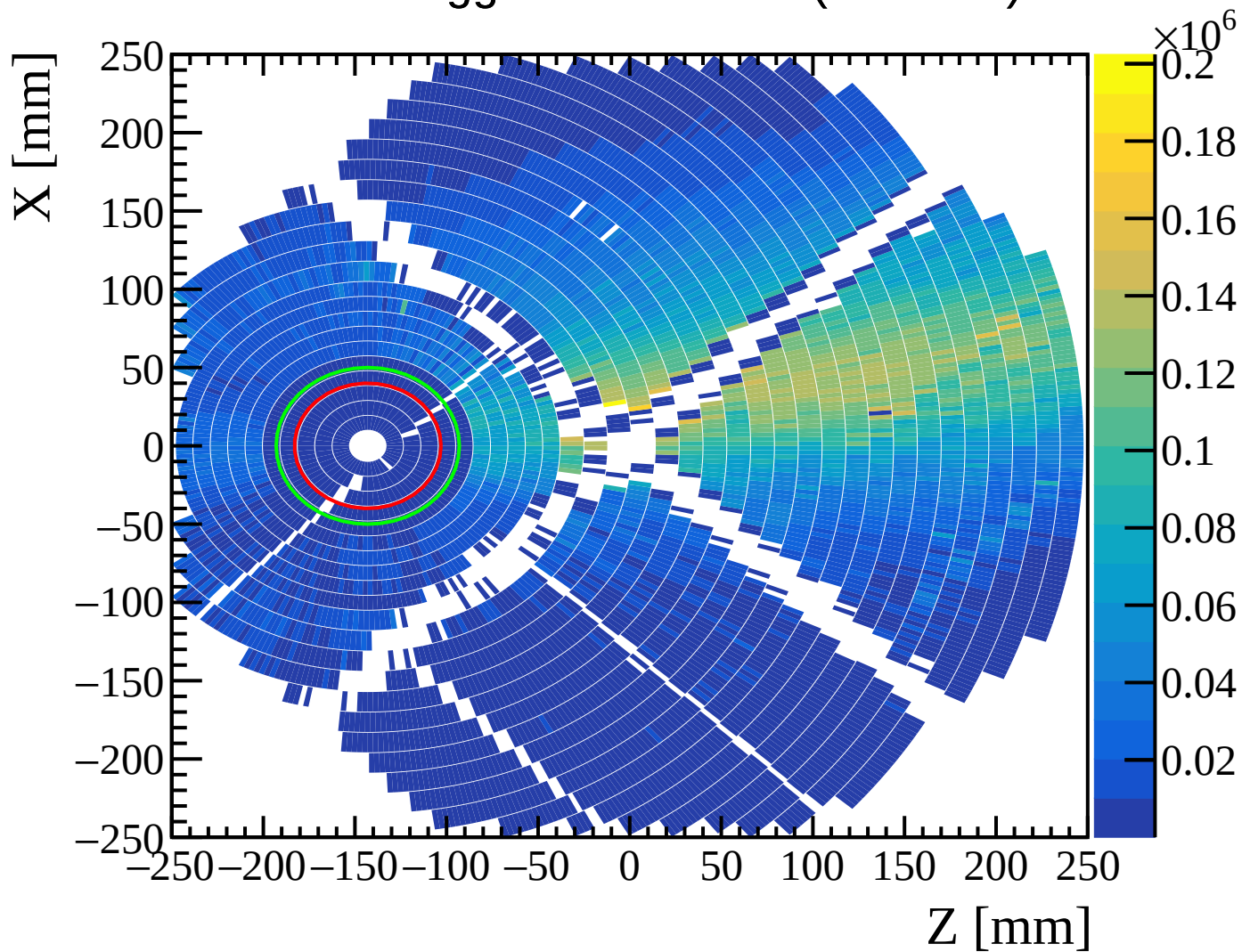
Pion PID, $Z_{\text{prod}} \geq -220$ mm



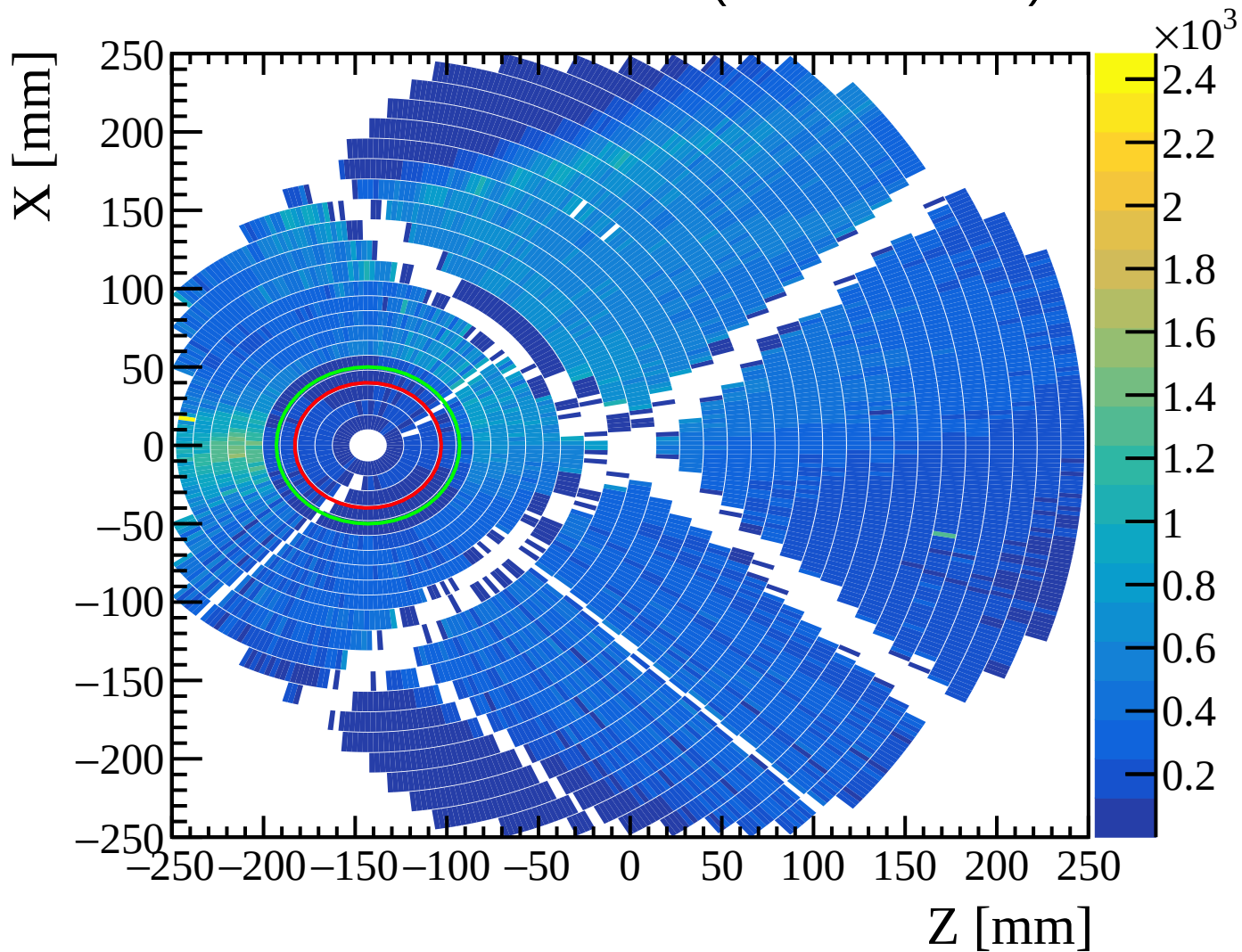
K18-tagged TPC tracks



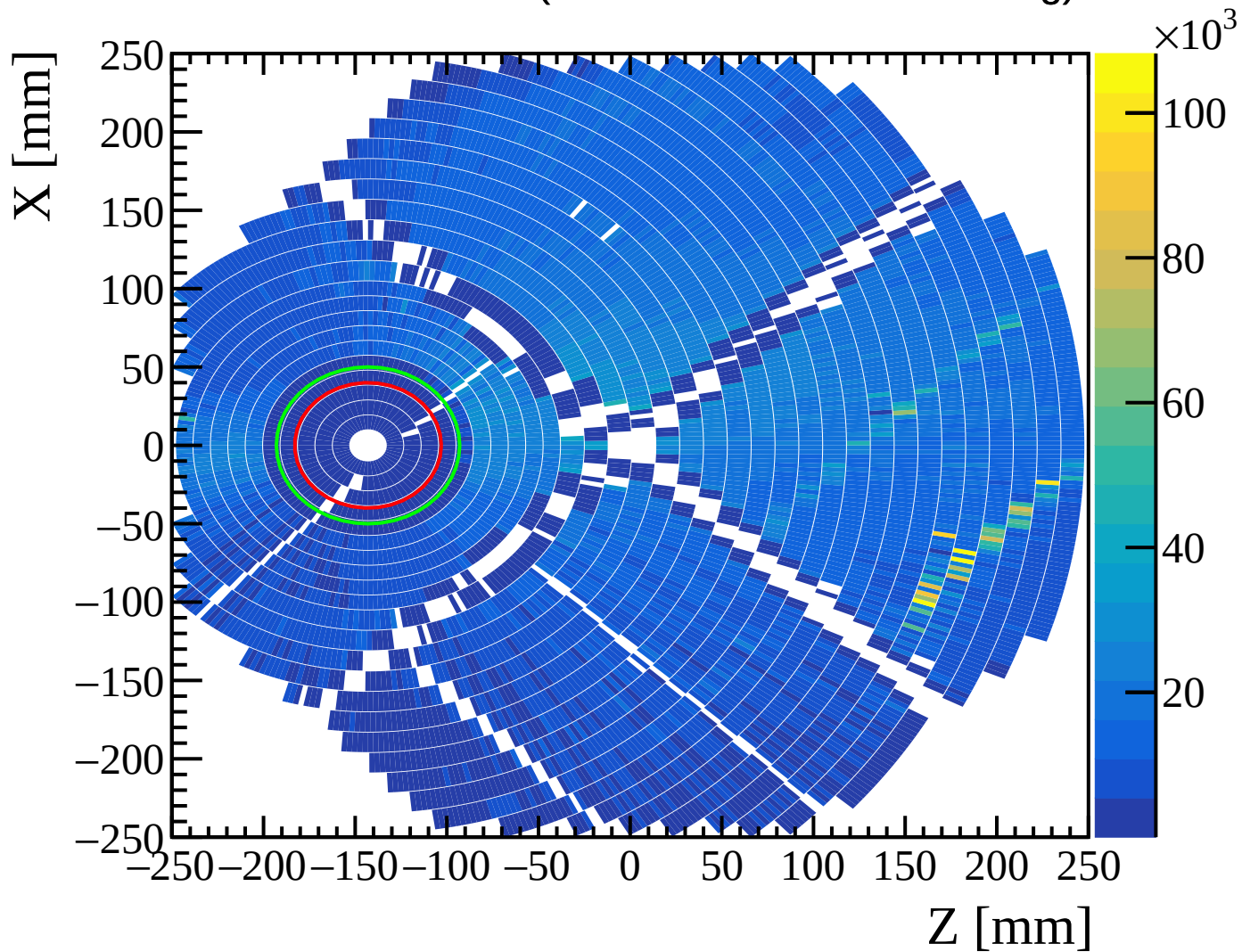
Beam-tagged TPC tracks (non-K18)



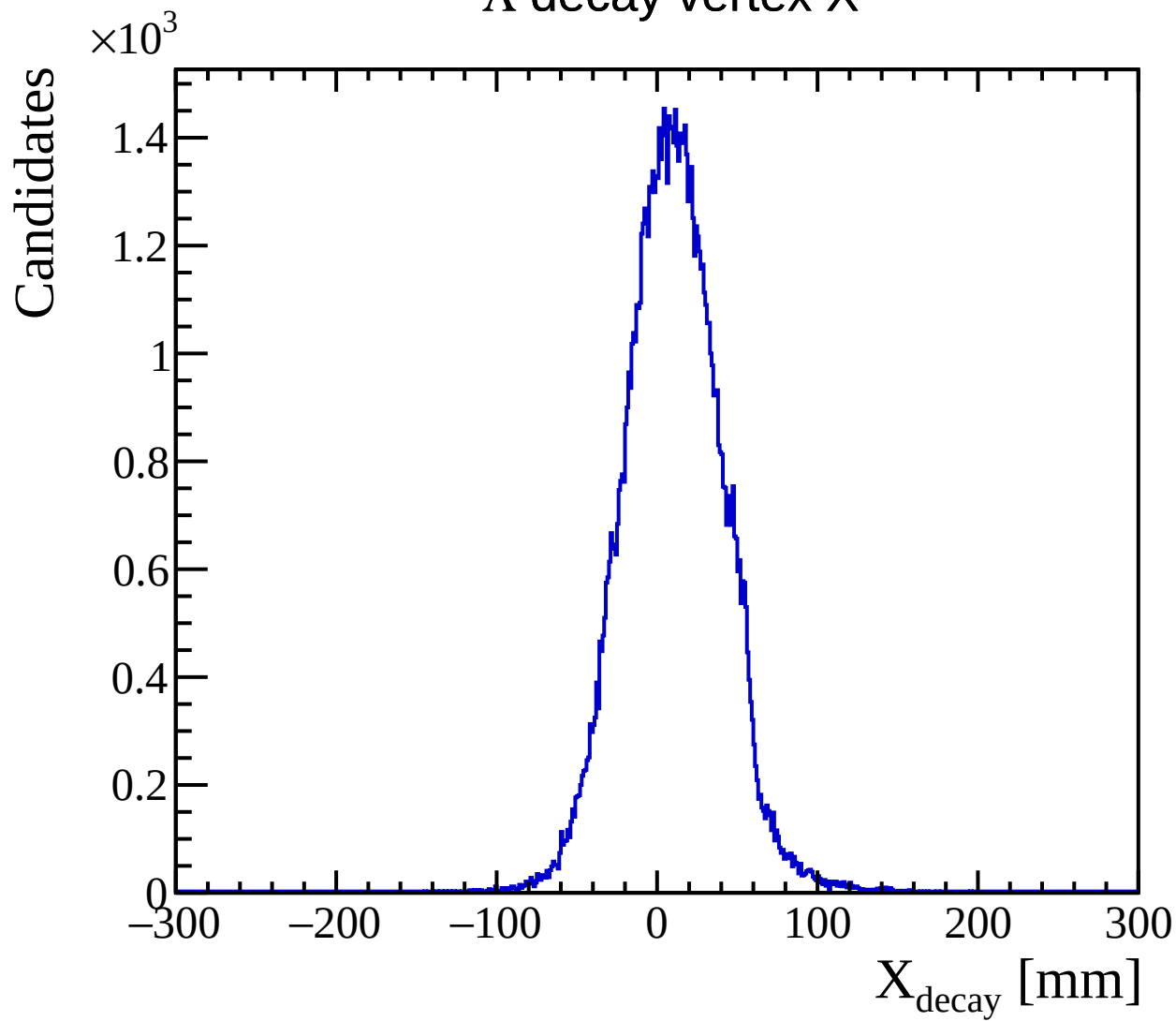
Accidental TPC tracks (non-K18/beam)



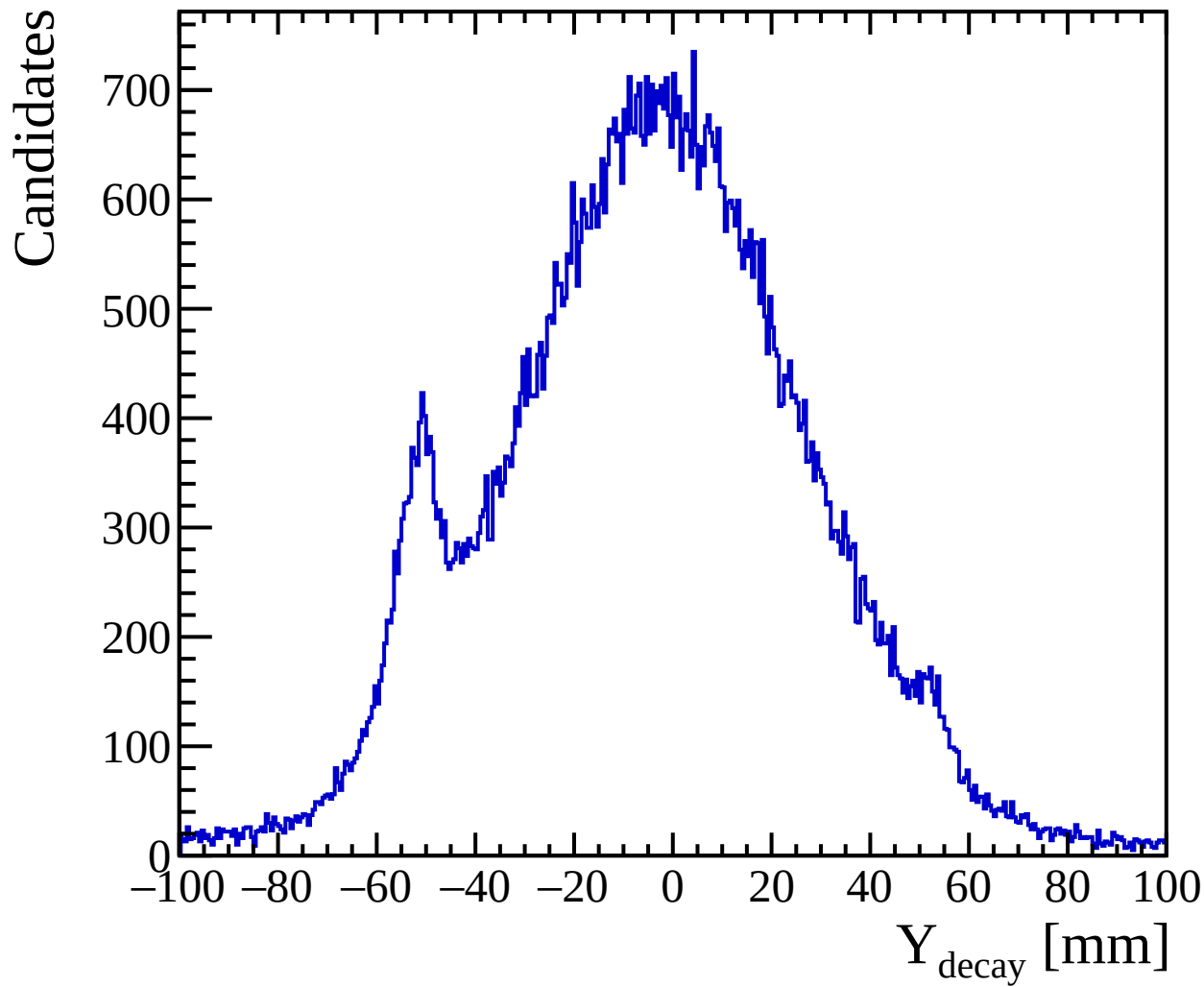
Other TPC tracks (no K18/beam/accidental flag)



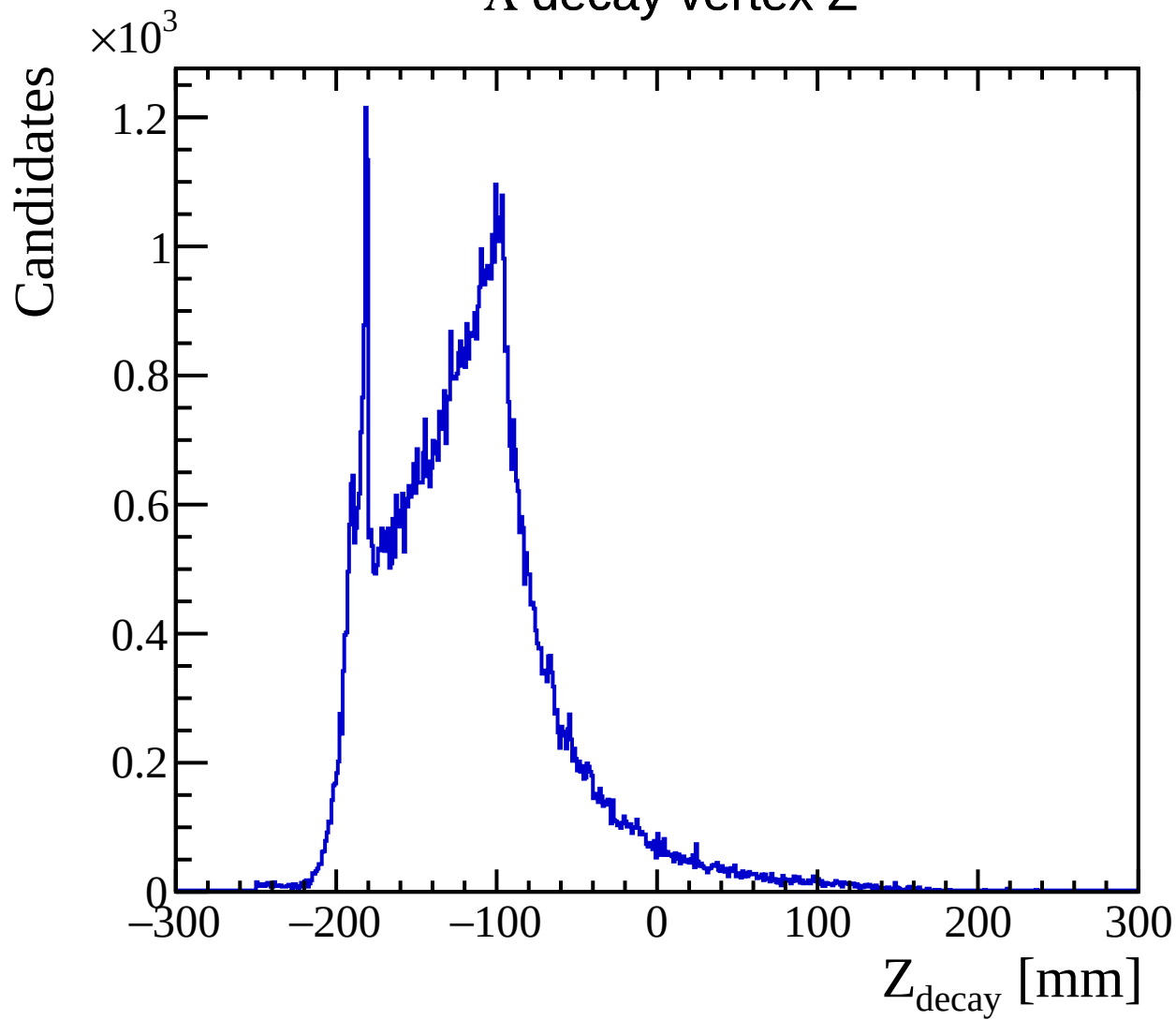
Λ decay vertex X



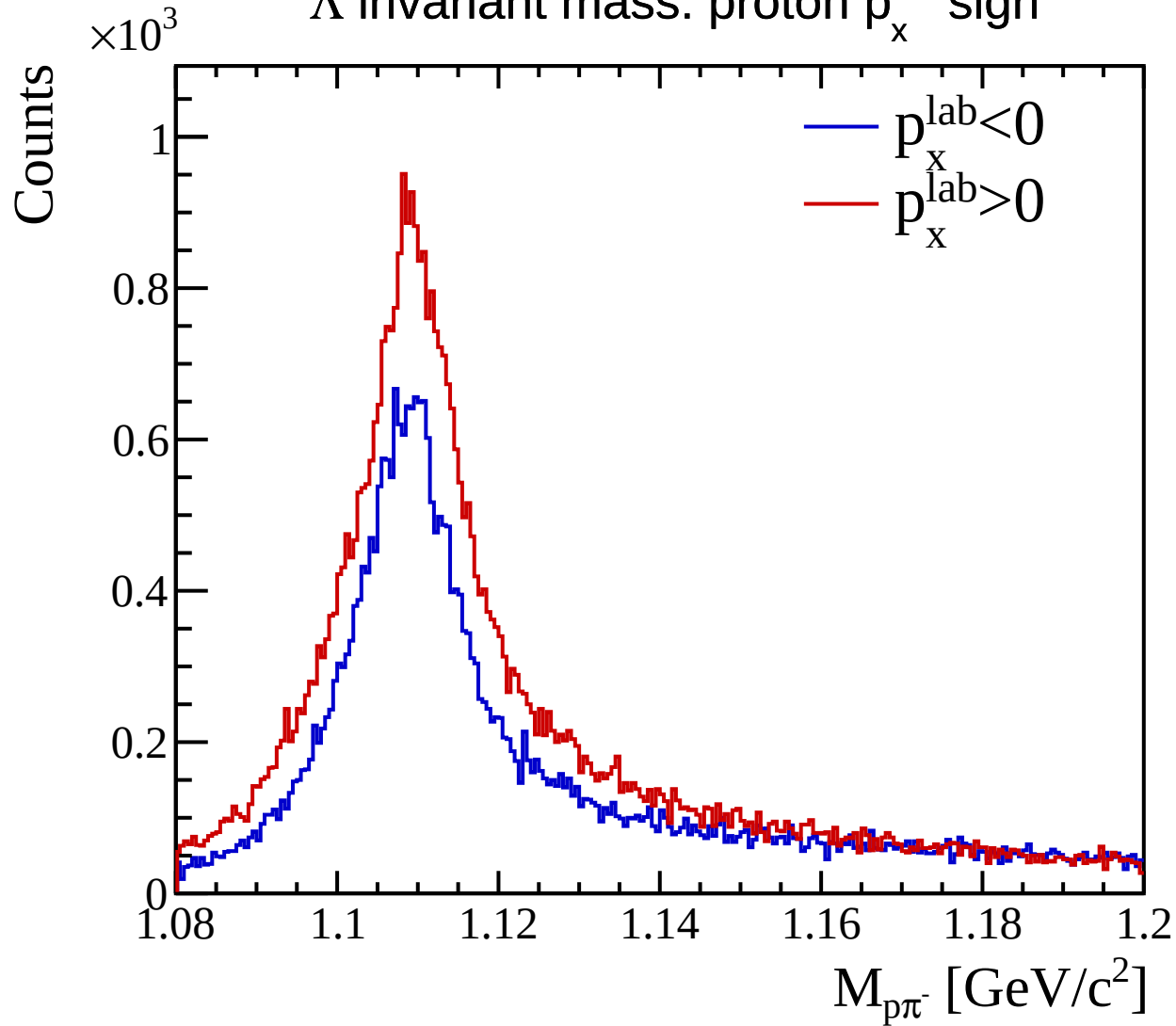
Λ decay vertex Y



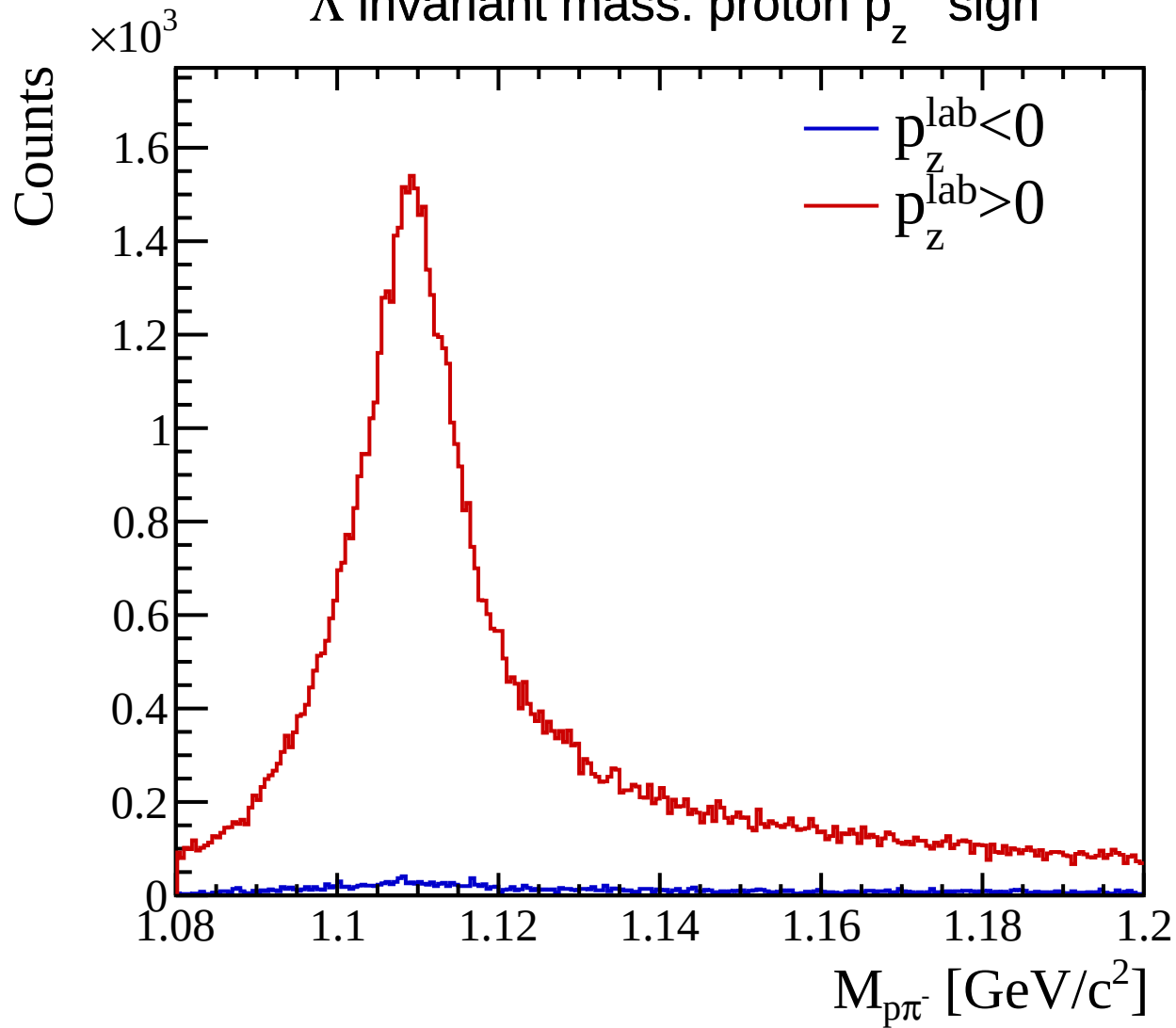
Λ decay vertex Z



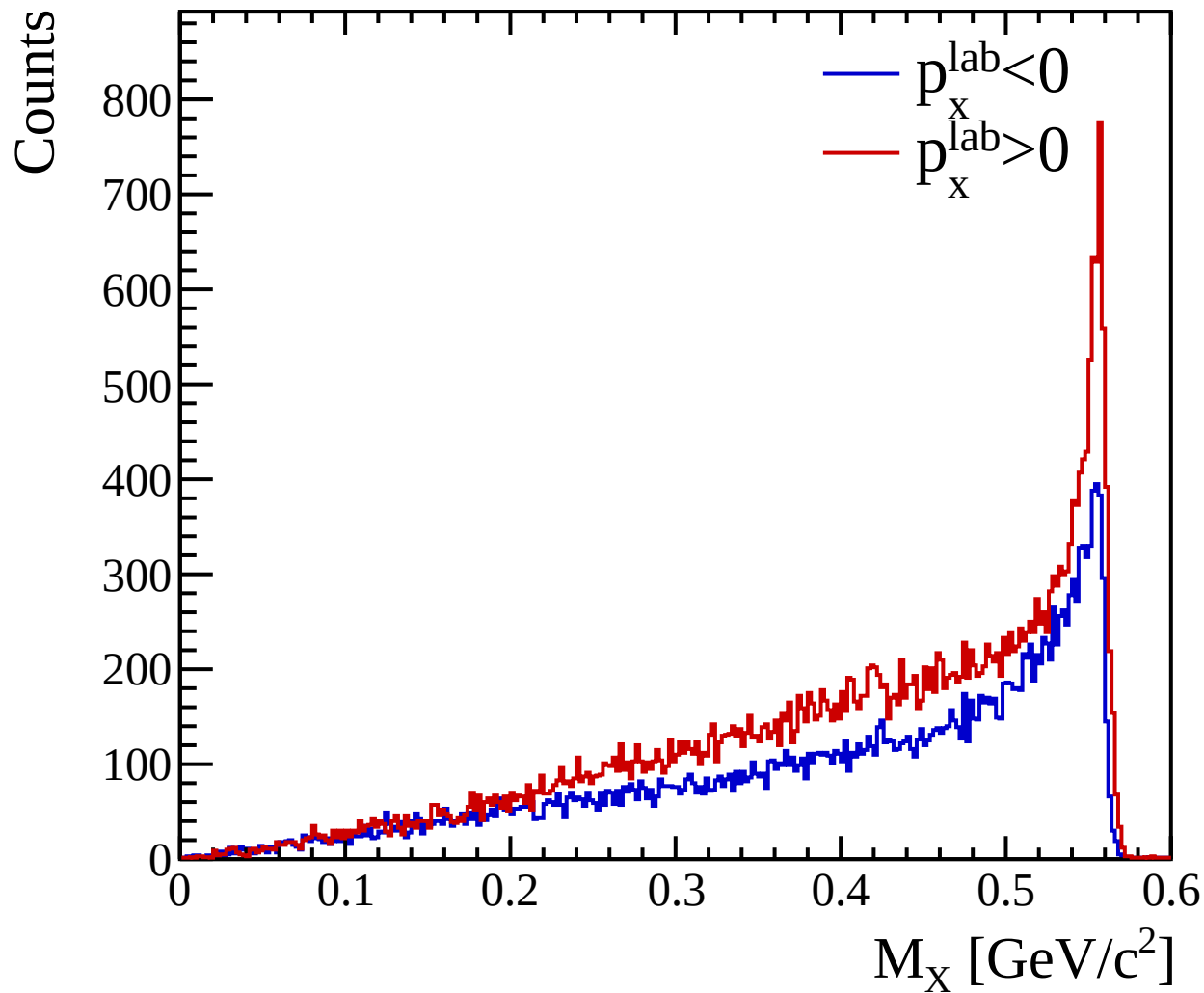
Λ invariant mass: proton p_x^{lab} sign



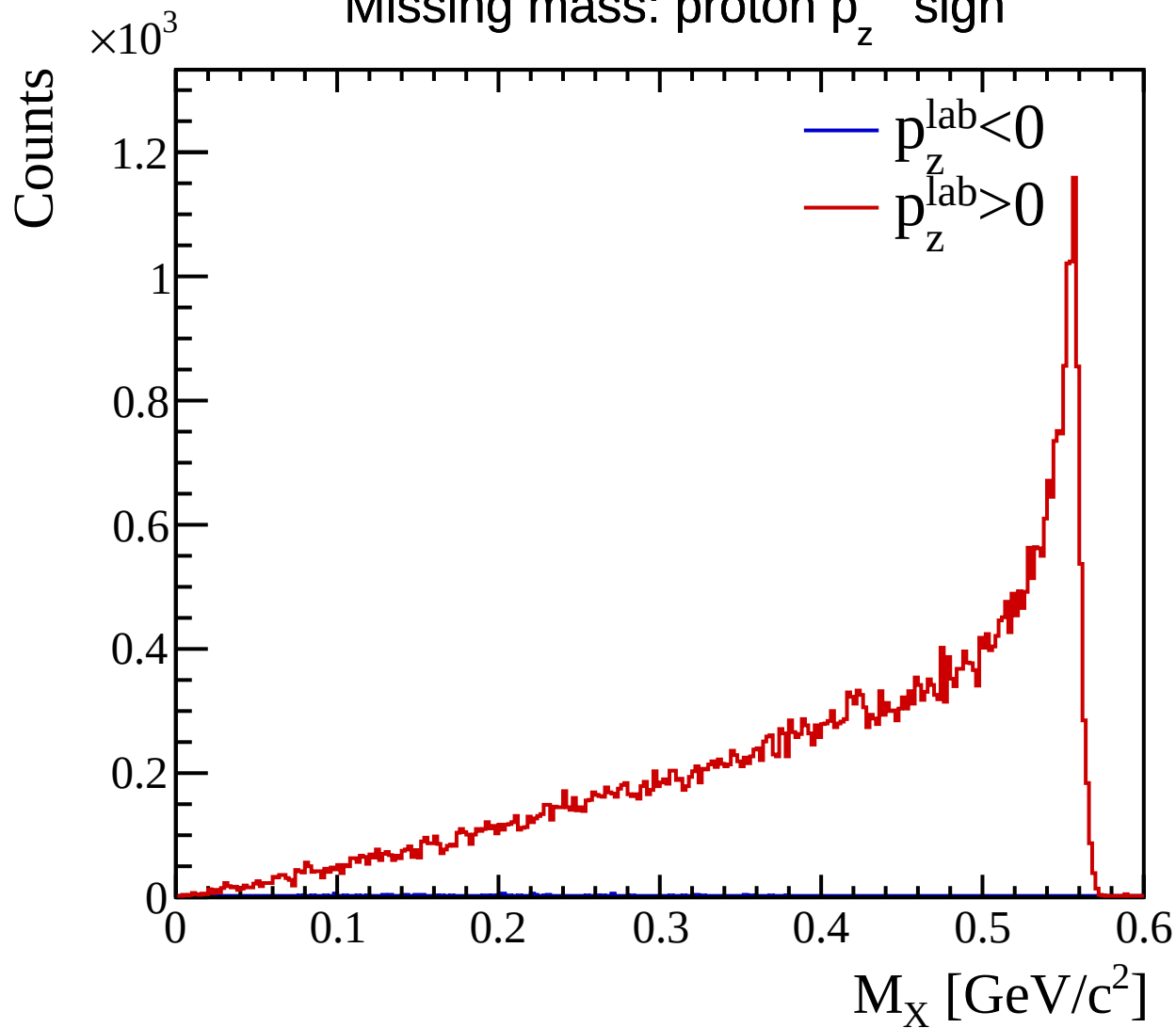
Λ invariant mass: proton p_z^{lab} sign



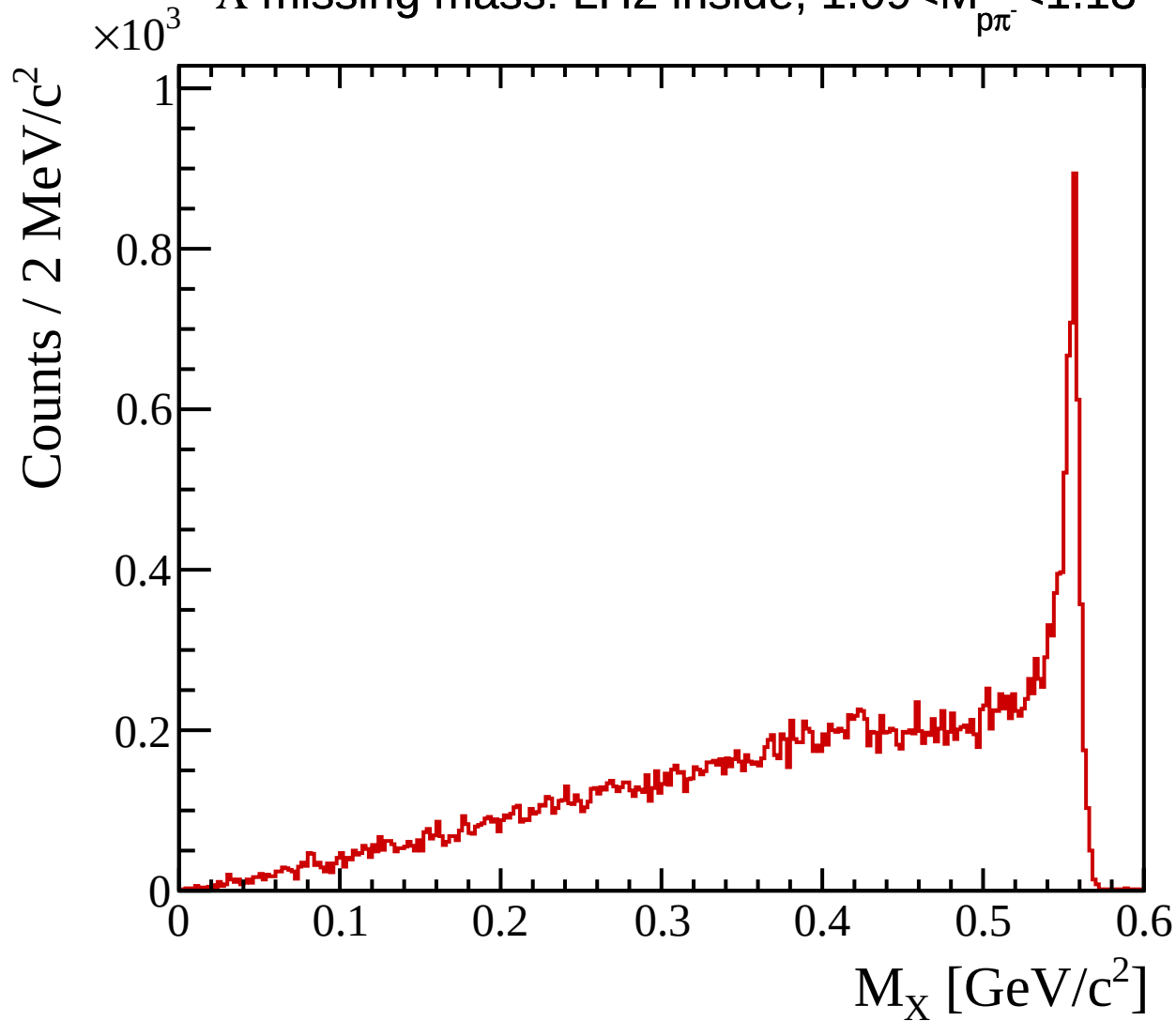
Missing mass: proton p_x^{lab} sign



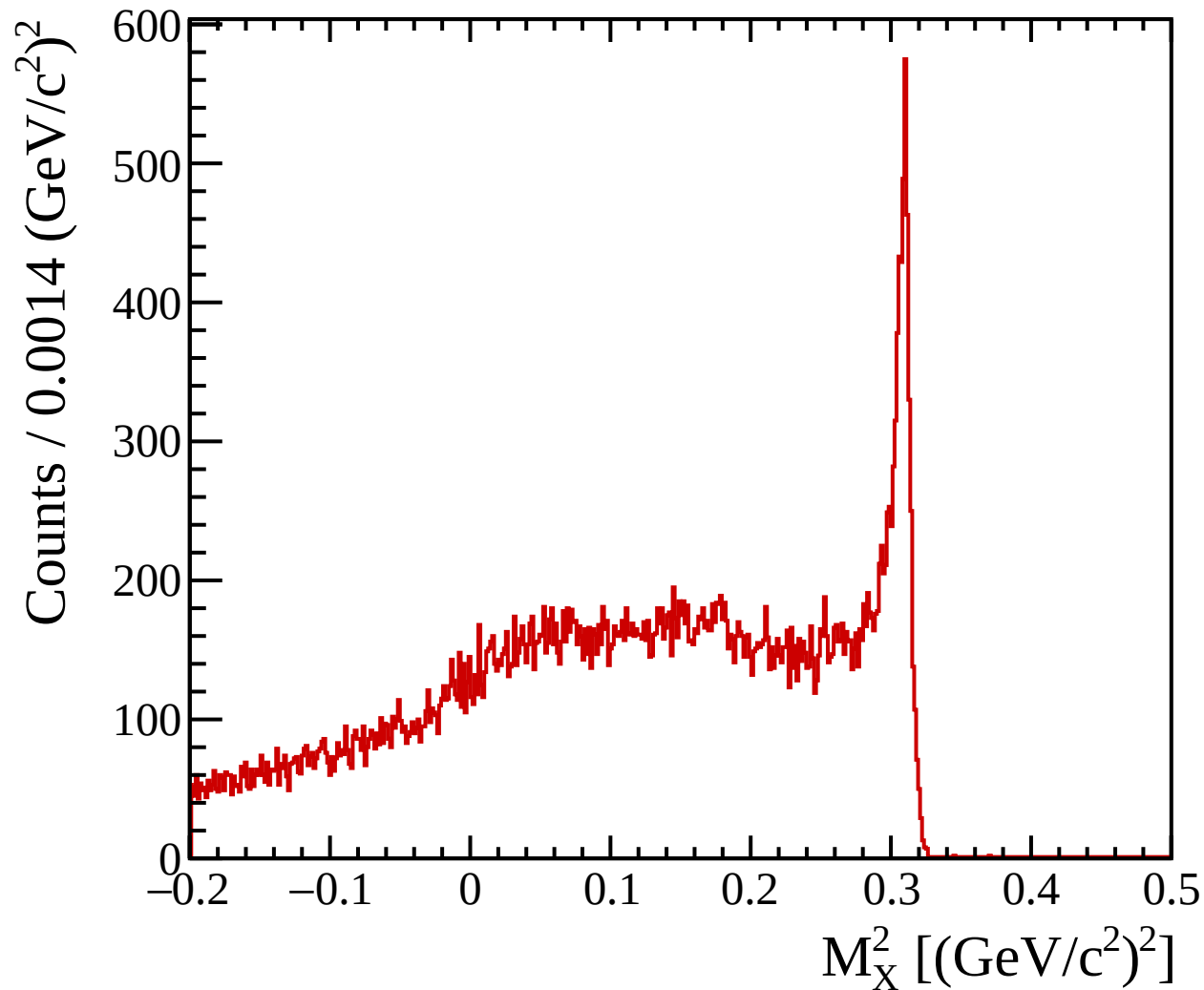
Missing mass: proton p_z^{lab} sign

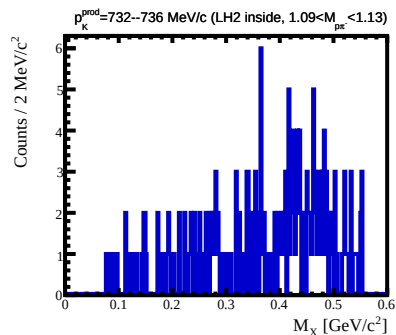
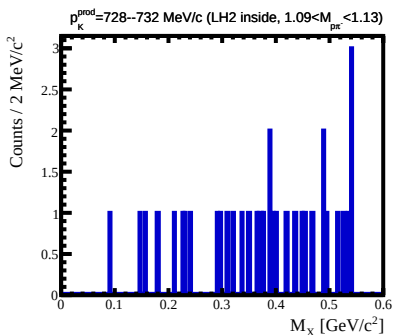
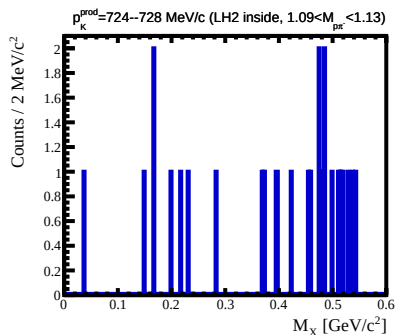
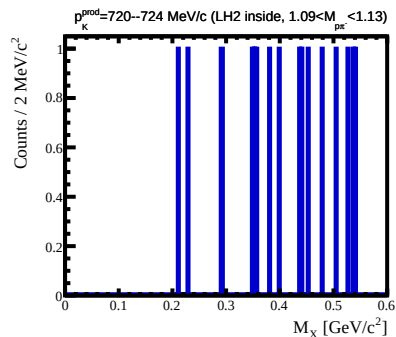
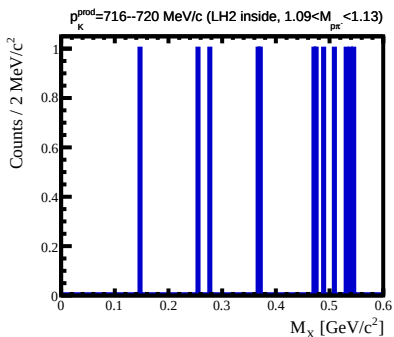
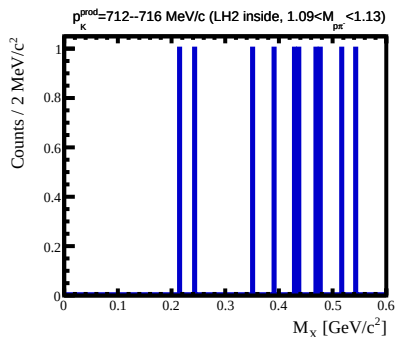
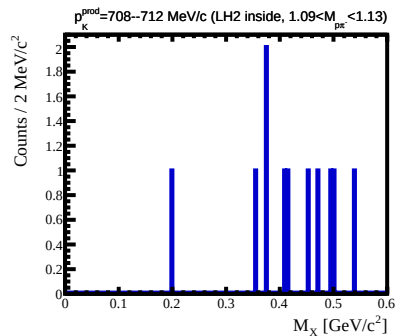
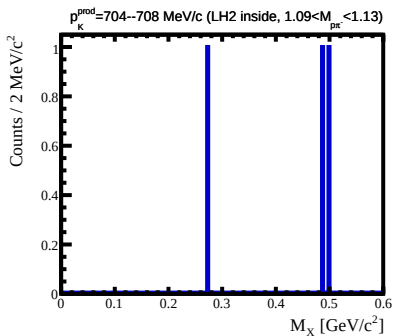
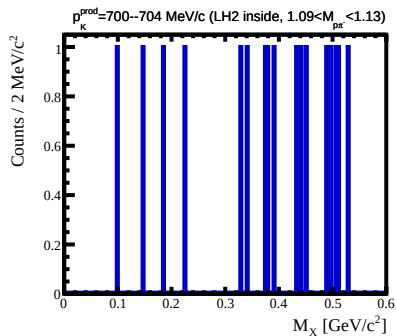


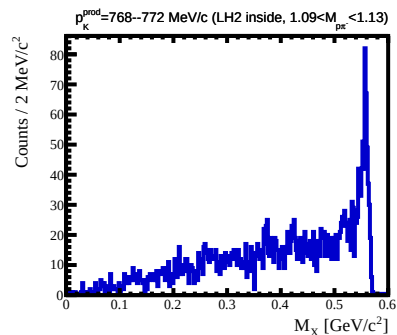
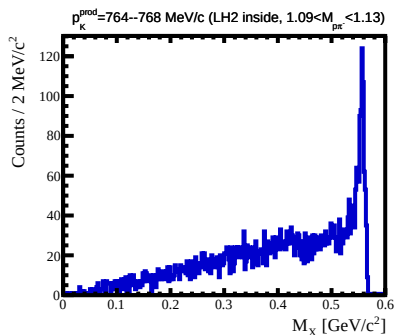
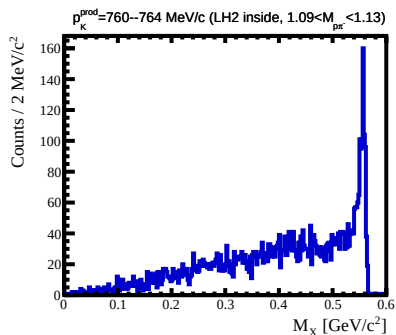
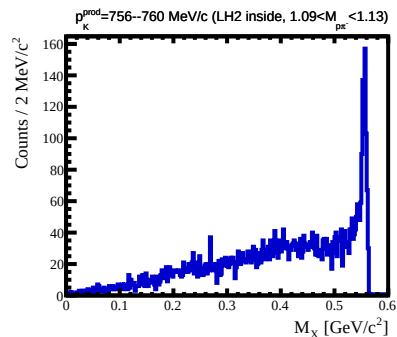
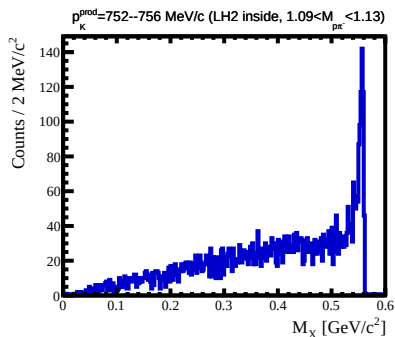
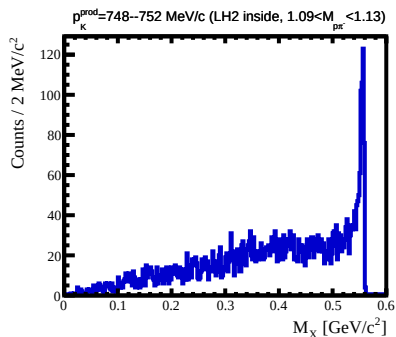
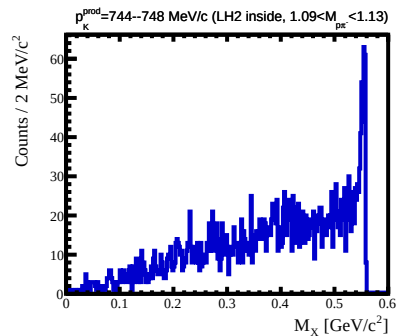
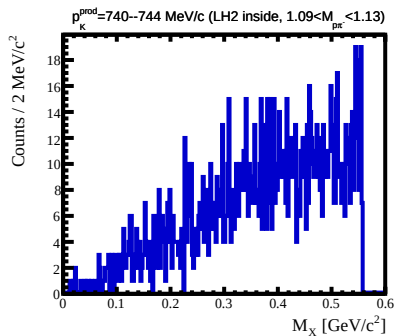
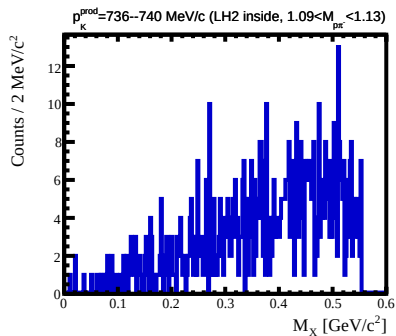
Λ missing mass: LH2 inside, $1.09 < M_{p\pi^-} < 1.13$

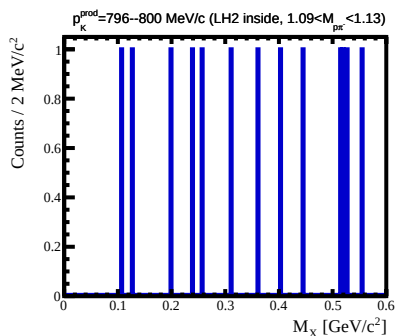
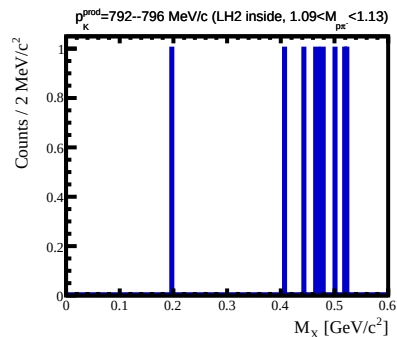
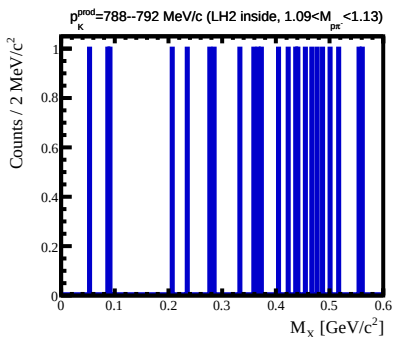
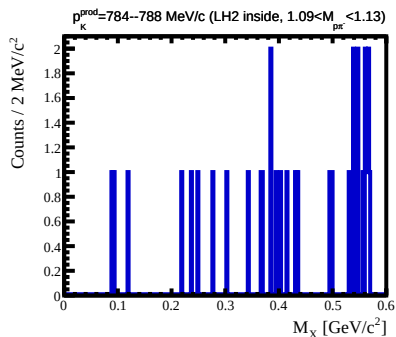
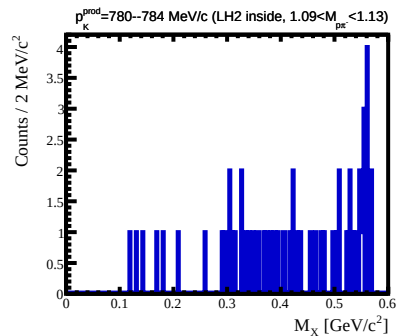
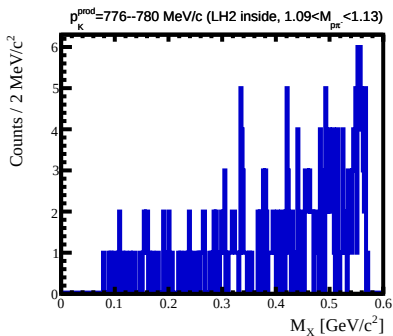
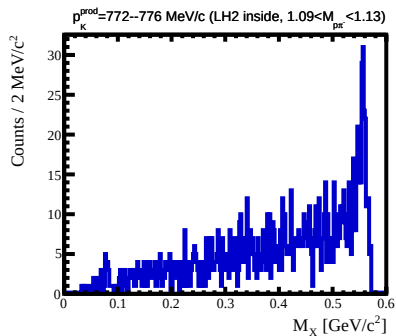


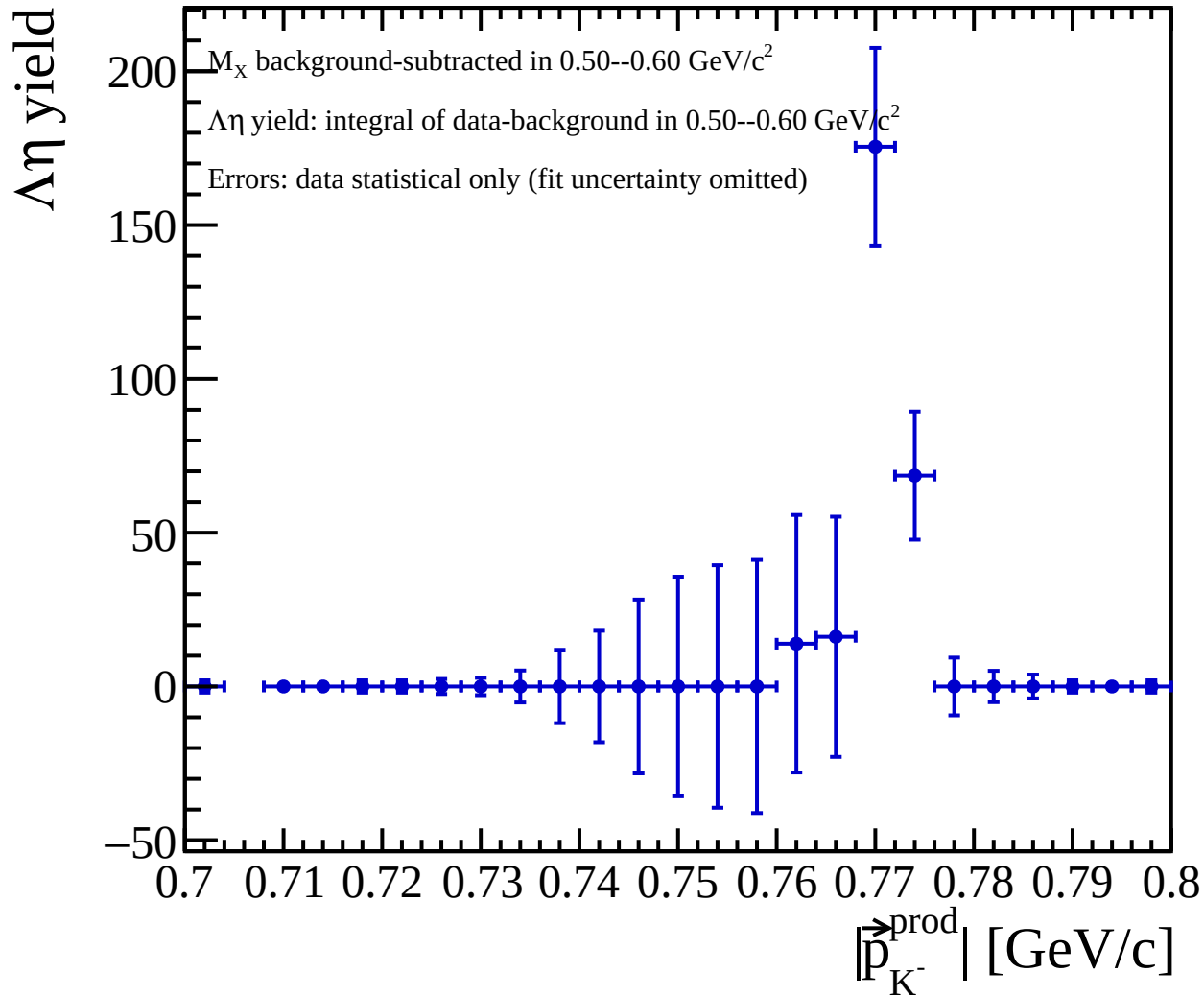
Λ missing mass squared: LH2 inside, $1.09 < M_{p\pi^-} < 1.13$



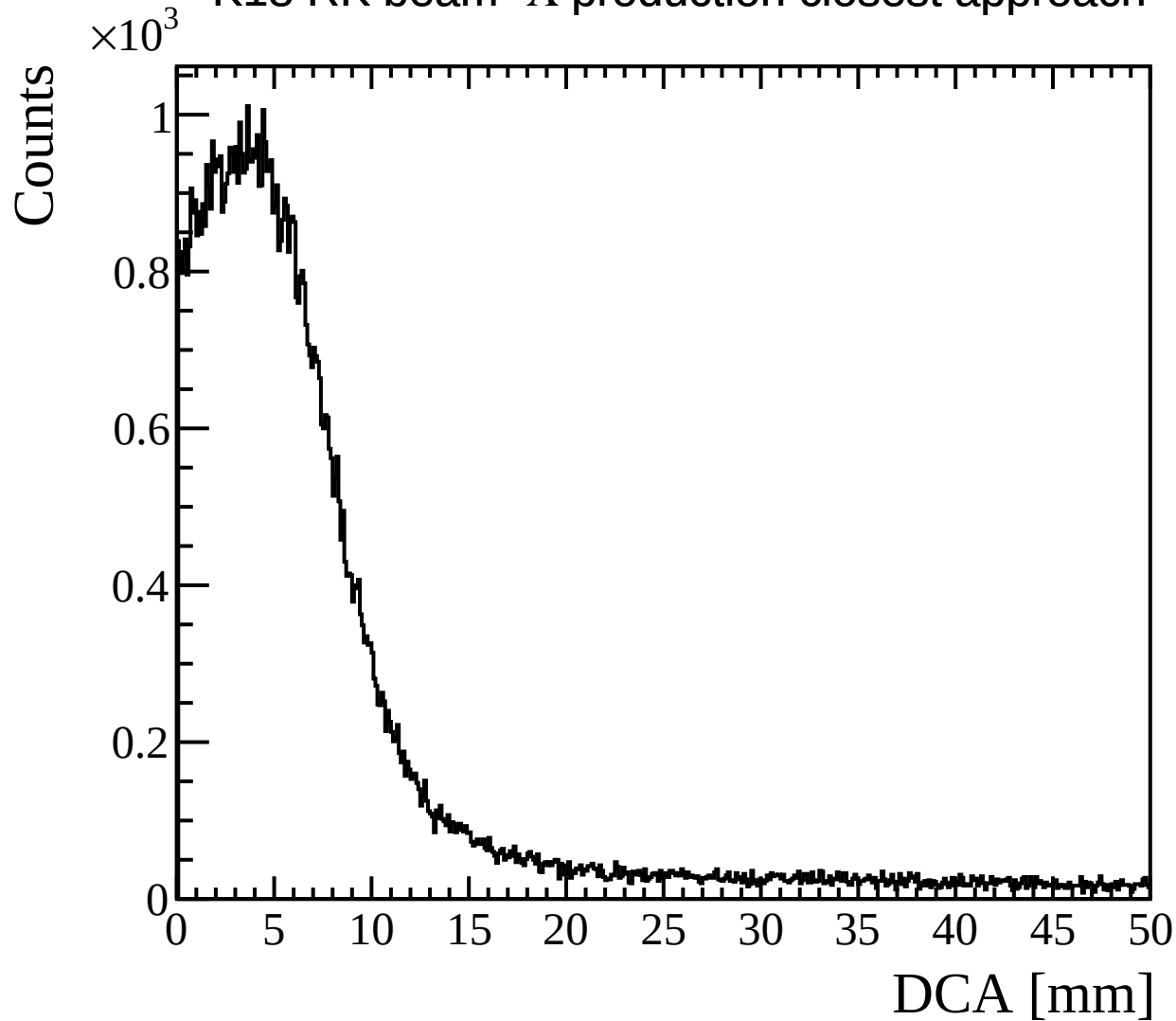




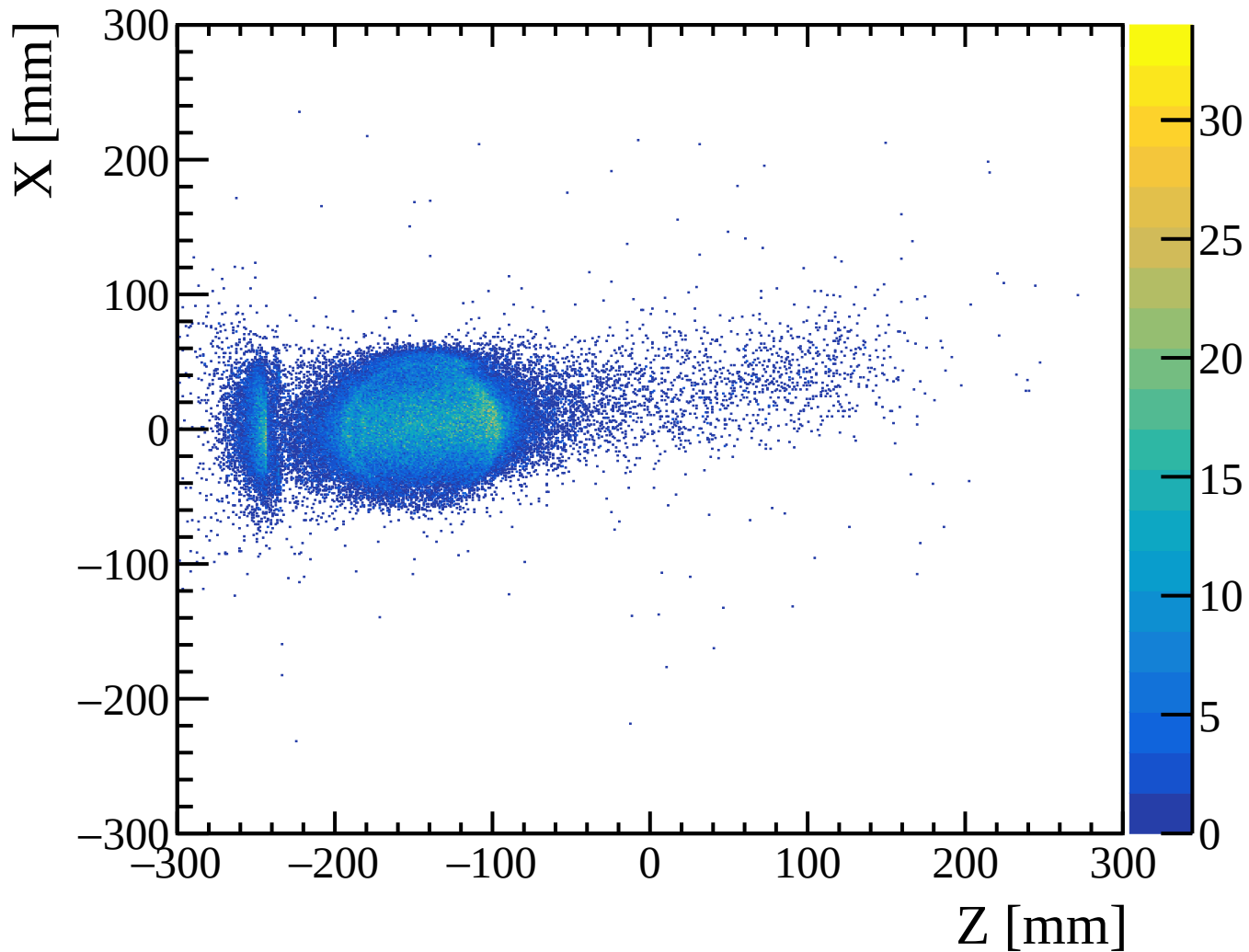




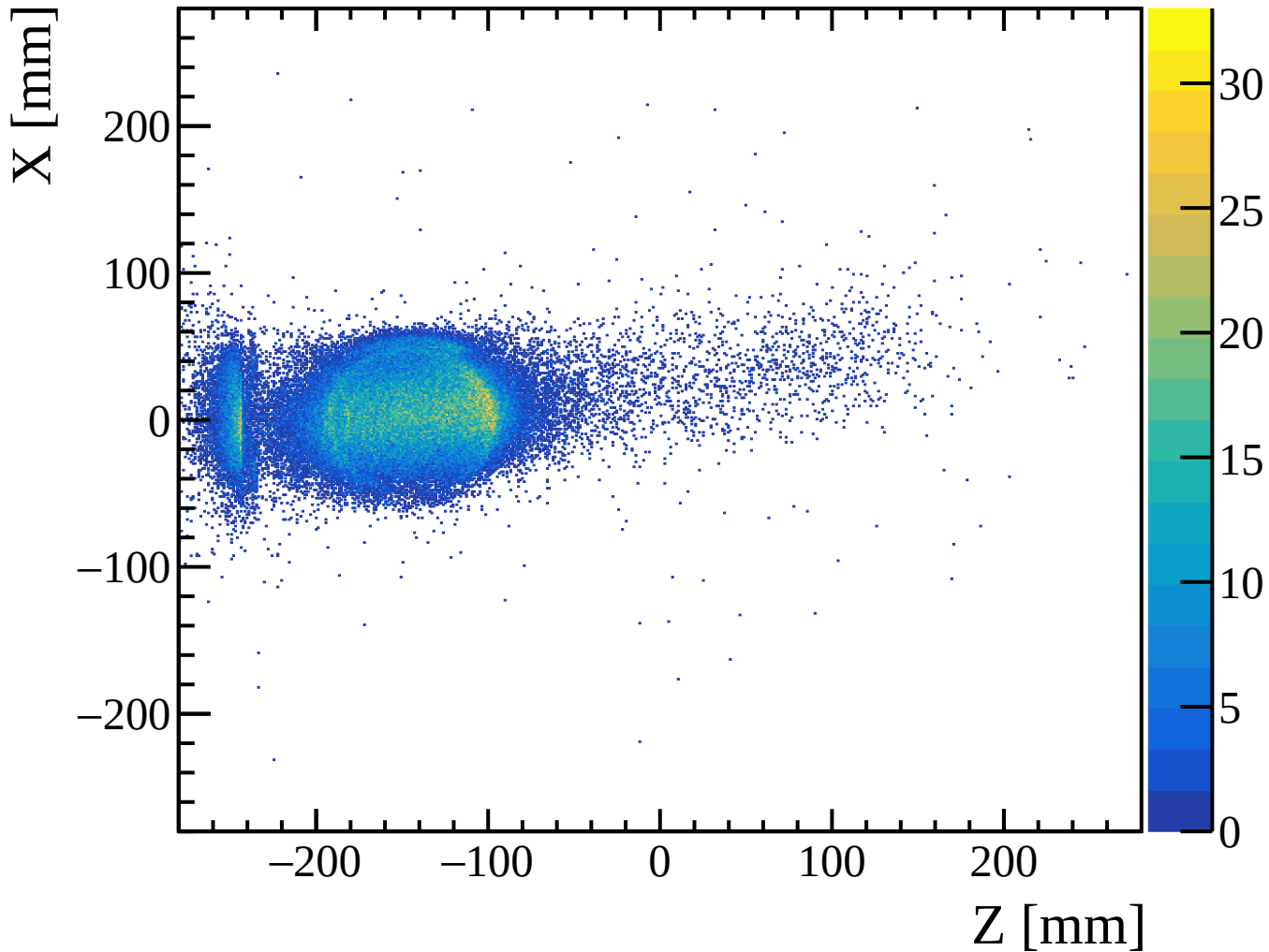
K18 RK beam-- Λ production closest approach



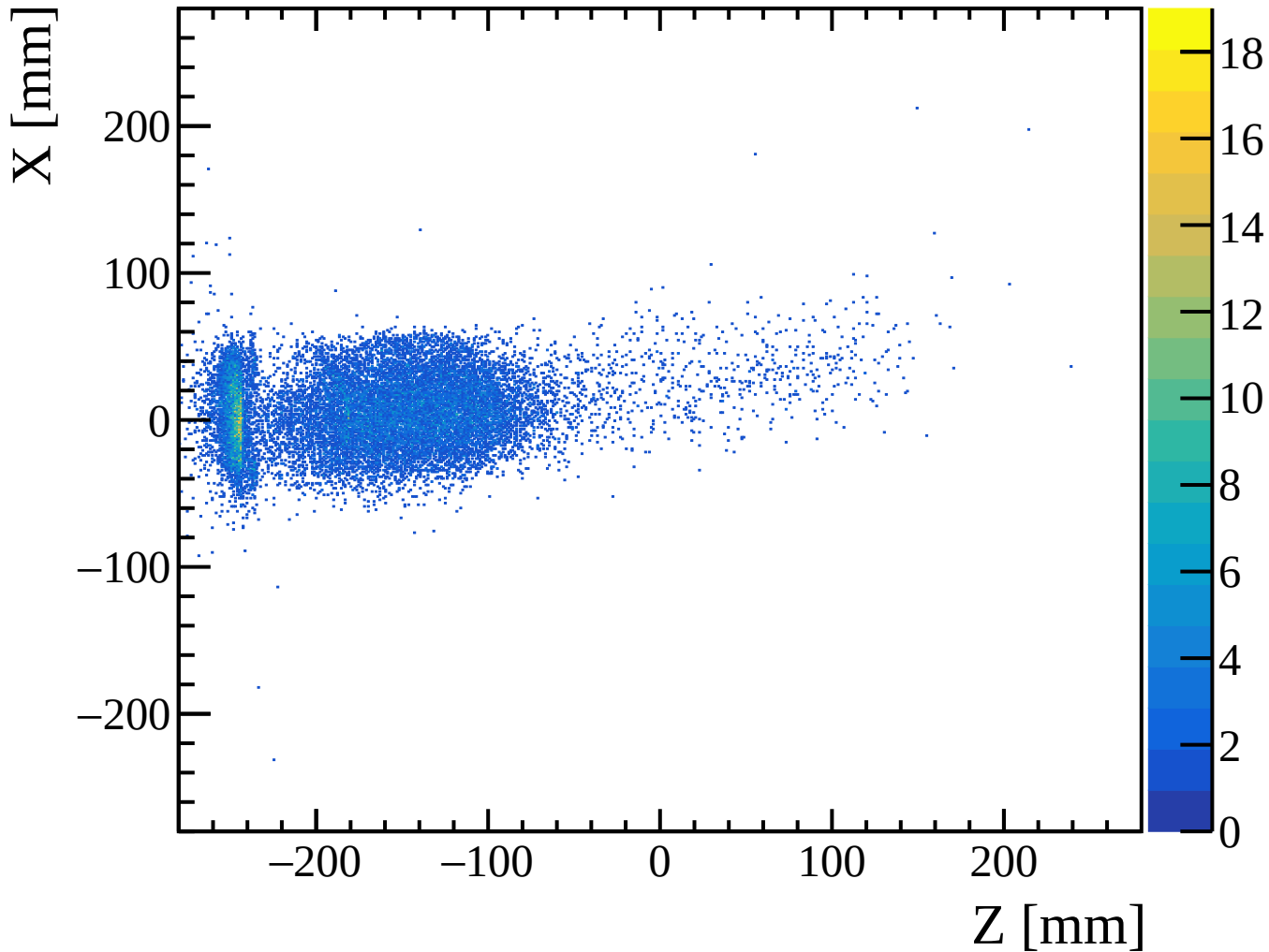
Reconstructed Λ production vertex (DST inclusive)



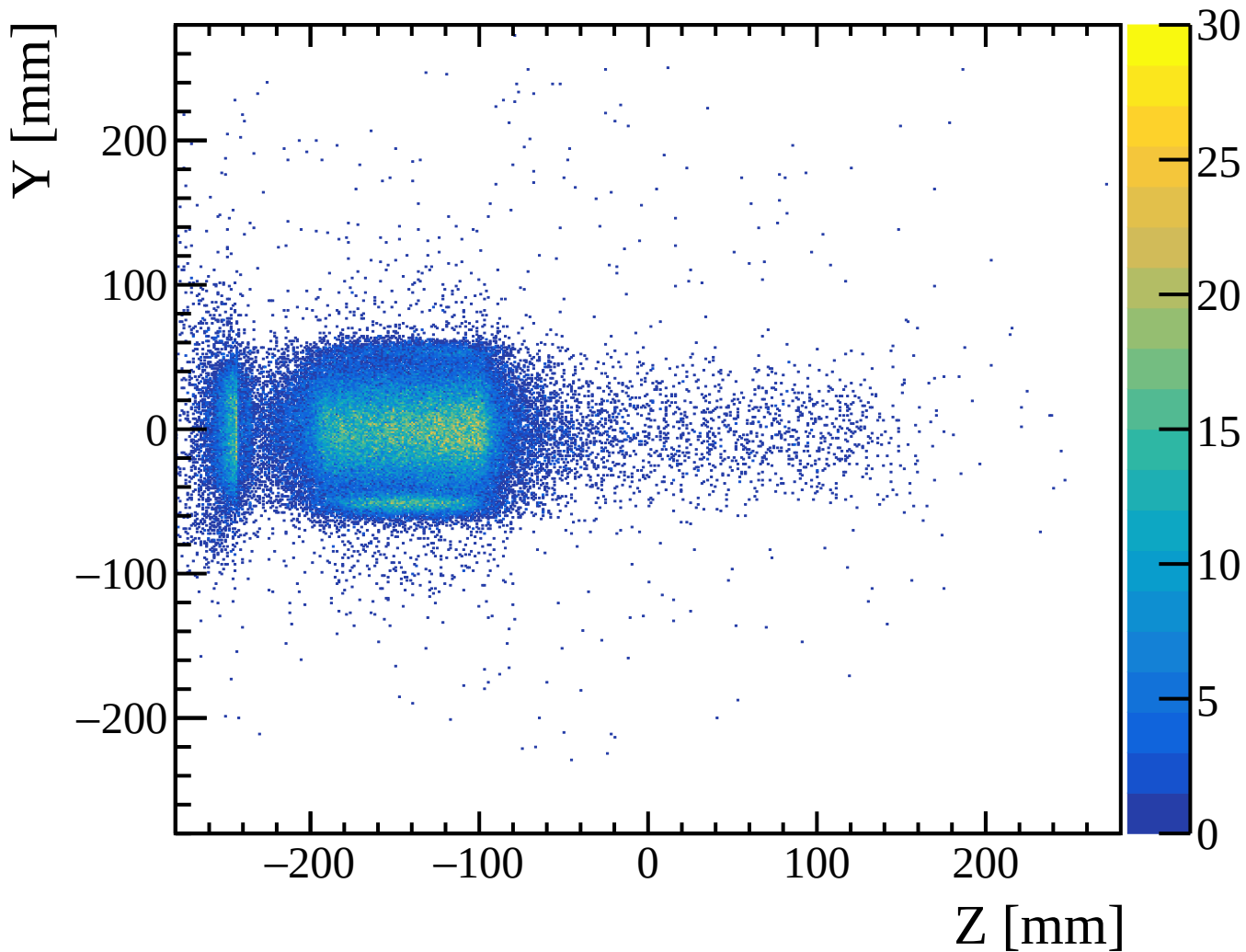
Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_x cut)



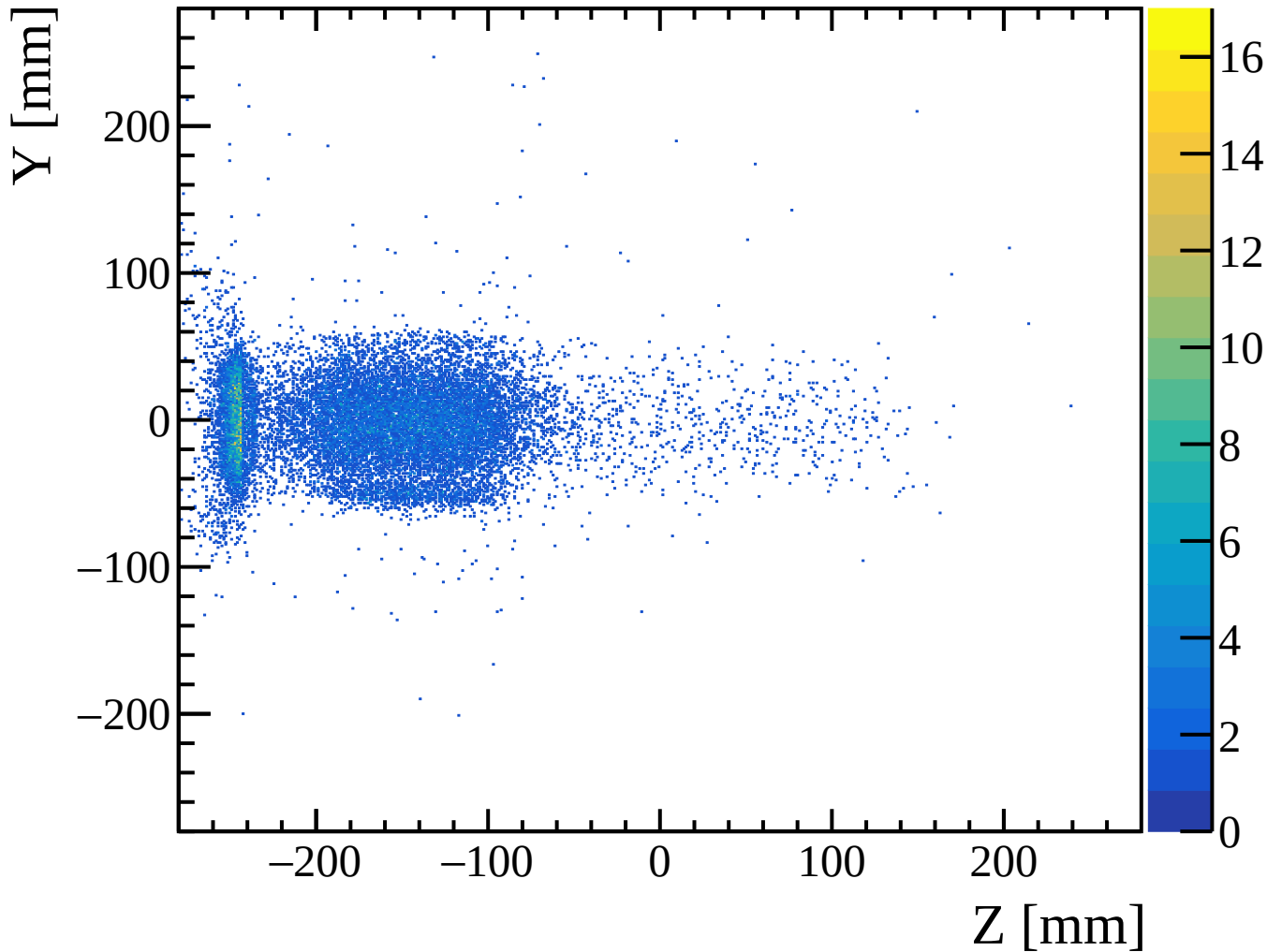
Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_X > 0.50$)



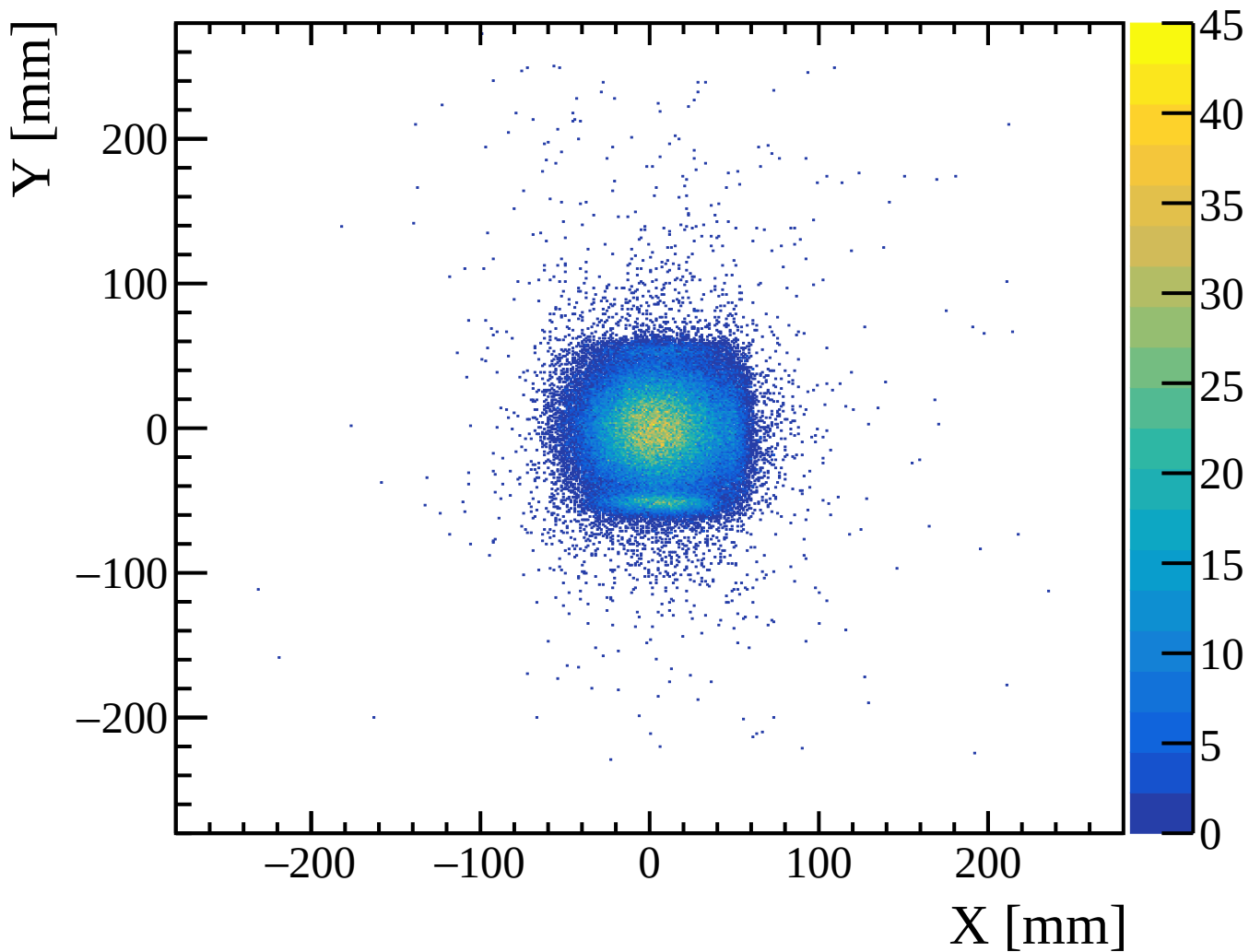
Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_x cut)



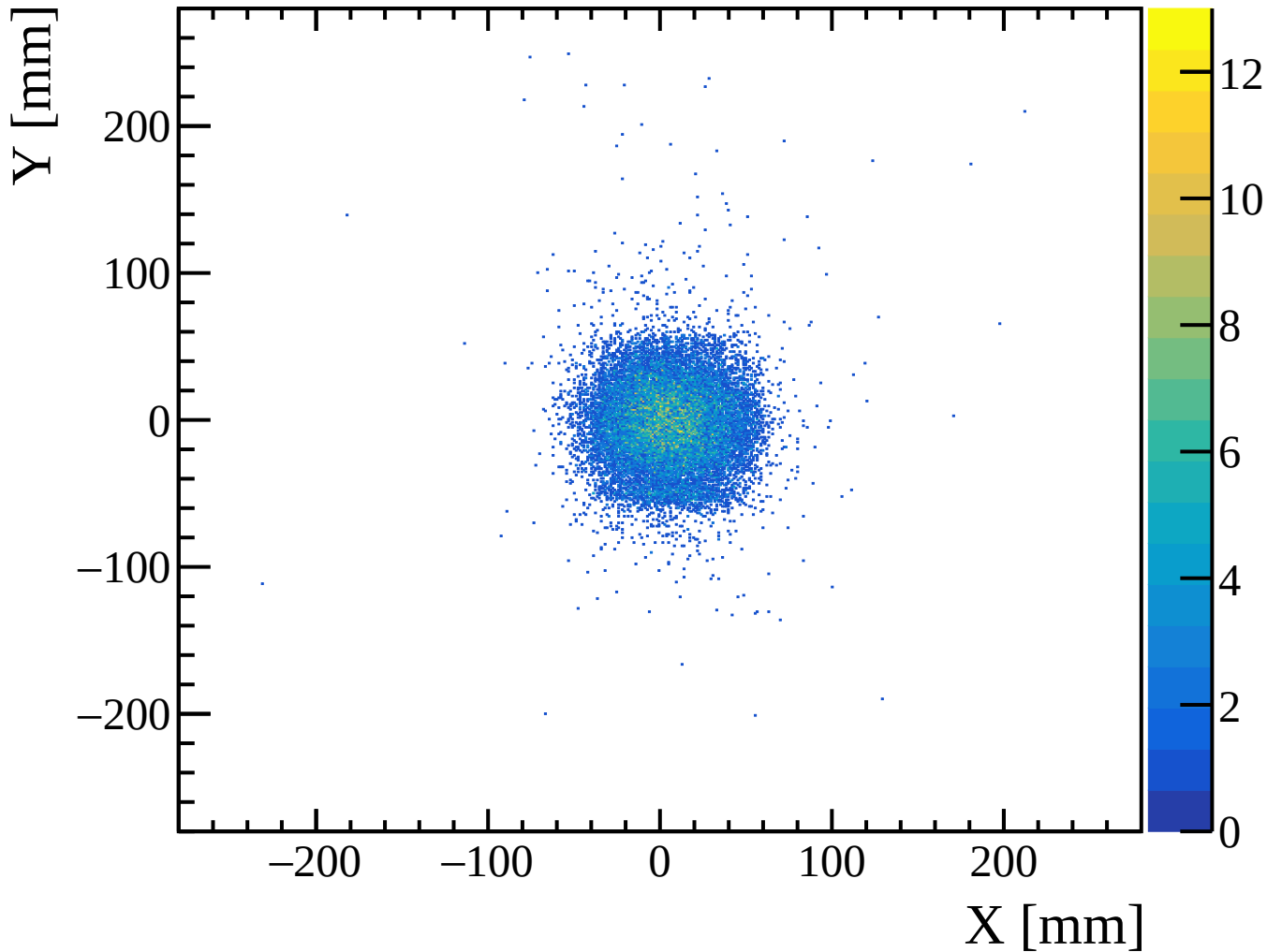
Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_x > 0.50$)

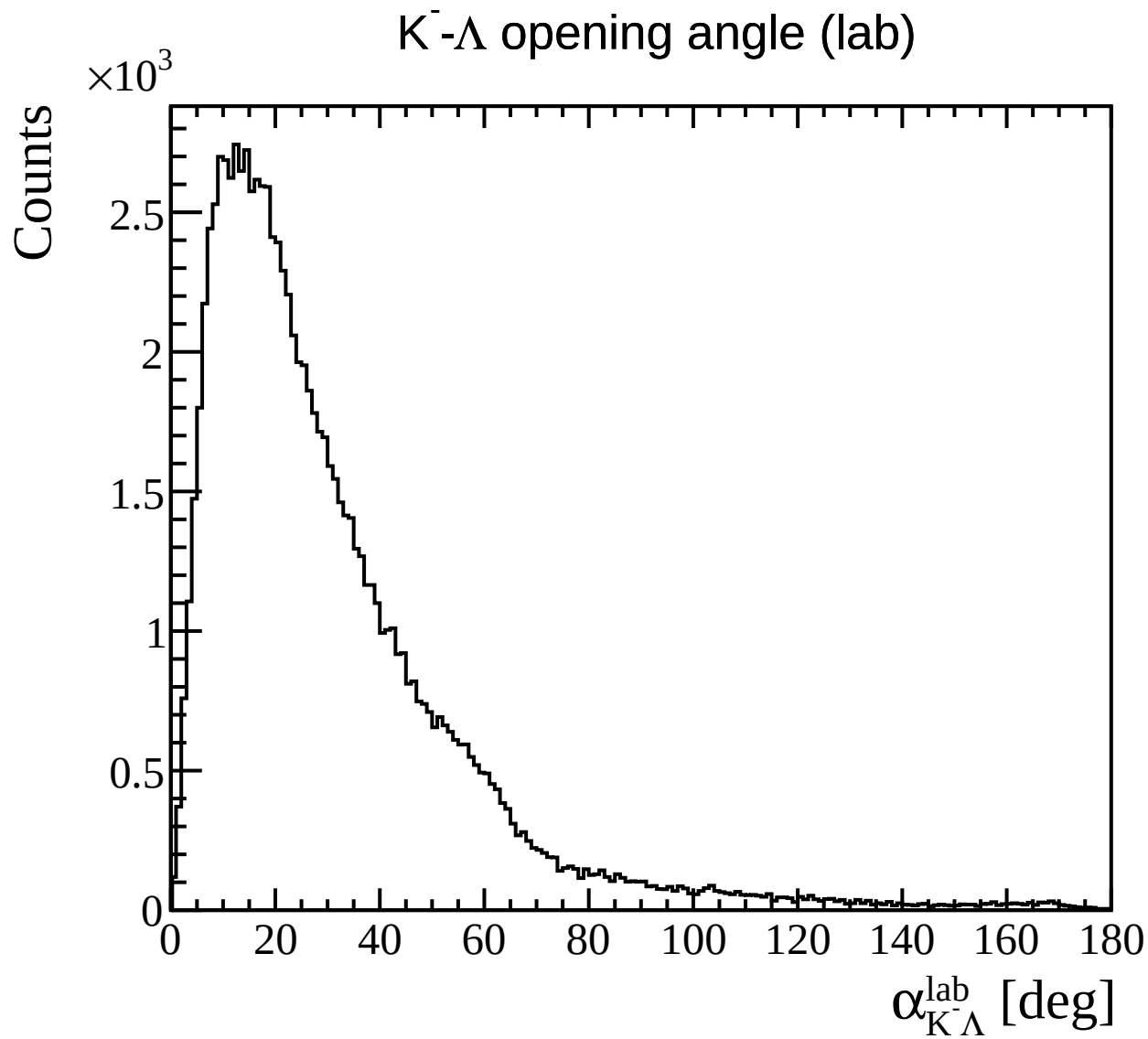


Reconstructed Λ production vertex (no LH2/Lambda-mass/ M_x cut)

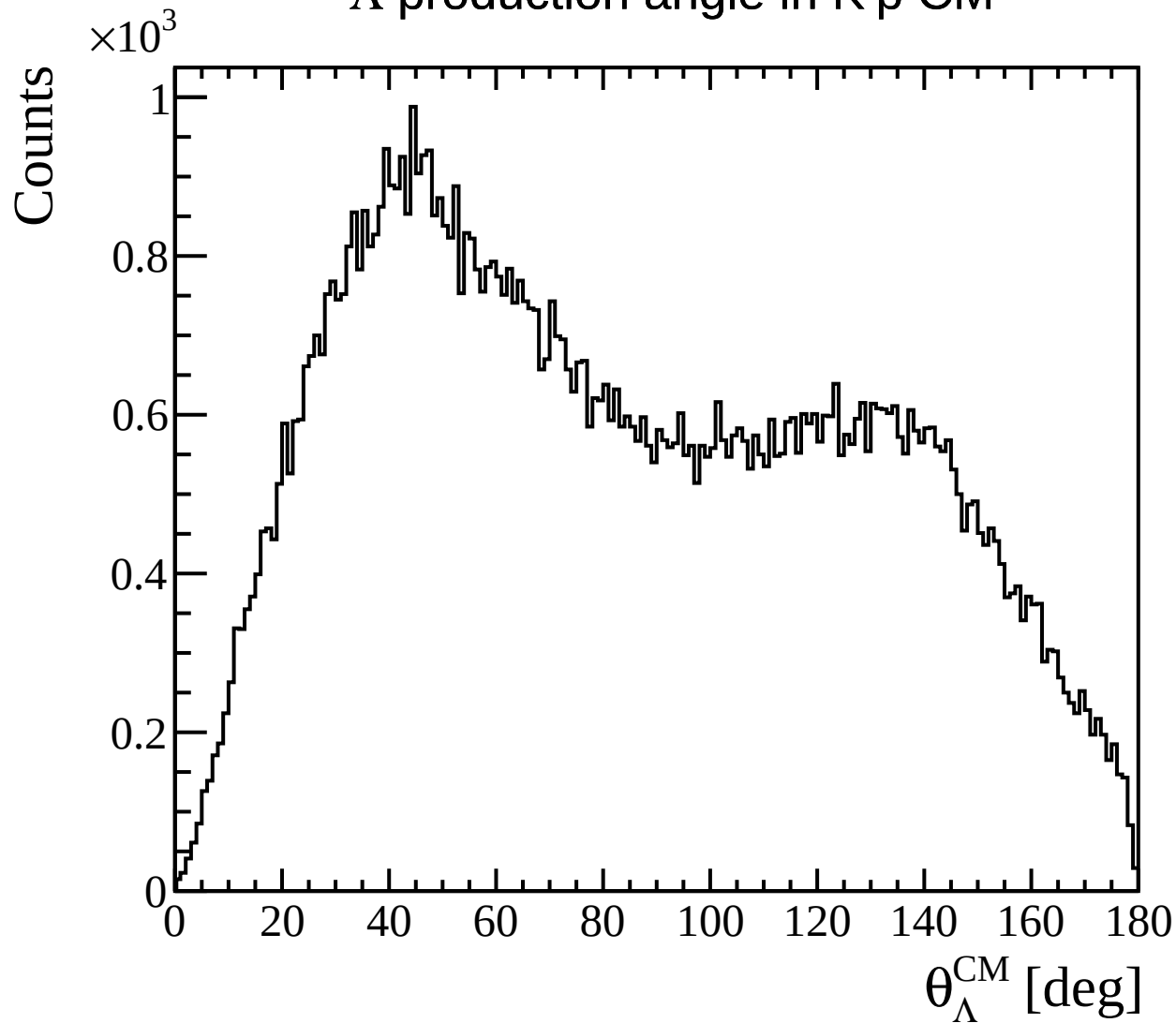


Reconstructed Λ production vertex (no LH2/Lambda-mass cut, $M_x > 0.50$)

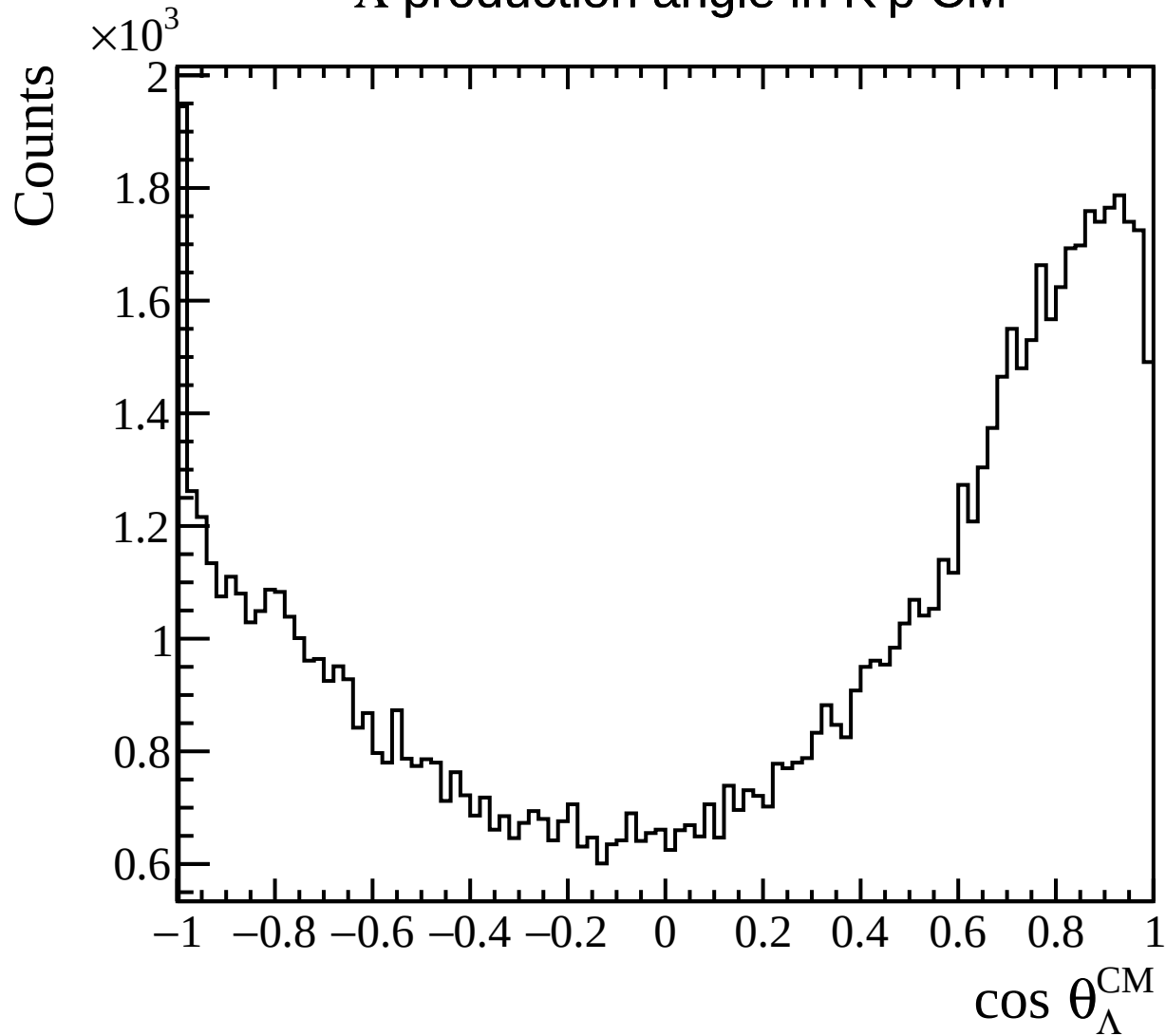




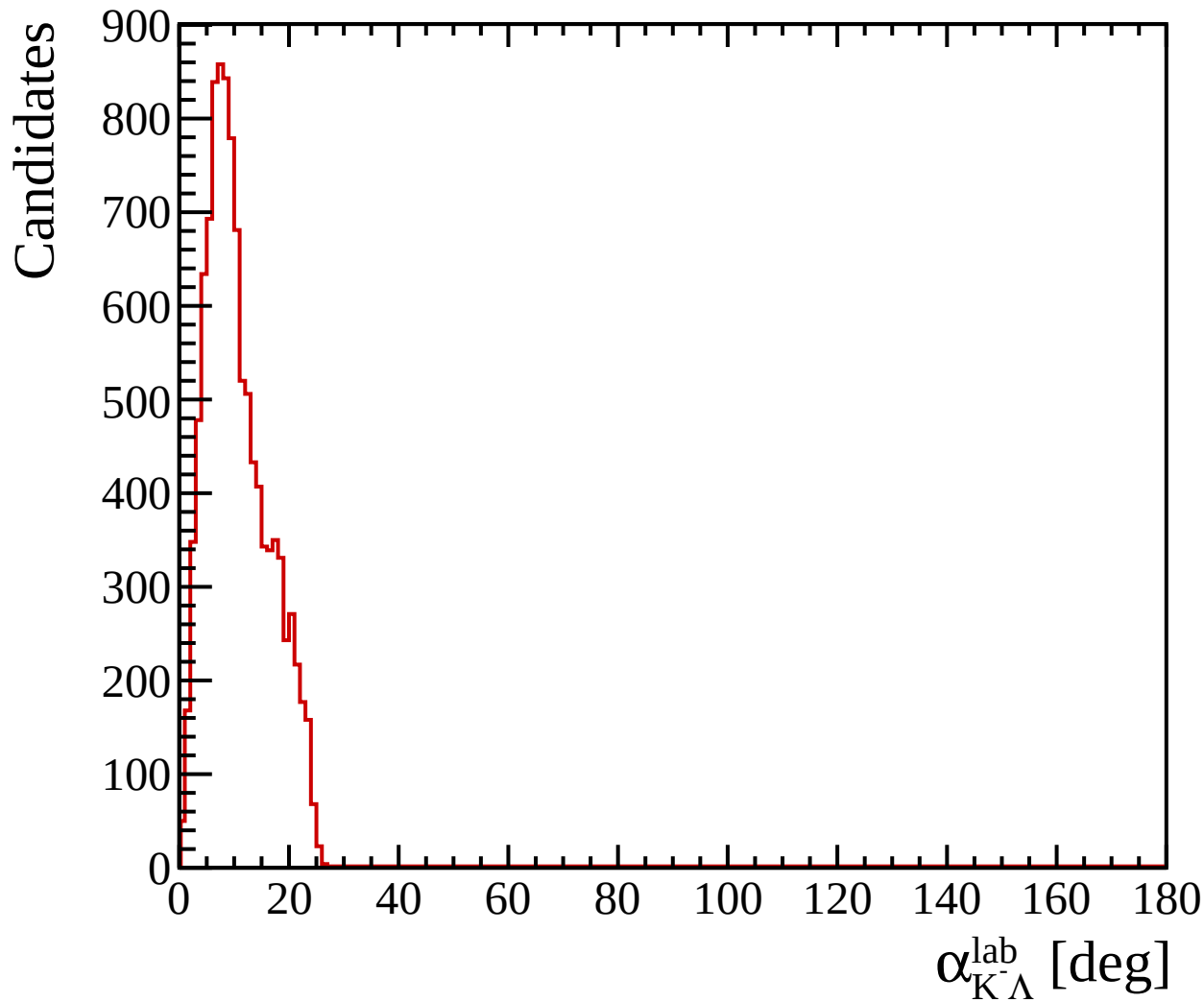
Λ production angle in K^-p CM



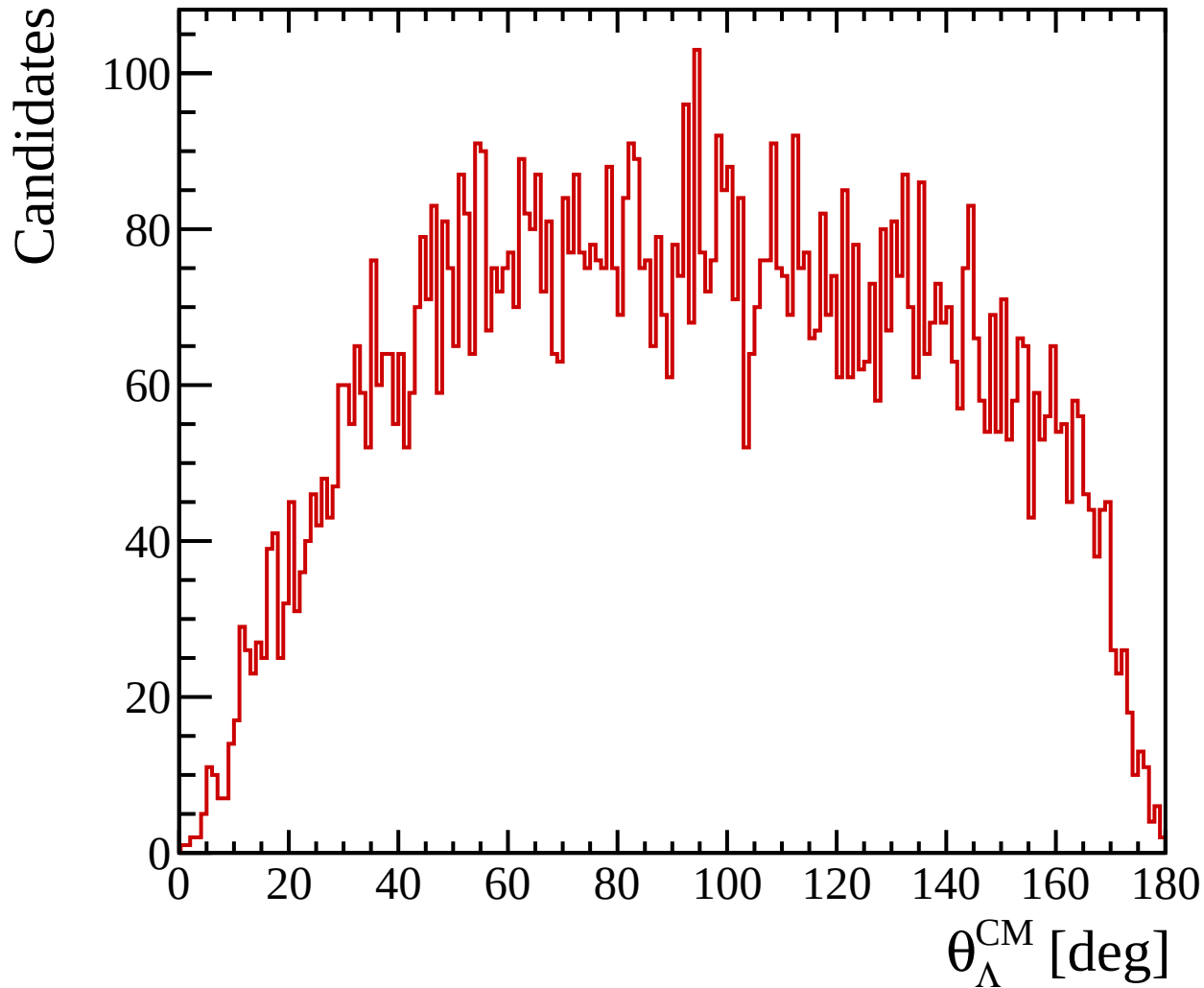
Λ production angle in K^-p CM



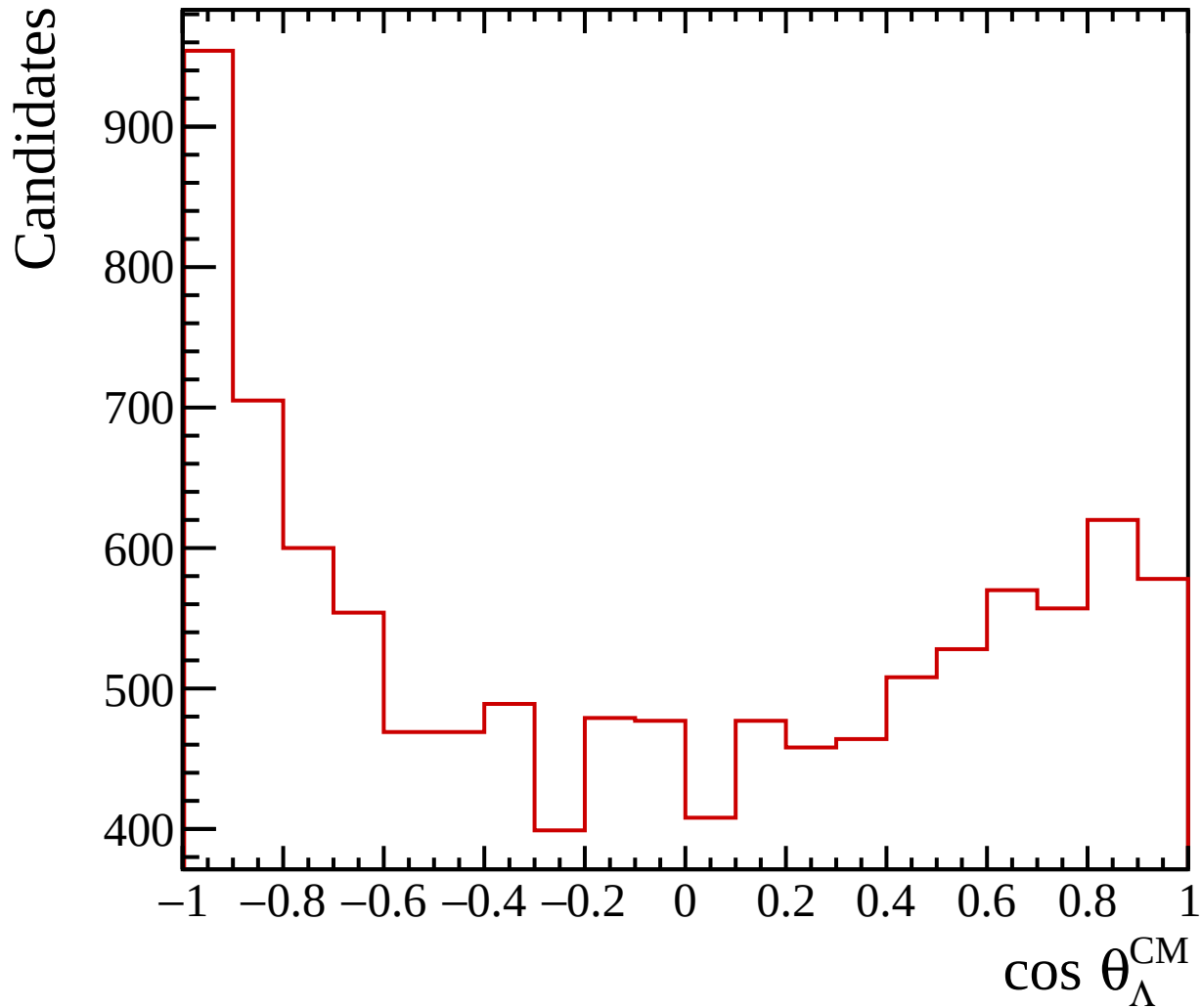
Selected Λ production angle ($M_x > 0.50$, no LH2 cut)



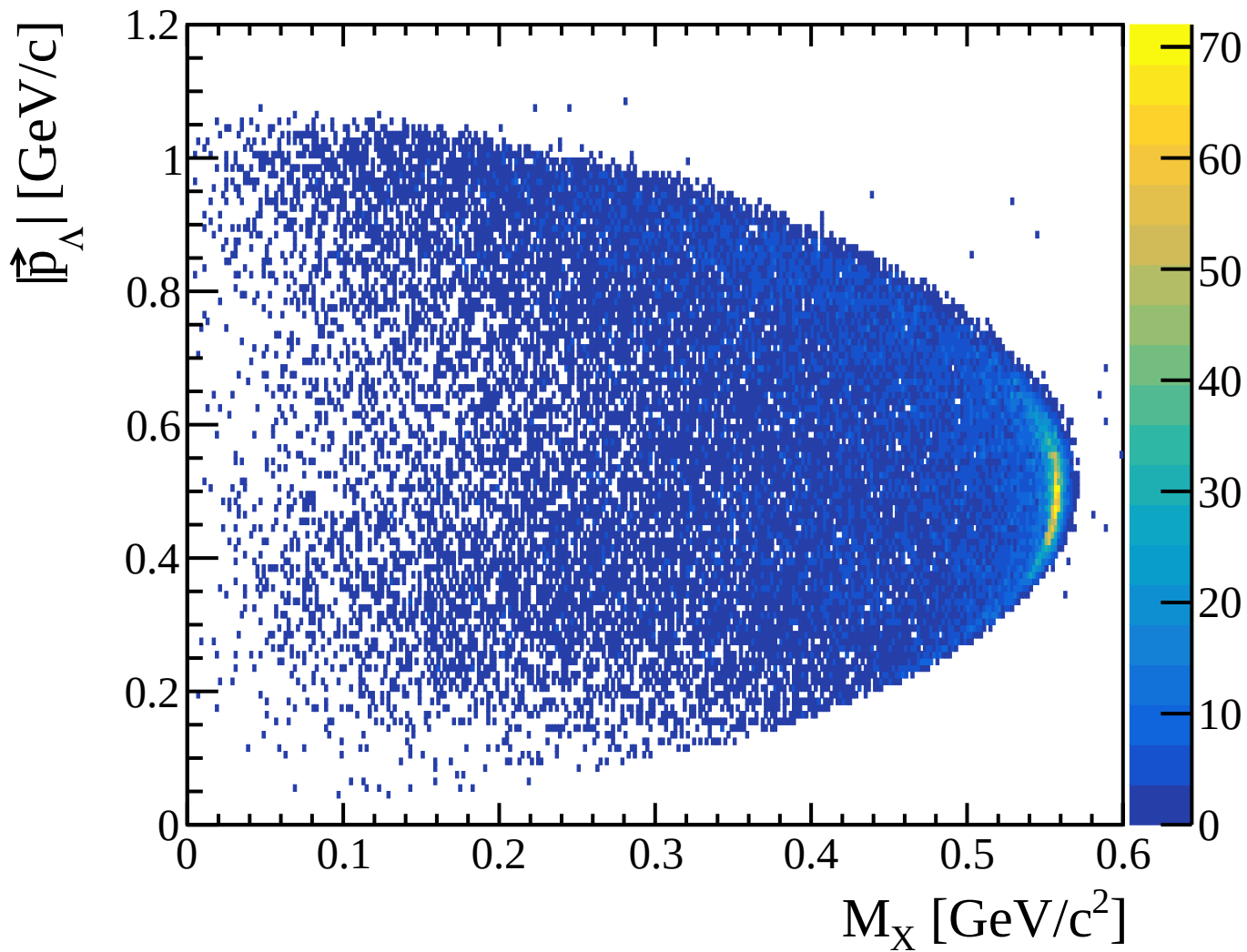
Selected Λ production angle ($M_x > 0.50$, no LH2 cut)



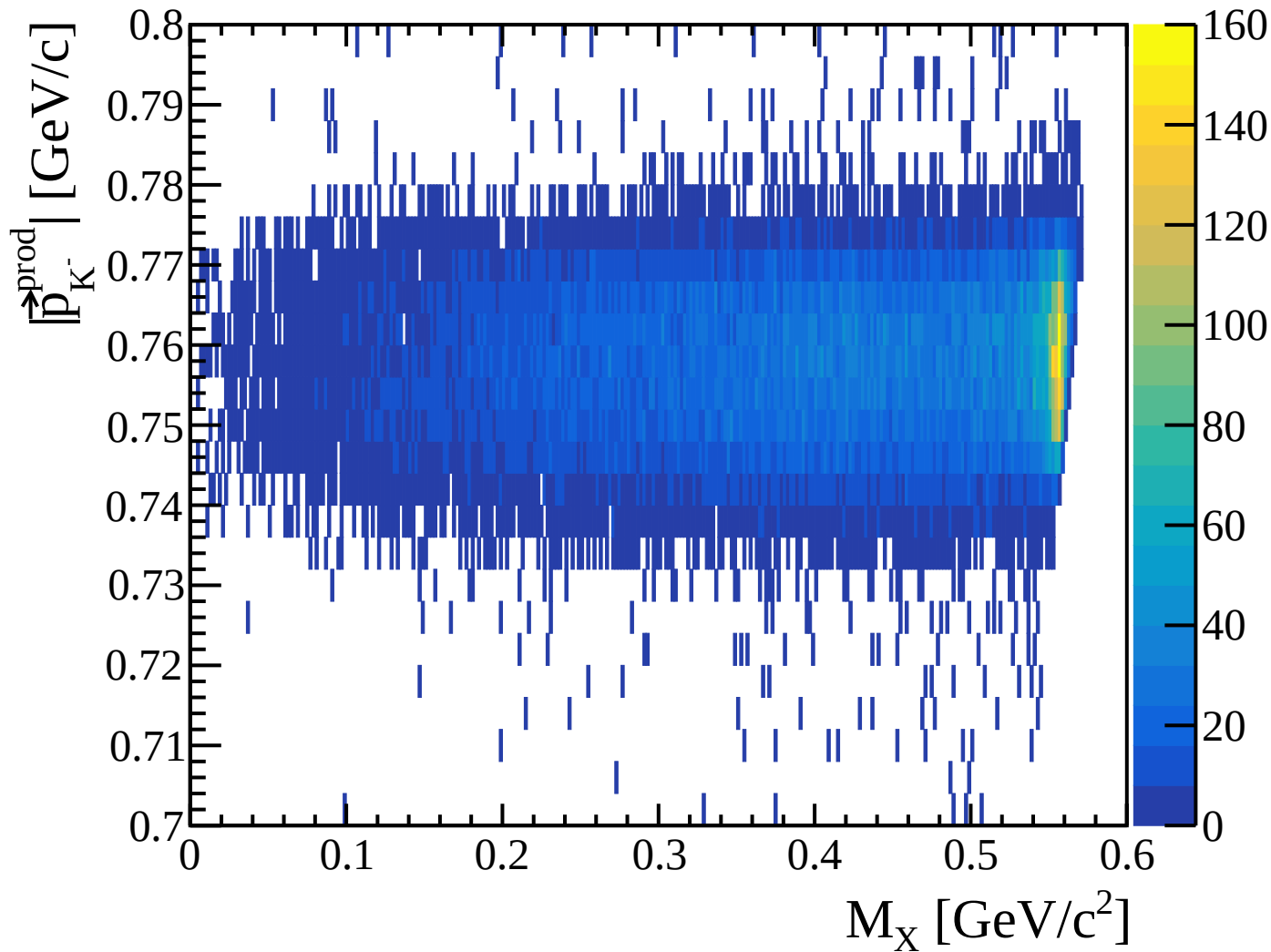
Selected Λ production angle ($M_x > 0.50$, no LH2 cut)



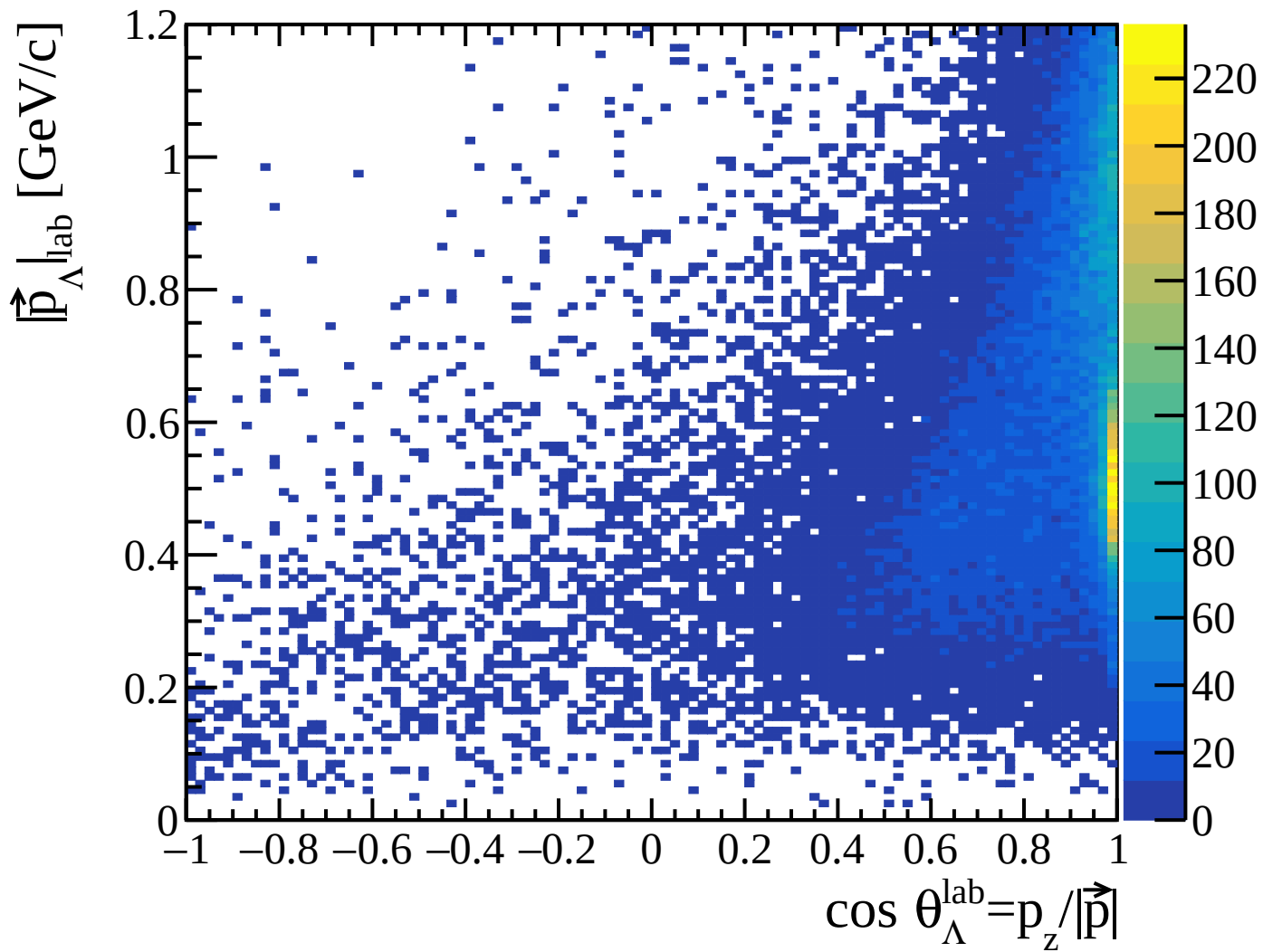
Missing mass vs reconstructed Λ momentum (no LH2 cut, no M_X cut)



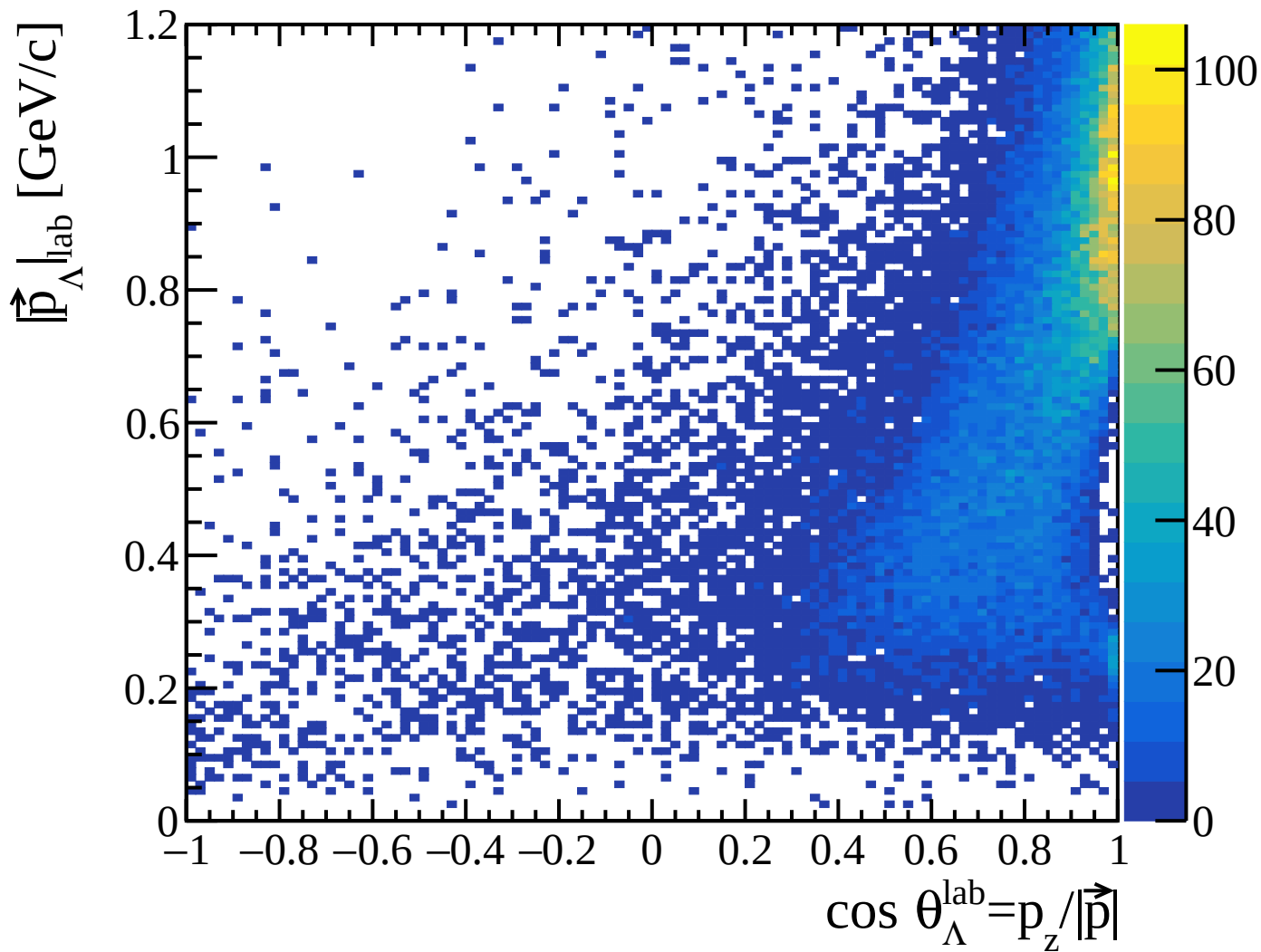
Missing mass vs production-vertex K18 momentum (no LH2 cut, no M_x cut)



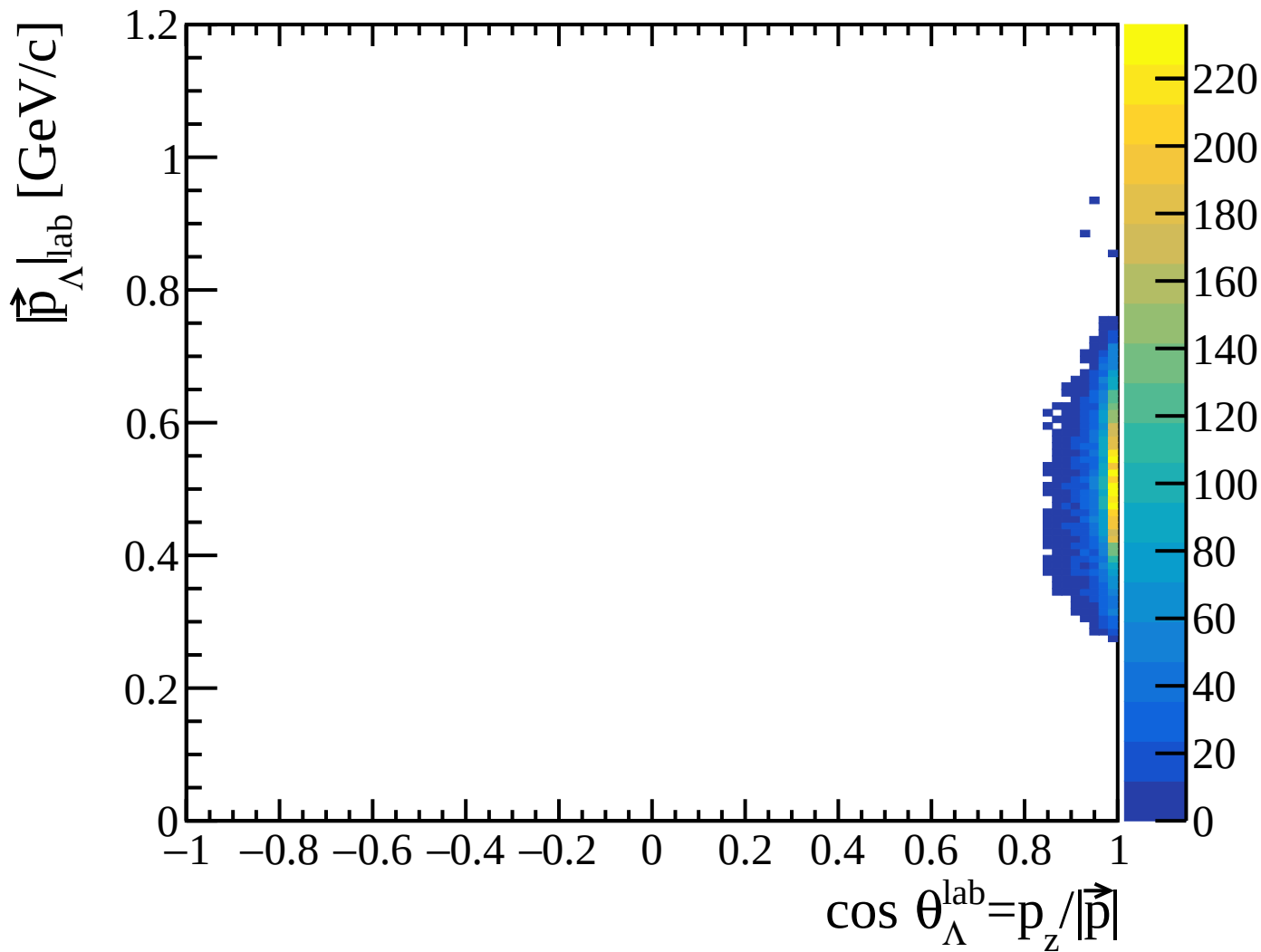
Selected Λ : lab $\cos\theta$ vs lab momentum (no LH2 cut, all M_X)



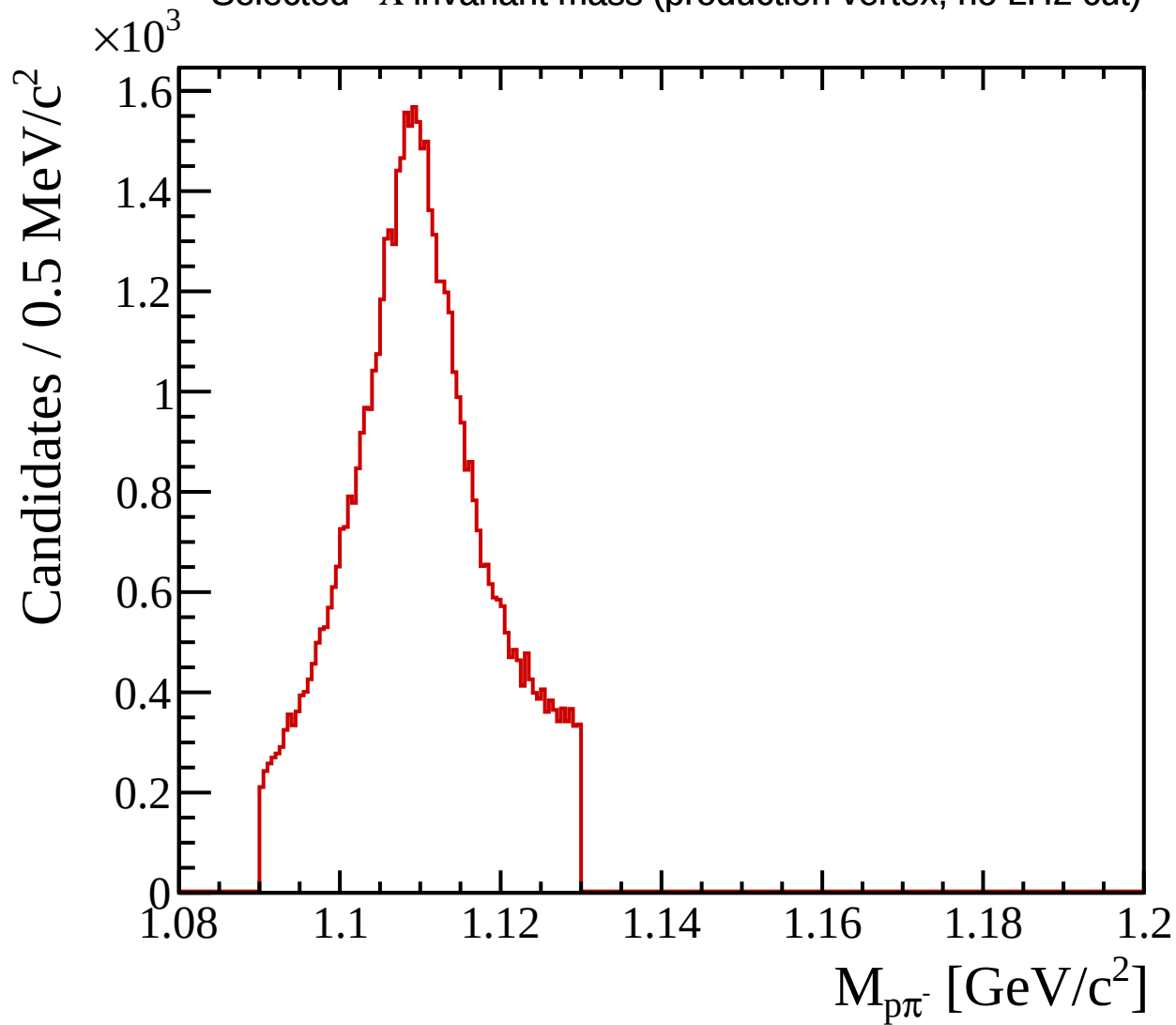
Selected Λ : lab $\cos\theta$ vs lab momentum (no LH2 cut, $M_x \leq 0.50 \text{ GeV}/c^2$)



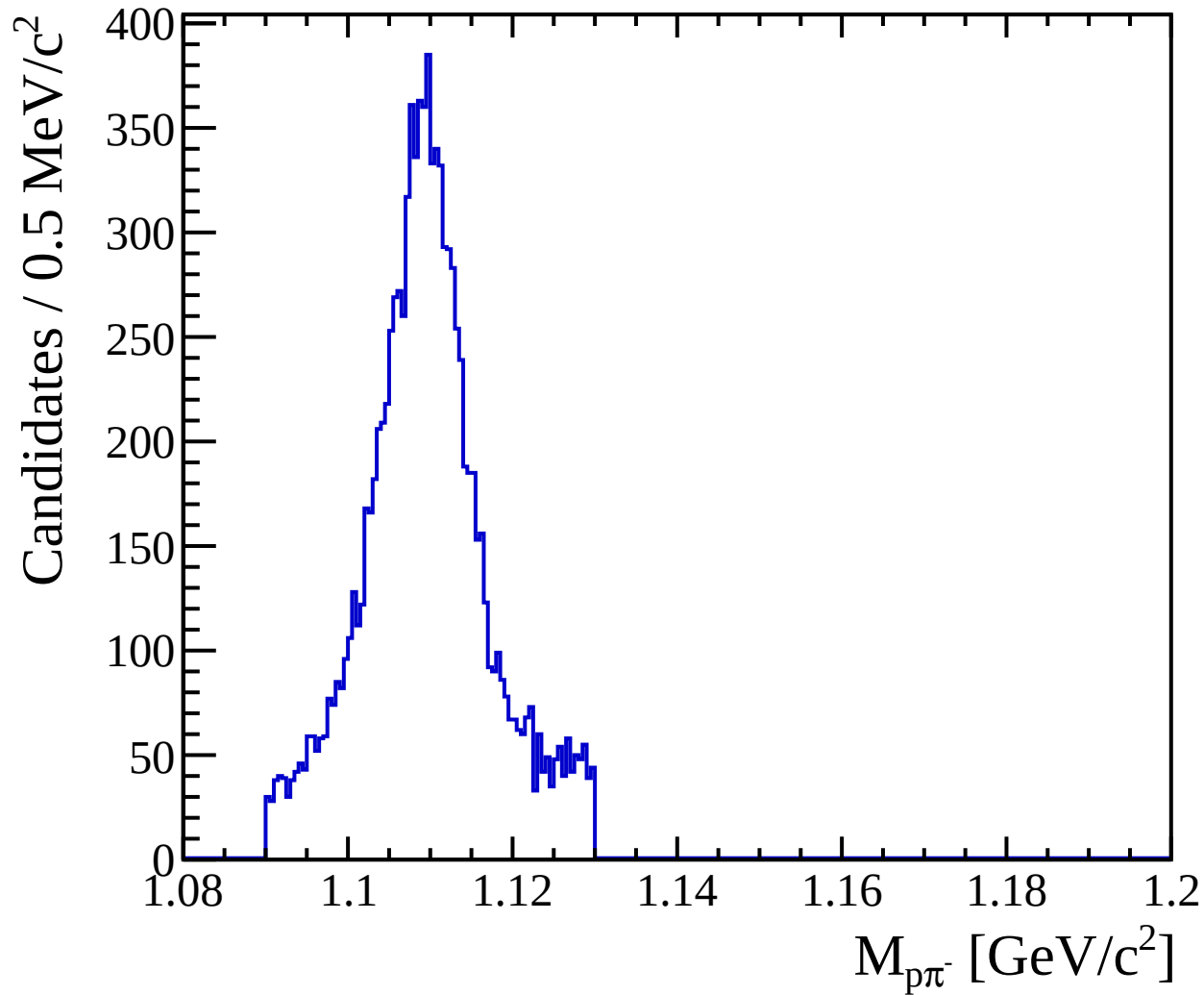
Selected Λ : lab $\cos\theta$ vs lab momentum (no LH2 cut, $M_X > 0.50 \text{ GeV}/c^2$)



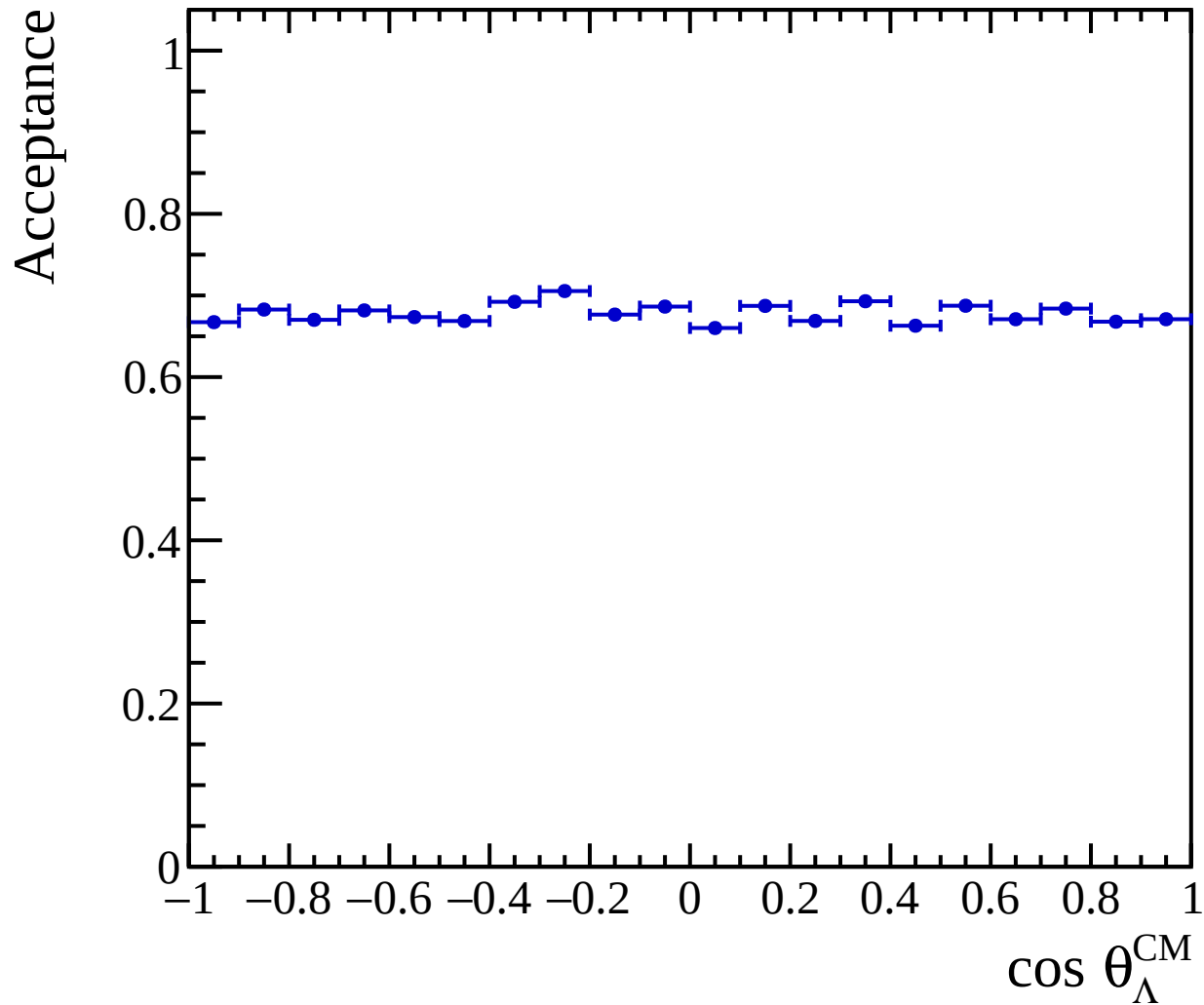
Selected Λ invariant mass (production vertex, no LH2 cut)



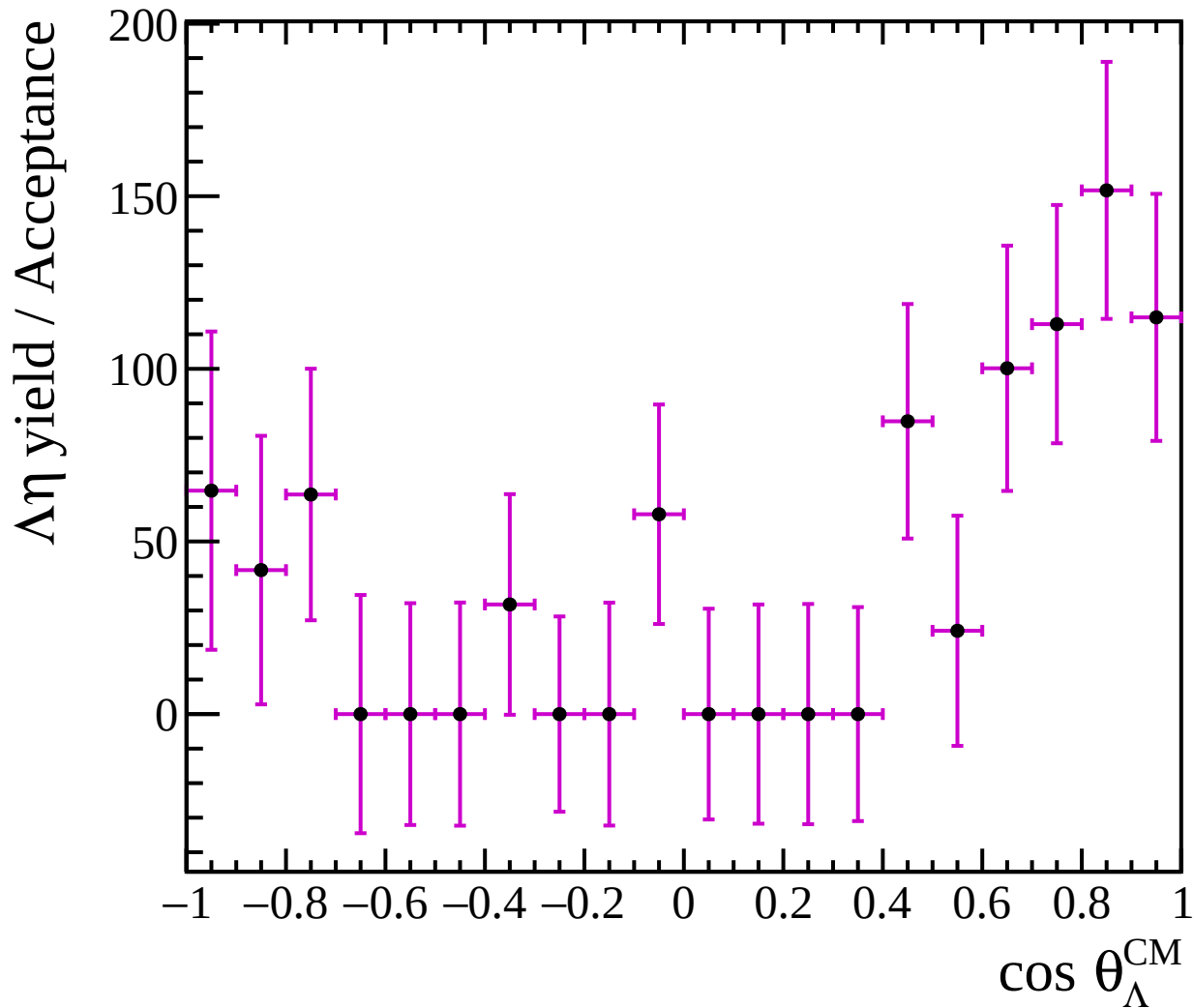
Selected Λ invariant mass (production vertex, no LH2 cut, $M_X > 0.50$)



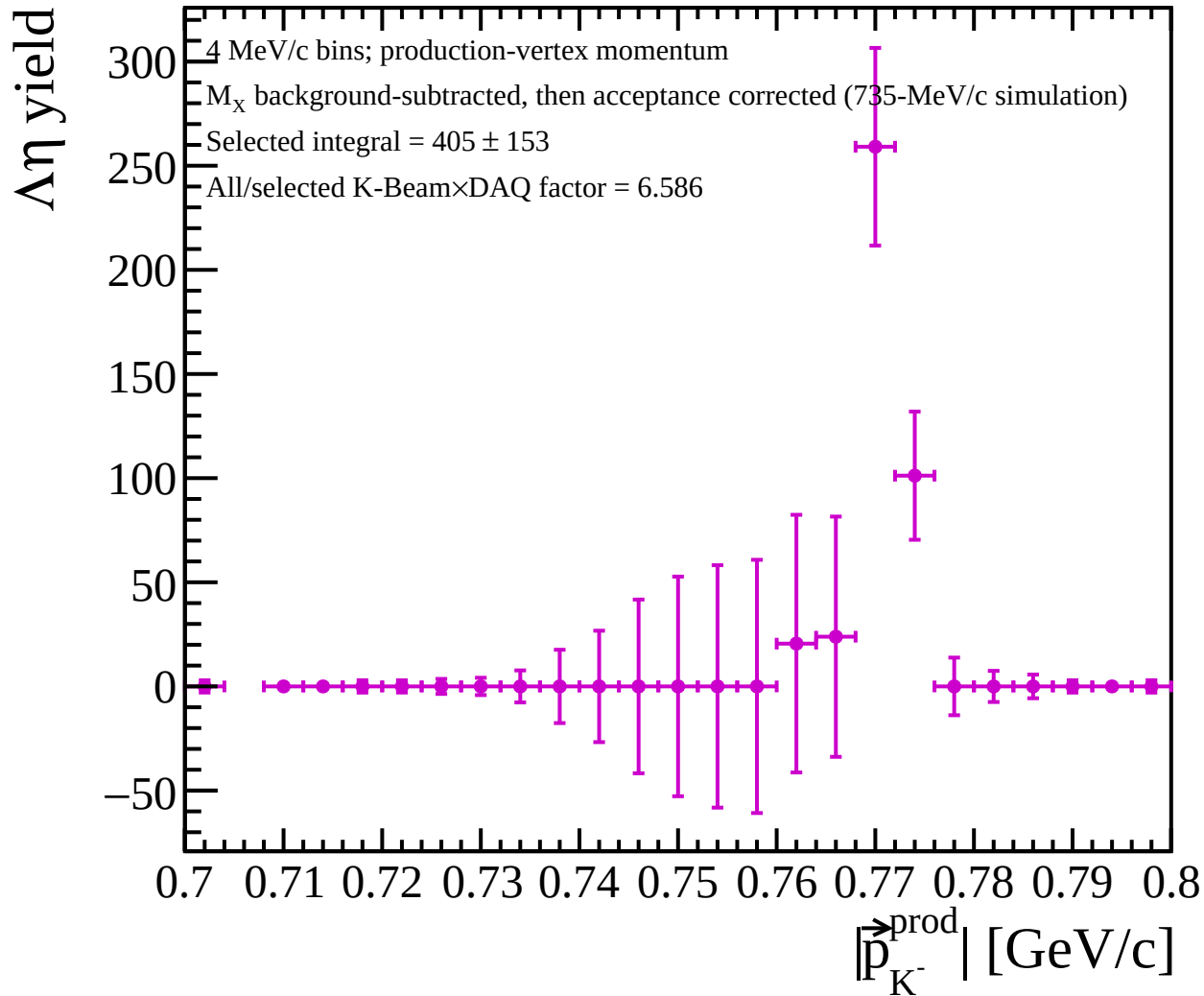
Trigger acceptance for generated $\Lambda\eta$



Background-subtracted $\Lambda\eta$ production angle, trigger-acceptance corrected (no LH2 cut, $1.09 < M_{p\pi^-} < 1.13$)



Background-subtracted $\Lambda\eta$ yield vs production-vertex K_{18} momentum (no LH2 cut, $1.09 < M_{p\pi^-} < 1.13$)



Expected $\Lambda\eta$ yield for all E72Physics runs (K-Beam scaled)

